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Zhou et al.

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(54) **CSI TRANSMISSION WITH MULTIPLE BANDWIDTH PARTS**

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See application file for complete search history.

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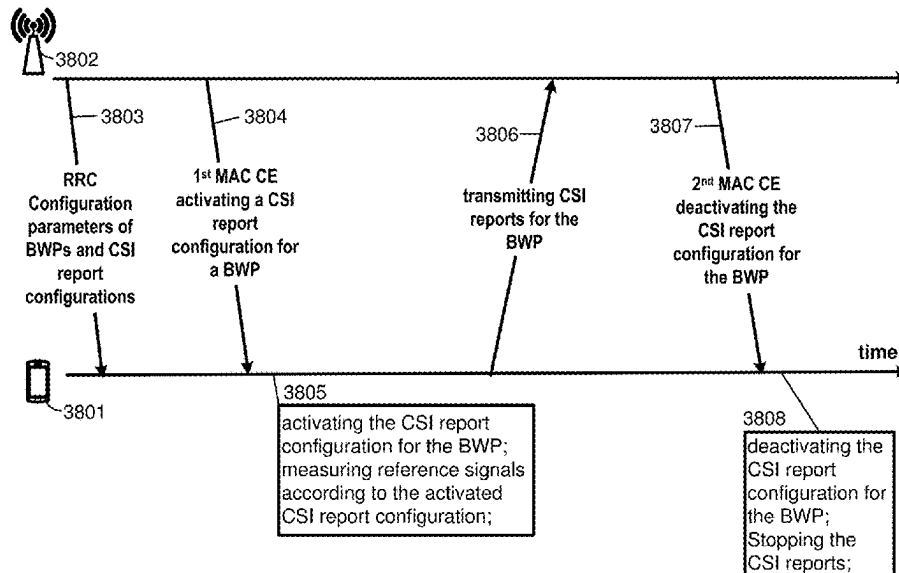
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(57) **ABSTRACT**

Systems, apparatuses, and methods are described for wireless communications. A base station may send, to a wireless device, a medium access control (MAC) control element (CE) comprising a bandwidth part identifier field and a channel state information (CSI) report configuration field. Based on the MAC CE, the wireless device may activate the CSI report configuration for a bandwidth part indicated by the bandwidth part identifier field. The wireless device may transmit one or more CSI reports based on the CSI report configuration.

30 Claims, 56 Drawing Sheets



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Zte et al: "QCL design for UL and DL MIMO", 3GPP Draft; R1-1715453 Qcl Design for UL and DL MIMO, 3rd Generation Partnership Project (3GPP), Mobile Competence Centre ; 650, Route Des Lucioles; F-06921 Sophia-Antipolis Cedex ; France. 3GPP TSG RAN WG1 Meeting #91, Reno, USA, Nov. 27-Dec. 1, 2017, Source: MCC Support, Title: Draft Report of 3GPP TSG RAN WG1 #90bis v0.1.0.

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Extended European Search Report mailed Jul. 11, 2025 in EP Patent Application No. 25166106.2.

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* cited by examiner

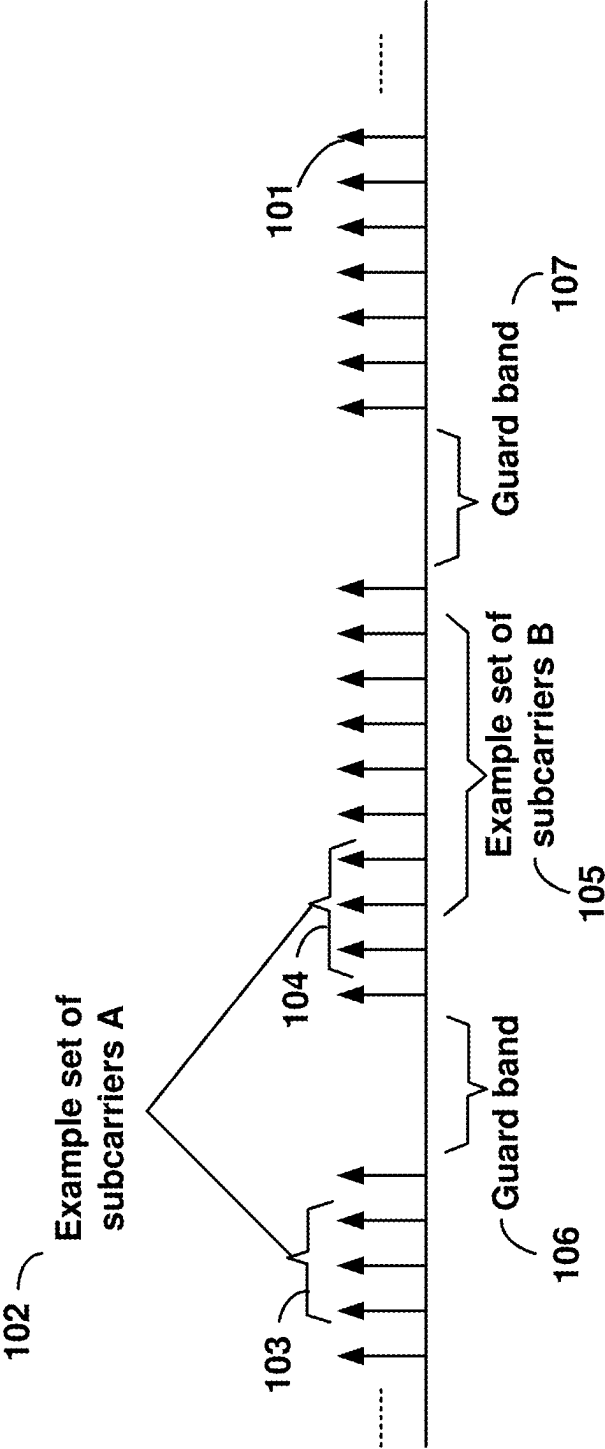


FIG. 1

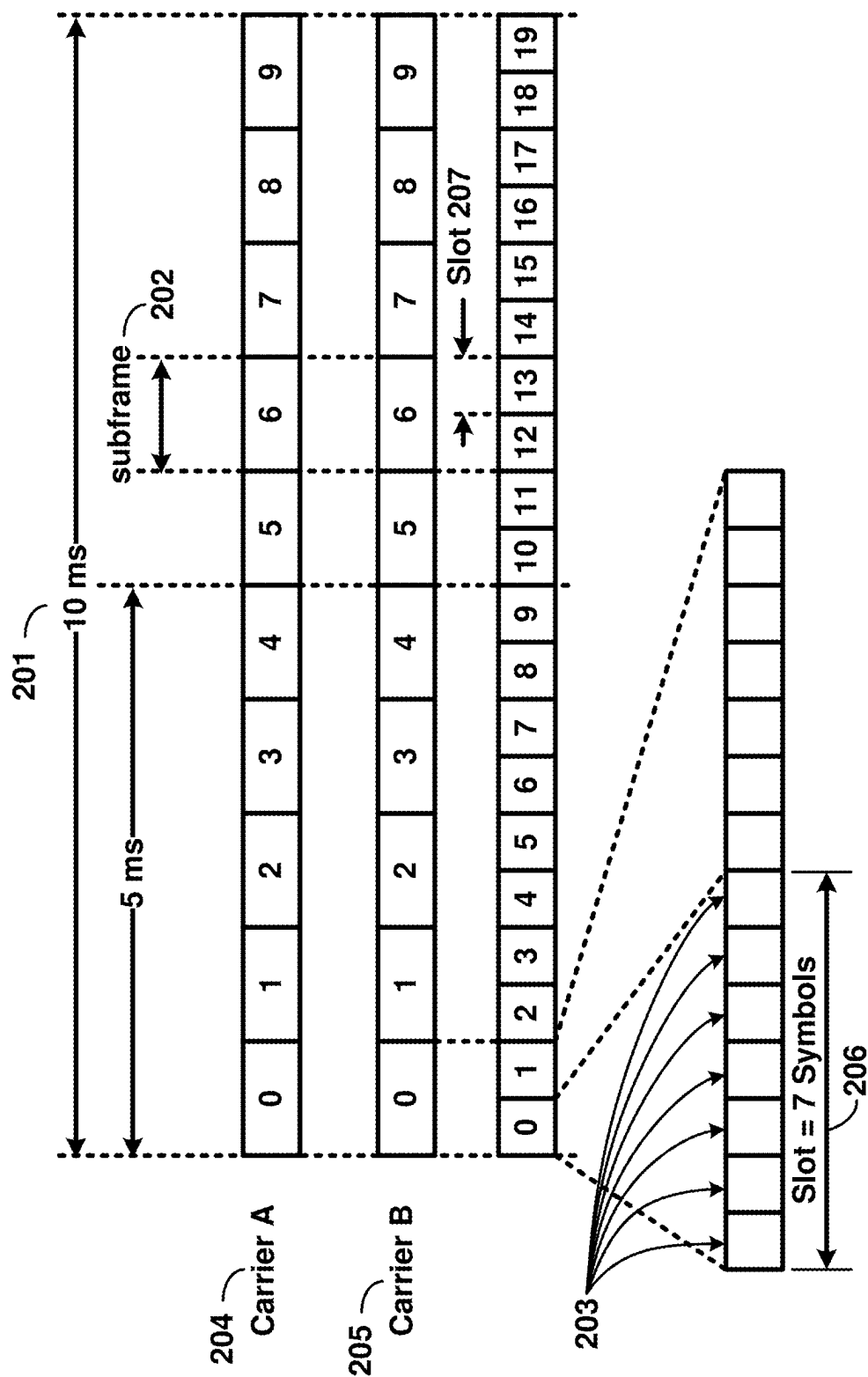


FIG. 2

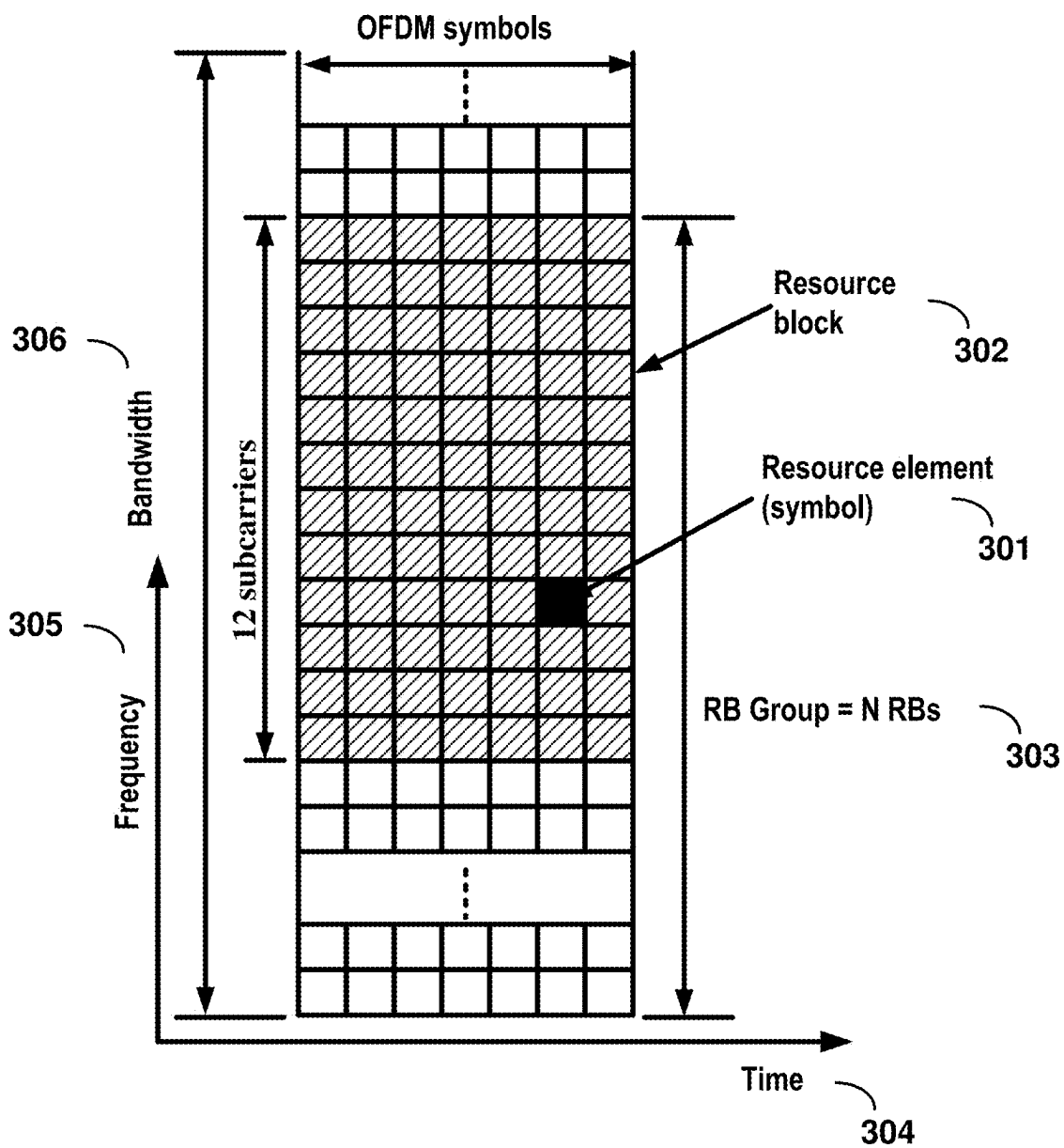


FIG. 3

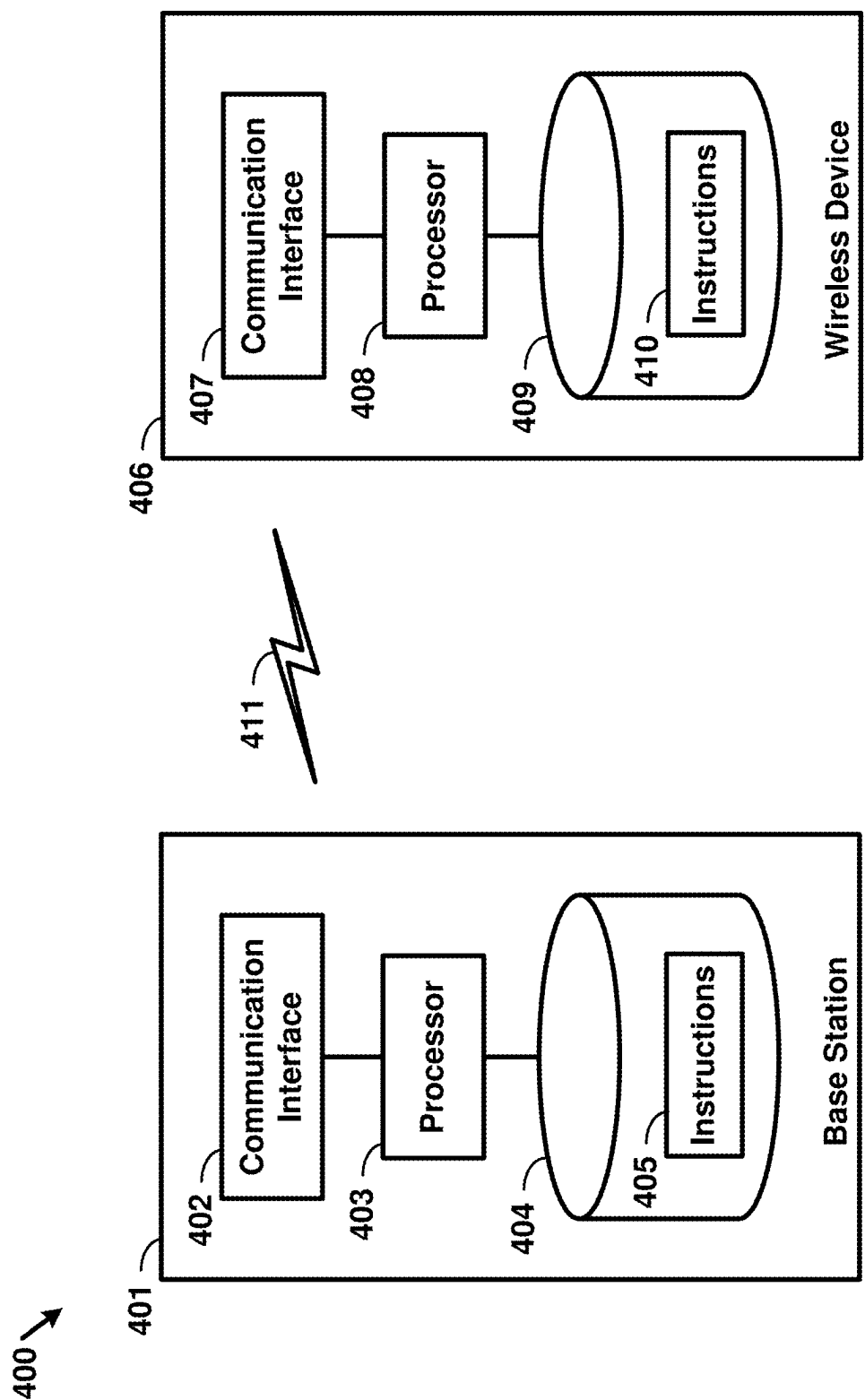


FIG. 4

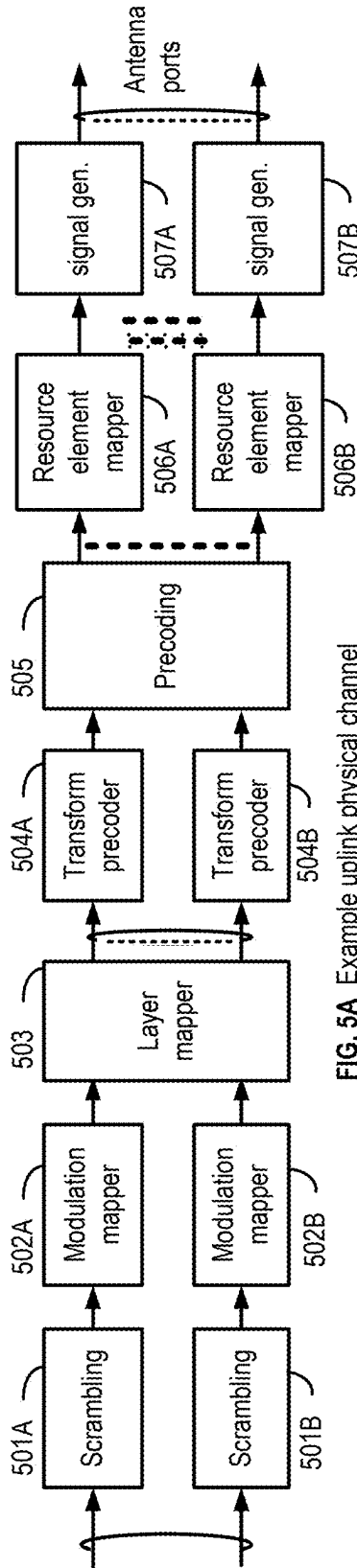


FIG. 5A Example uplink physical channel

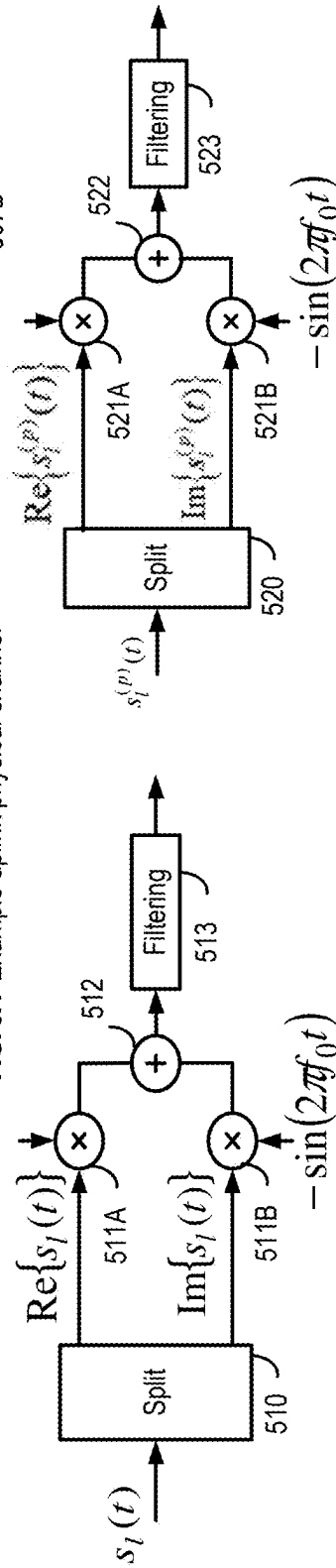


FIG. 5B Example uplink modulation

FIG. 5D Example downlink modulation

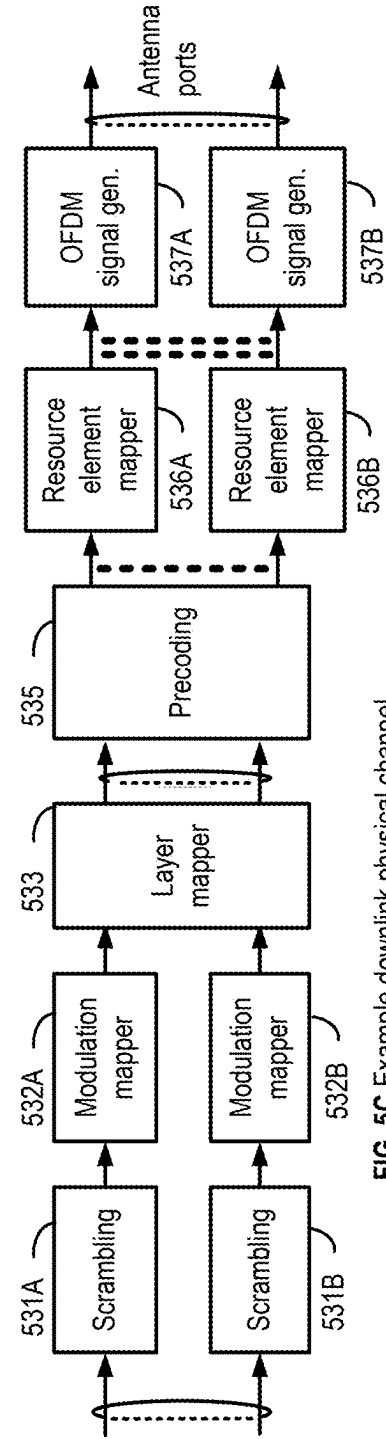


FIG. 5C Example downlink physical channel

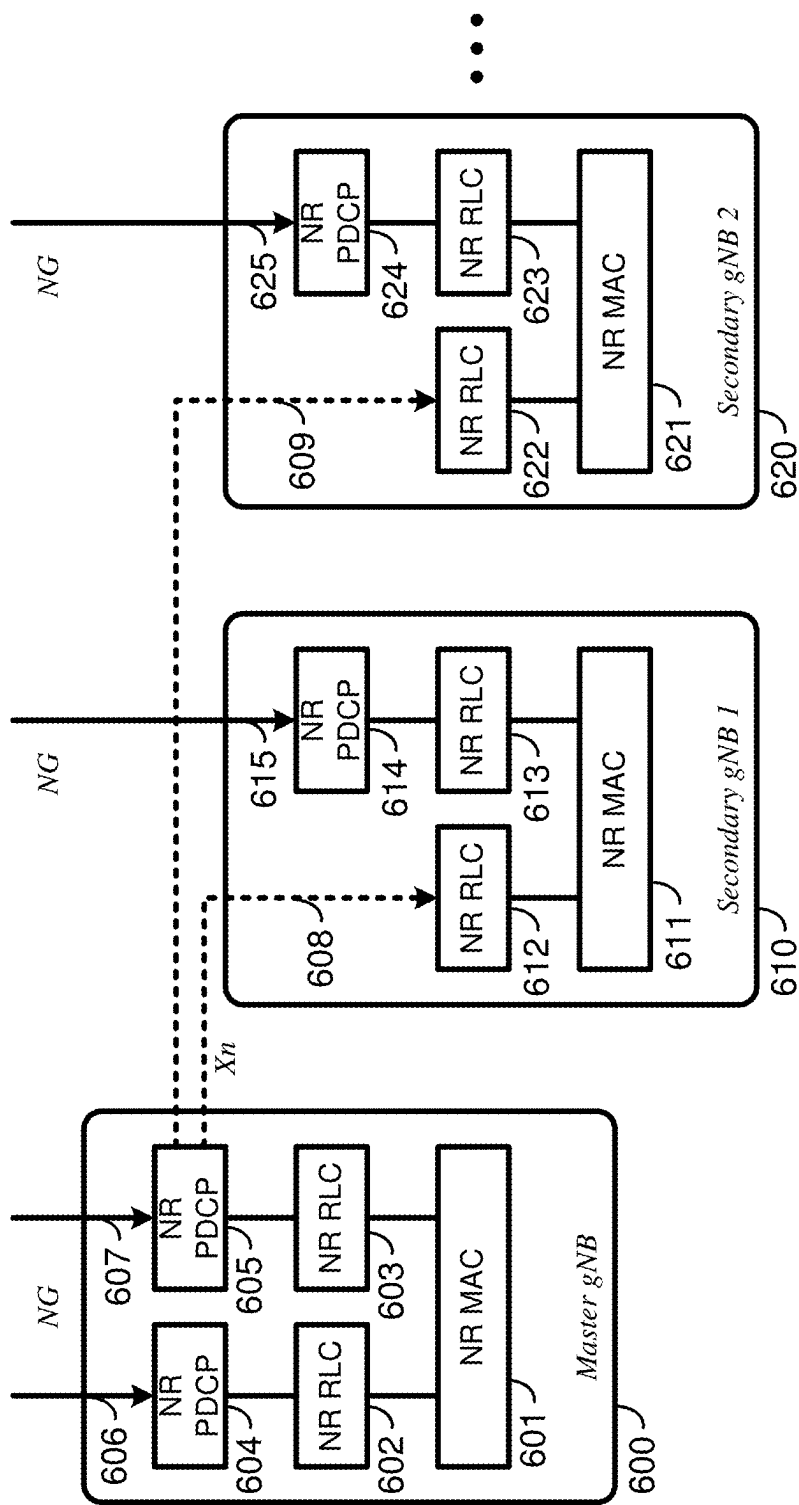
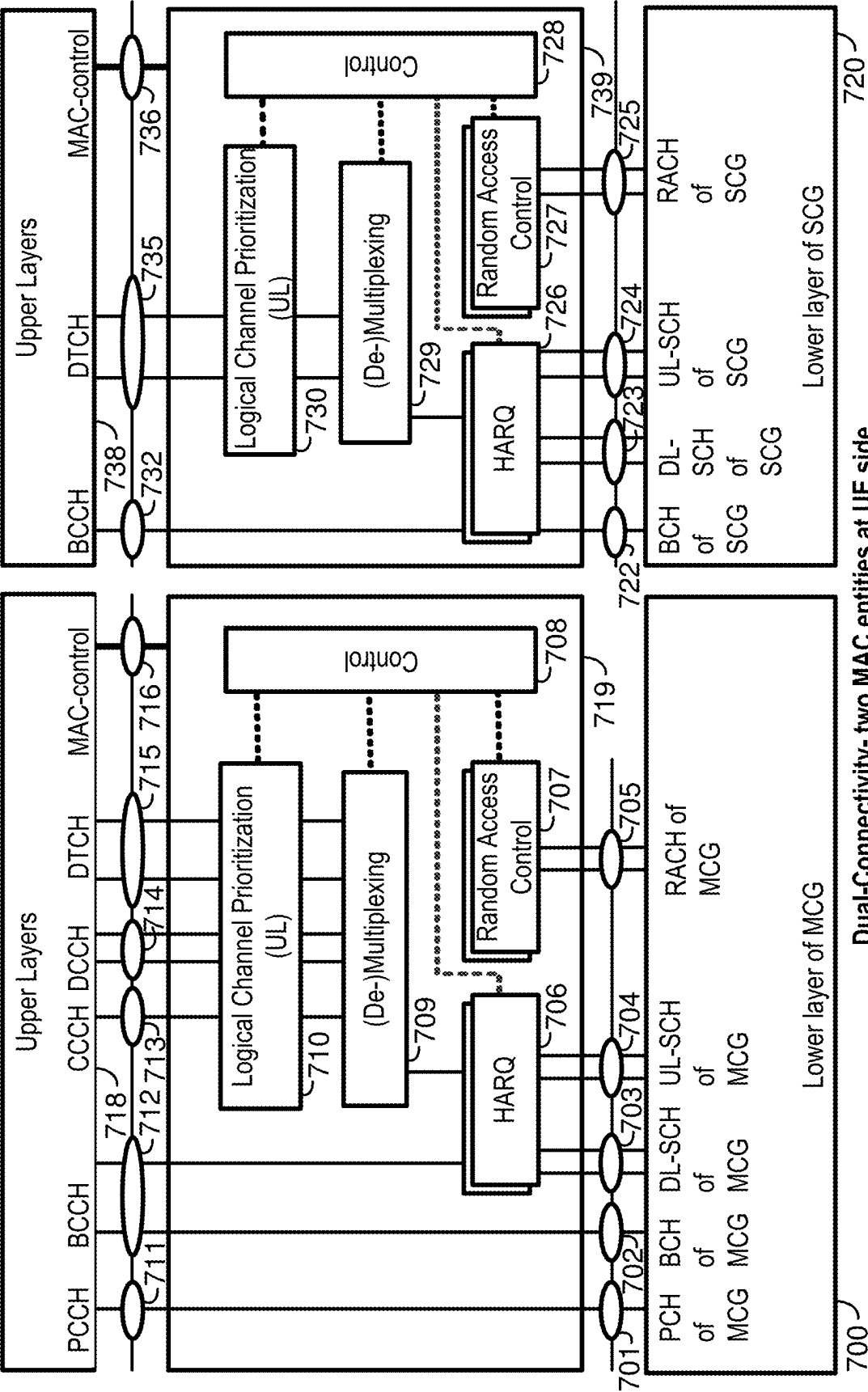


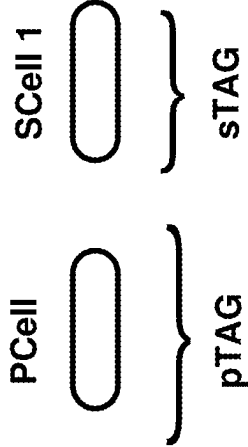
FIG. 6



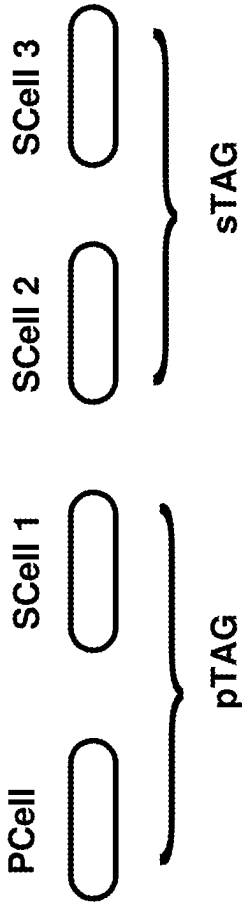
Dual-Connectivity- two MAC entities at UE side

FIG. 7

Example 1:



Example 2:



Example 3:

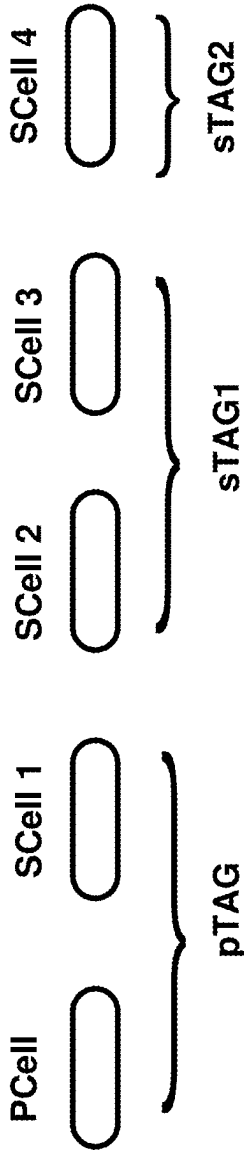


FIG. 8

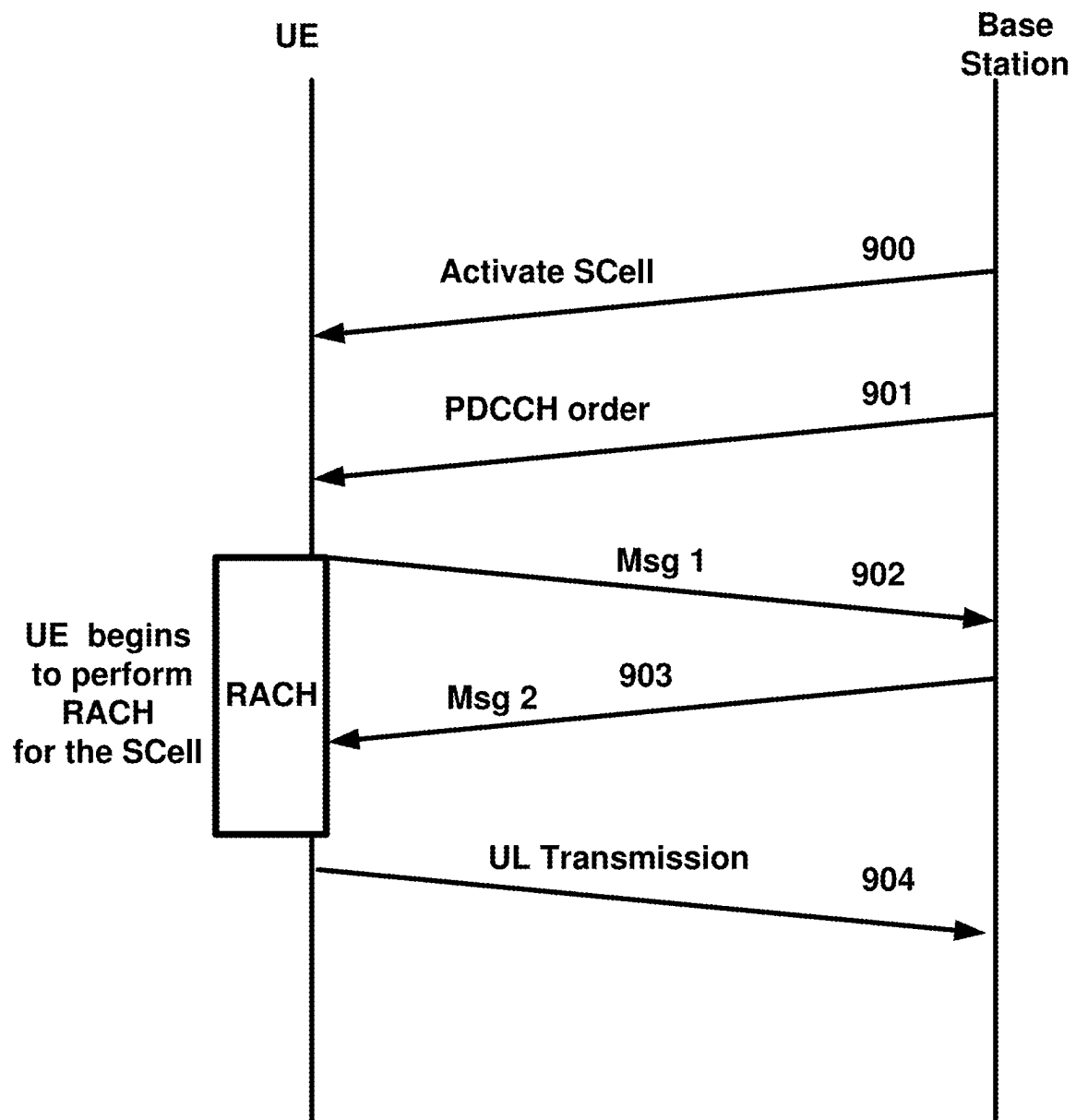


FIG. 9

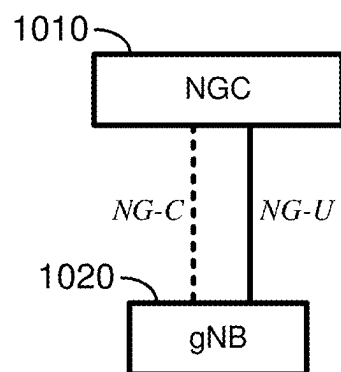


FIG. 10A gNB connected to NGC

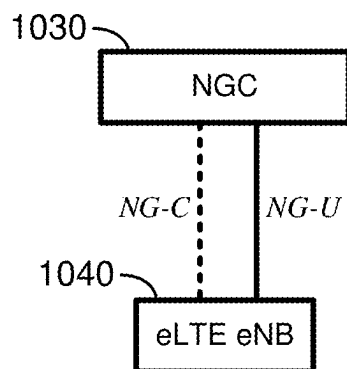


FIG. 10B eLTE eNB connected to NGC

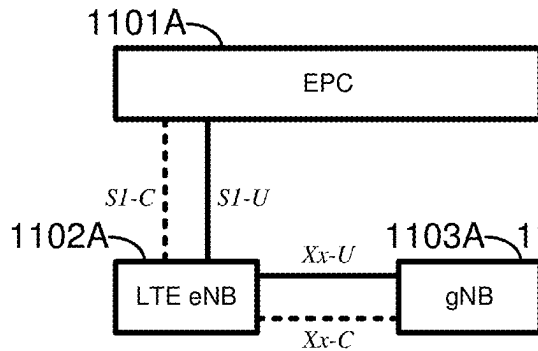


FIG. 11A LTE eNB connected to EPC with non-standalone gNB. gNB user plane connected to EPC via LTE eNB.

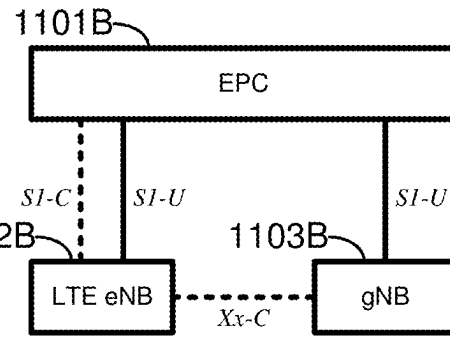


FIG. 11B LTE eNB connected to EPC with non-standalone gNB. gNB user plane connected to EPC directly.

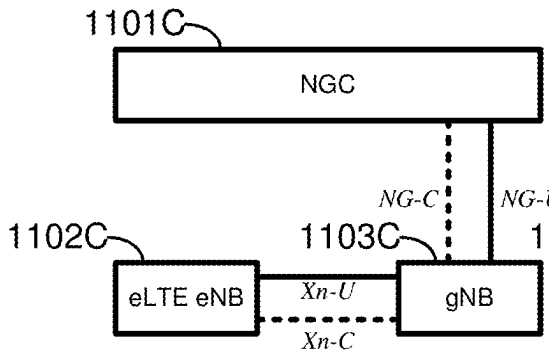


FIG. 11C gNB connected to NGC with non-standalone eLTE eNB. eLTE eNB user plane connected to NGC via gNB.

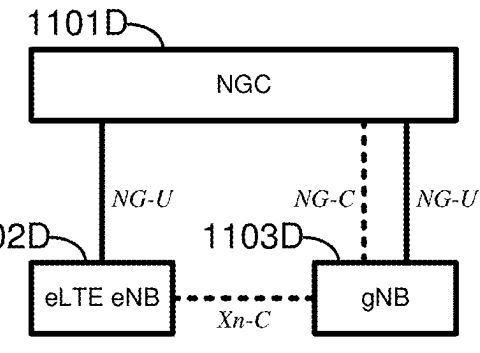


FIG. 11D gNB connected to NGC with non-standalone eLTE eNB. eLTE eNB user plane connected to NGC directly.

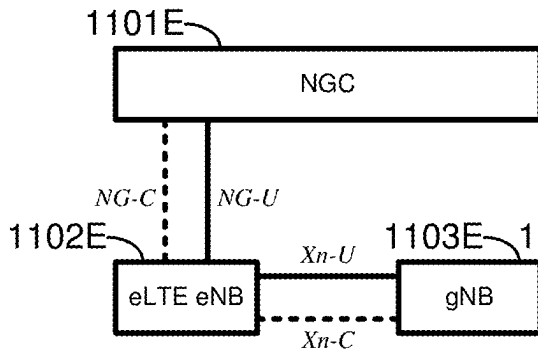


FIG. 11E eLTE eNB connected to NGC with non-standalone gNB. gNB user plane connected to NGC via eLTE eNB.

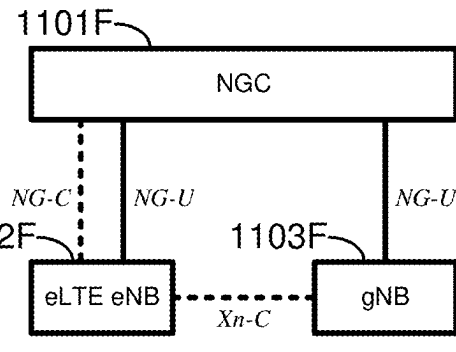


FIG. 11F eLTE eNB connected to NGC with non-standalone gNB. gNB user plane connected to NGC directly.

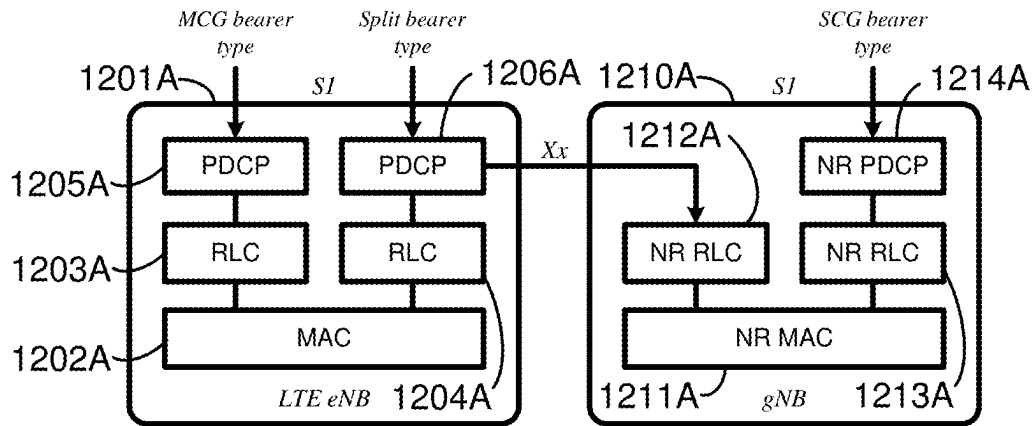


FIG. 12A Radio protocol architecture for split bearer and SCG bearer. LTE eNB connected to EPC with non-standalone gNB.

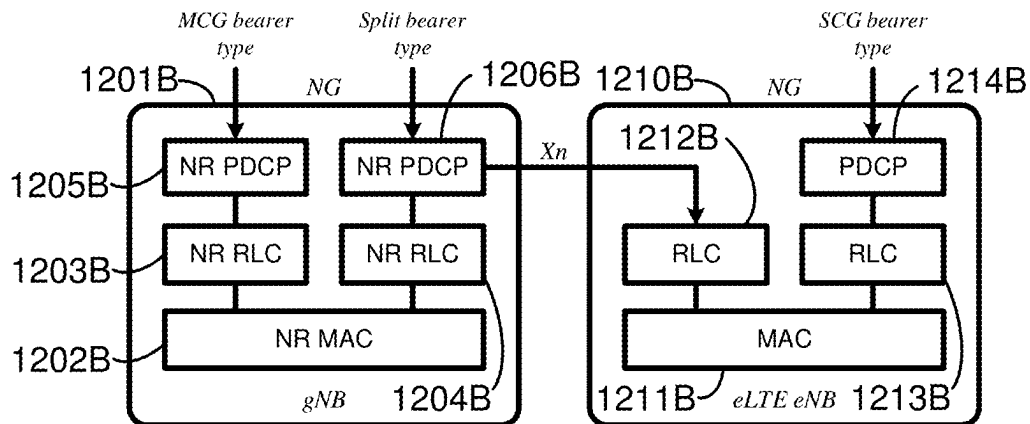


FIG. 12B Radio protocol architecture for split bearer and SCG bearer. gNB connected to NGC with non-standalone eLTE eNB.

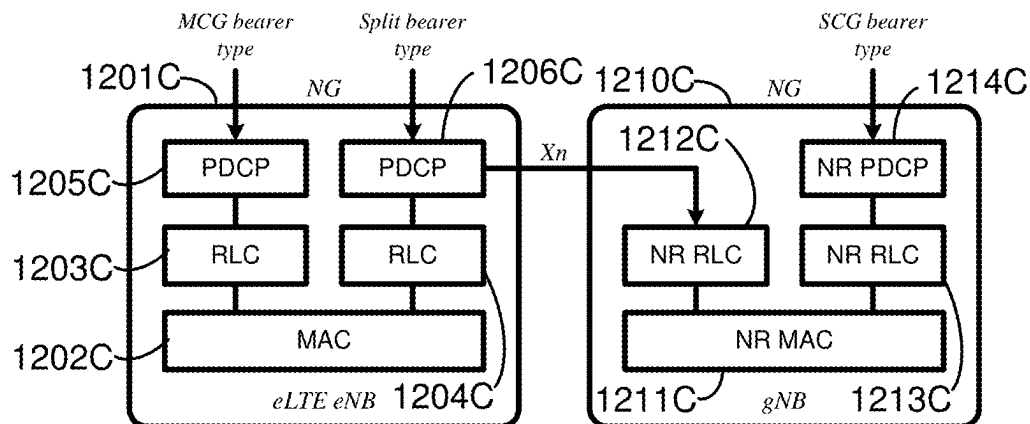


FIG. 12C Radio protocol architecture for split bearer and SCG bearer. eLTE eNB connected to NGC with non-standalone gNB.

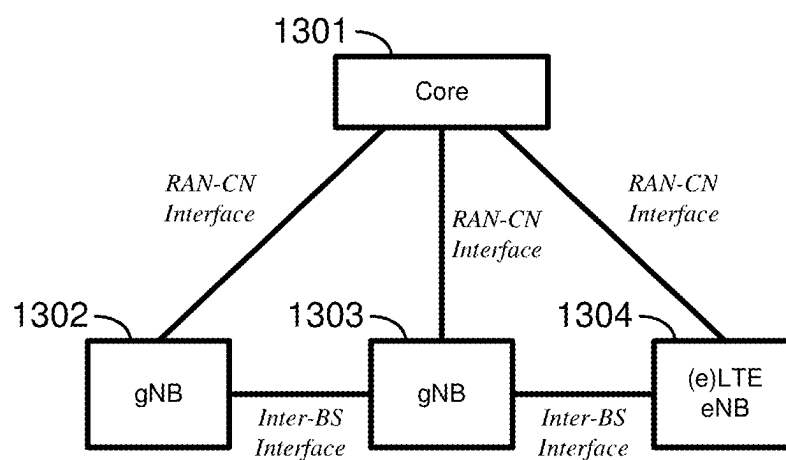


FIG. 13A Non-centralized deployment

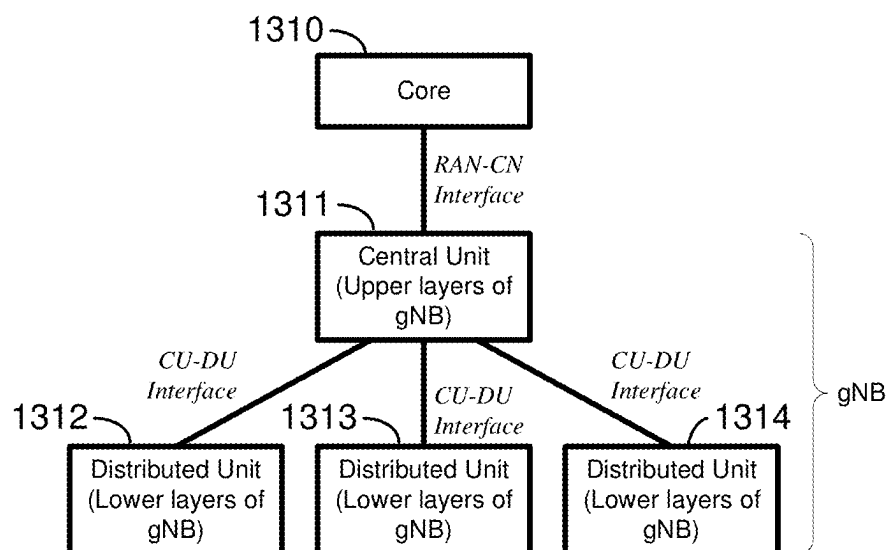


FIG. 13B Centralized deployment

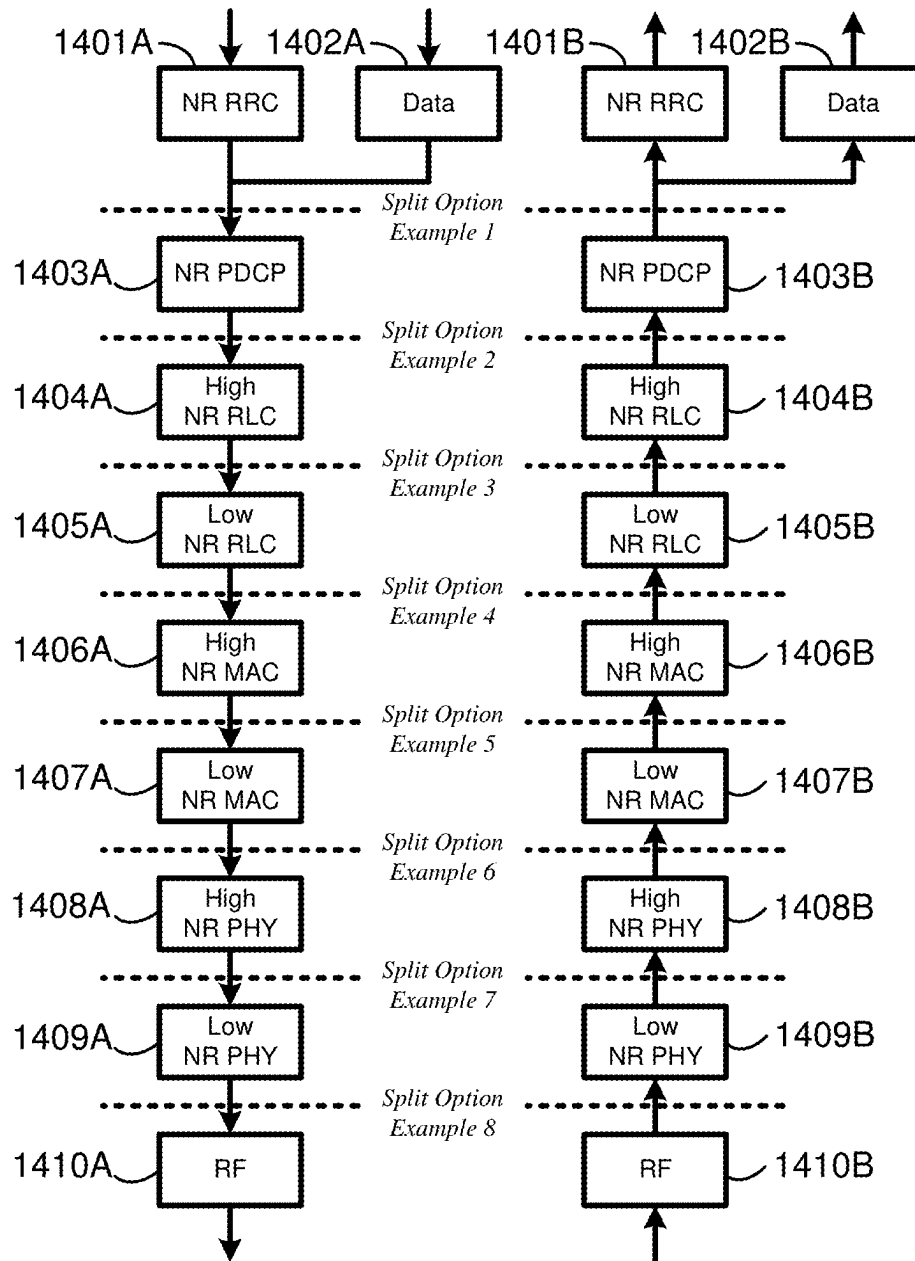


FIG. 14

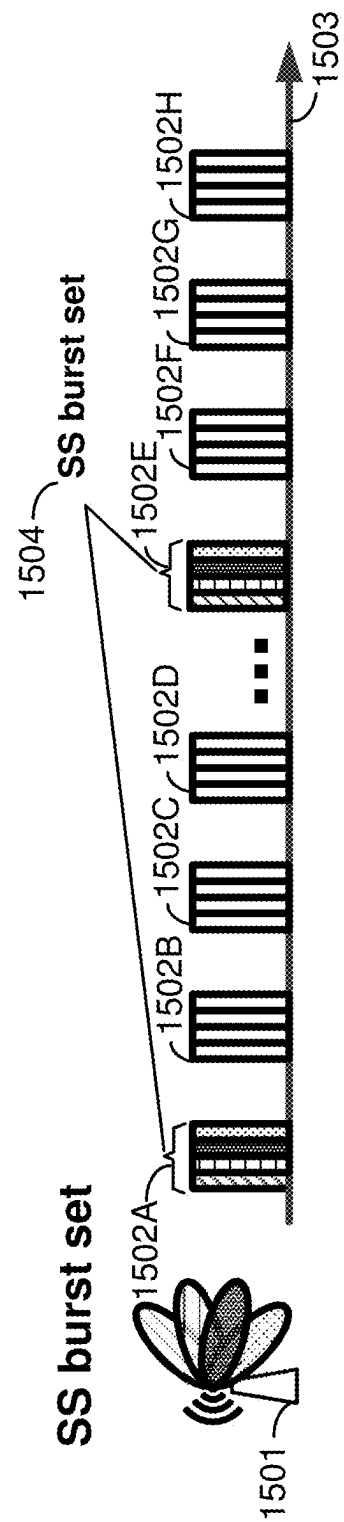


FIG. 15

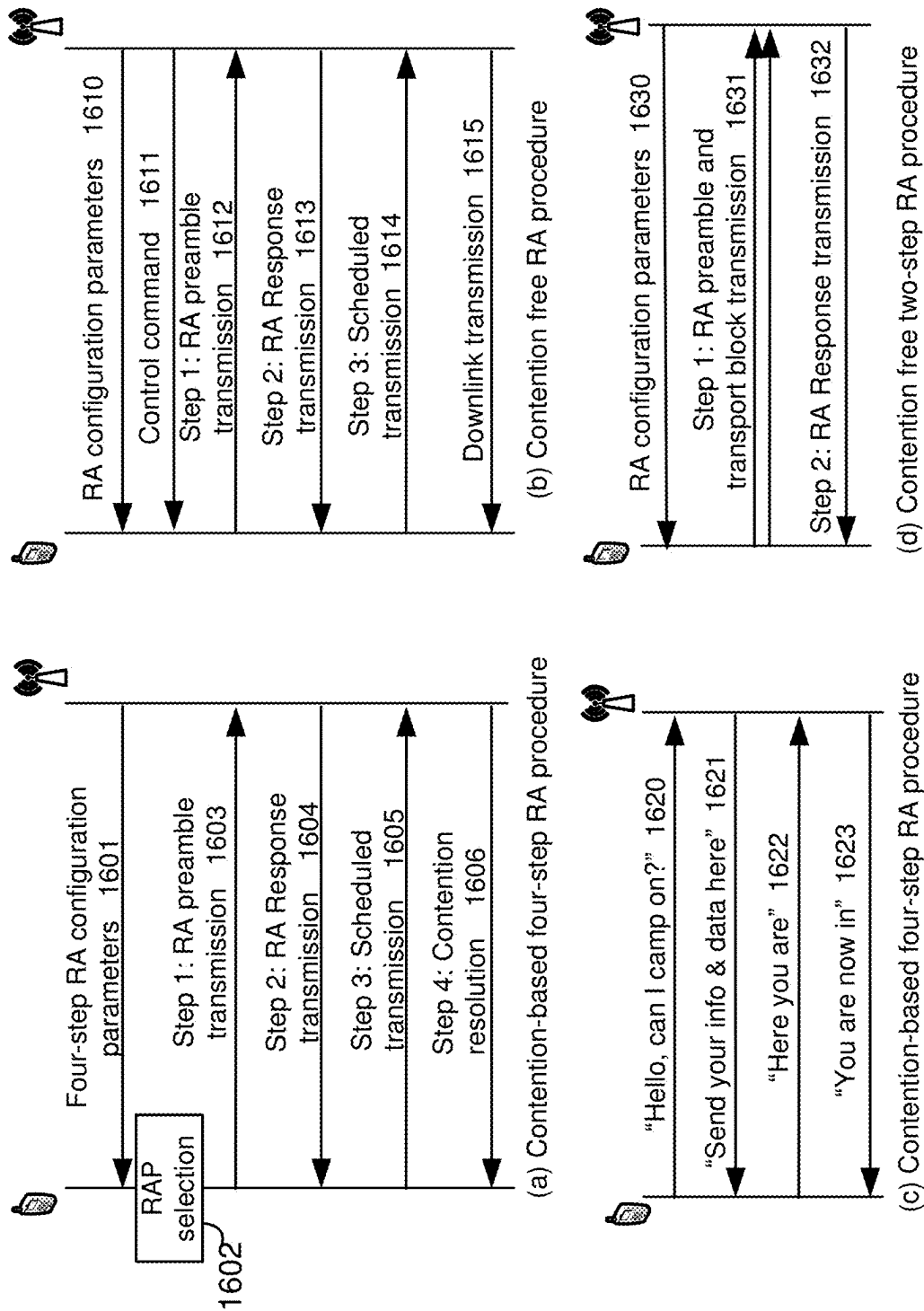


FIG. 16

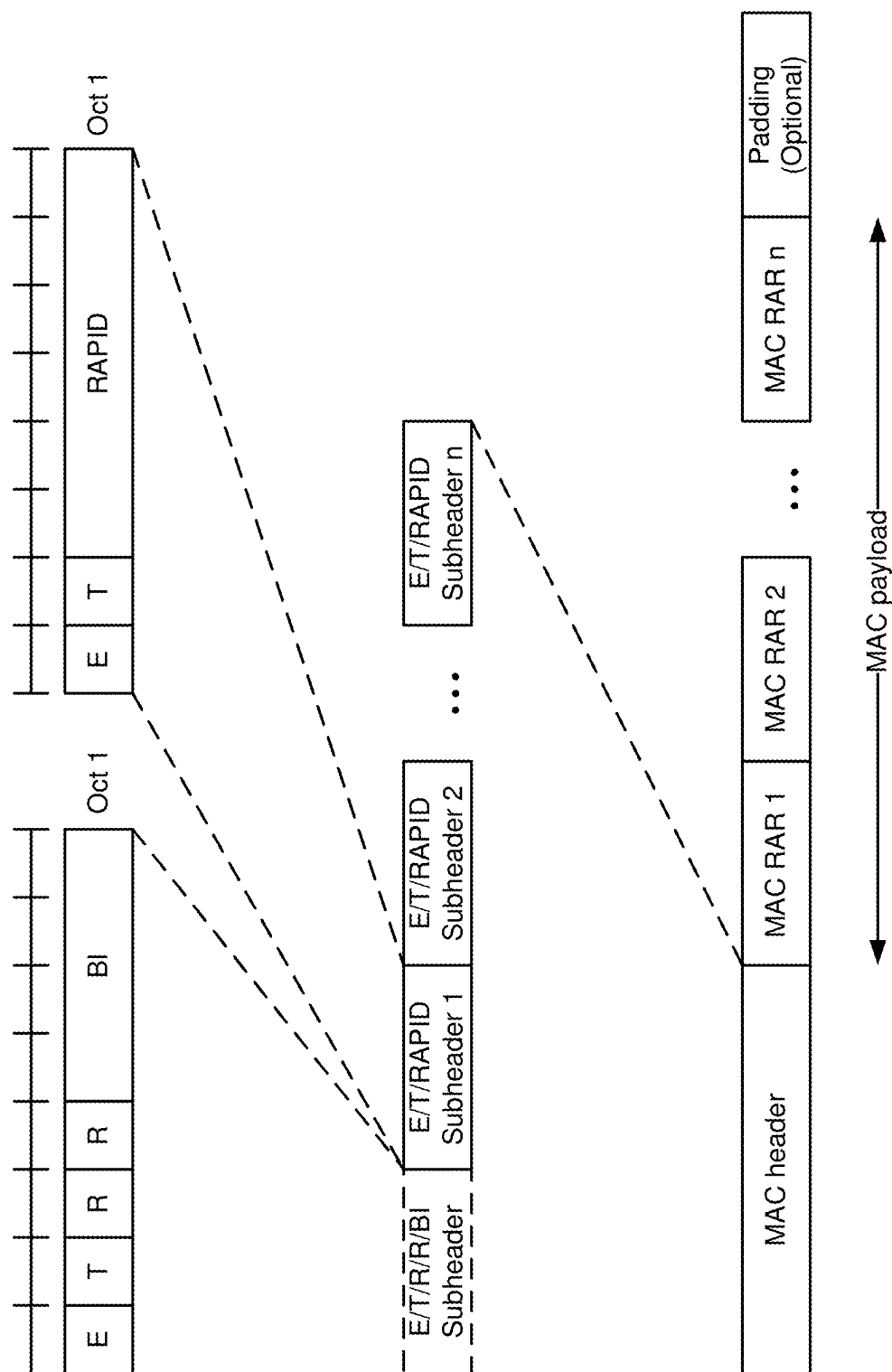


FIG. 17

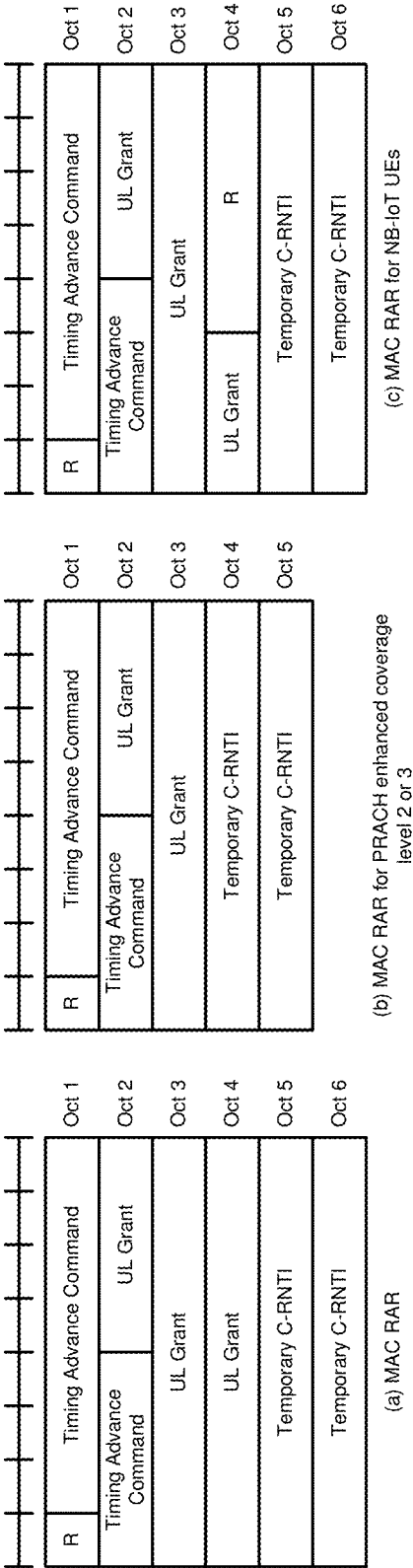


FIG. 18

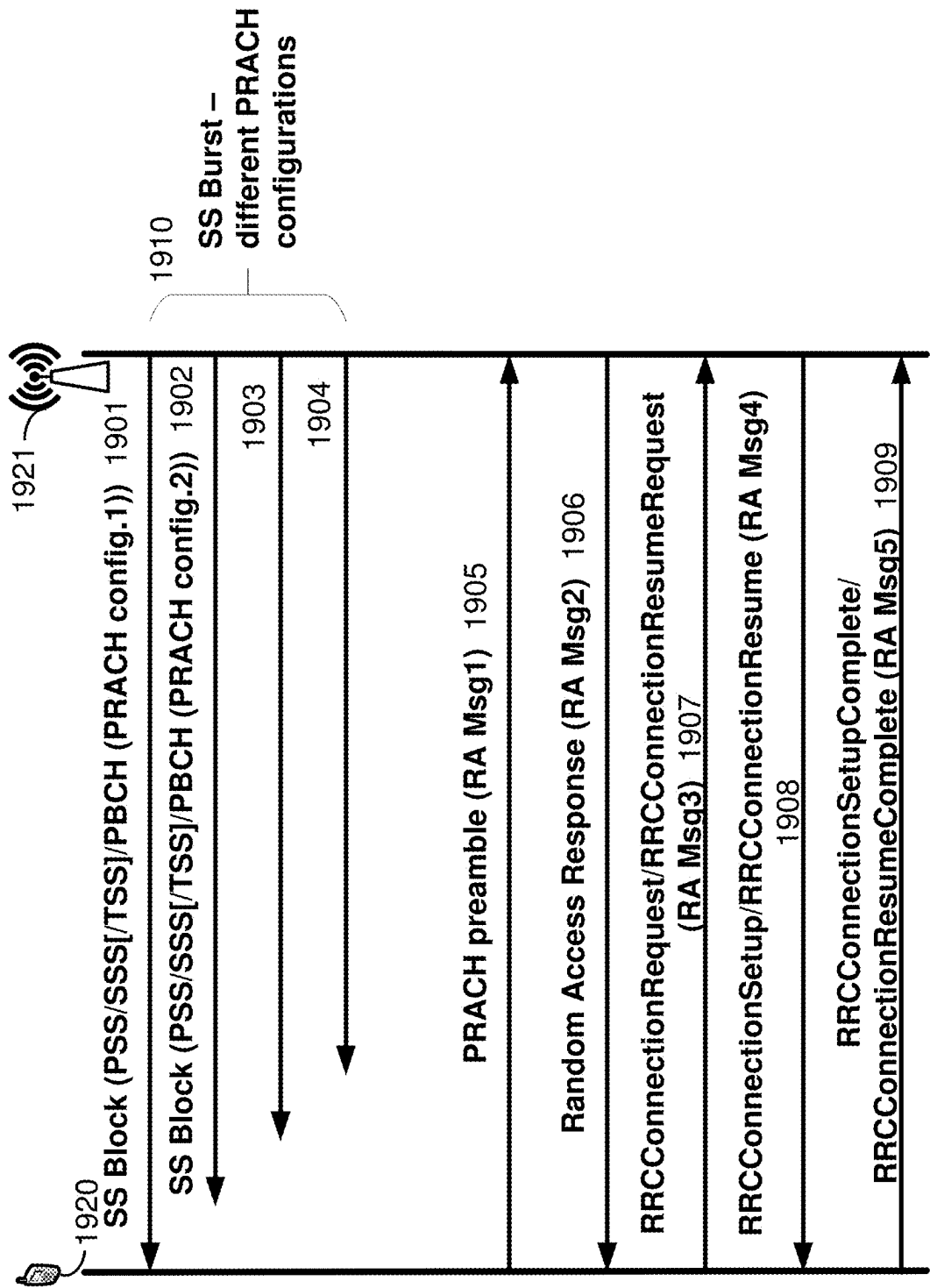


FIG. 19

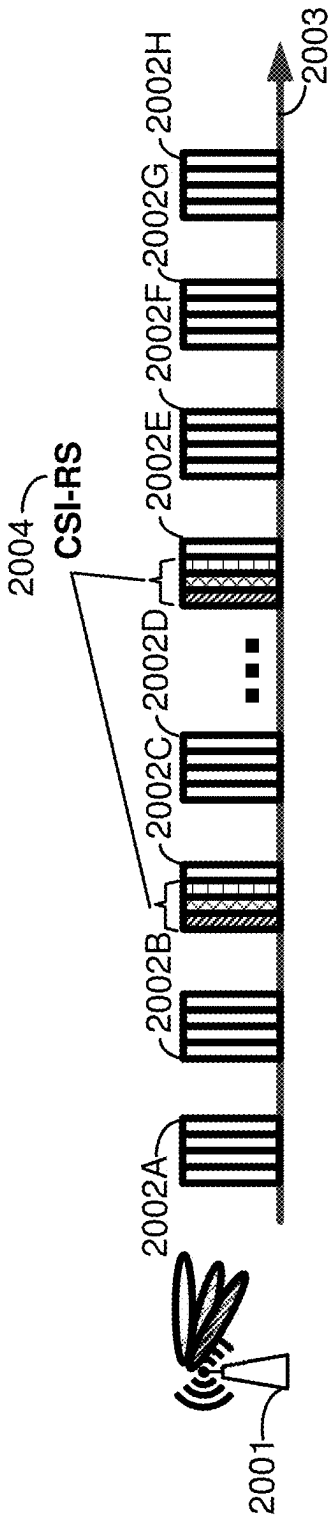


FIG. 20

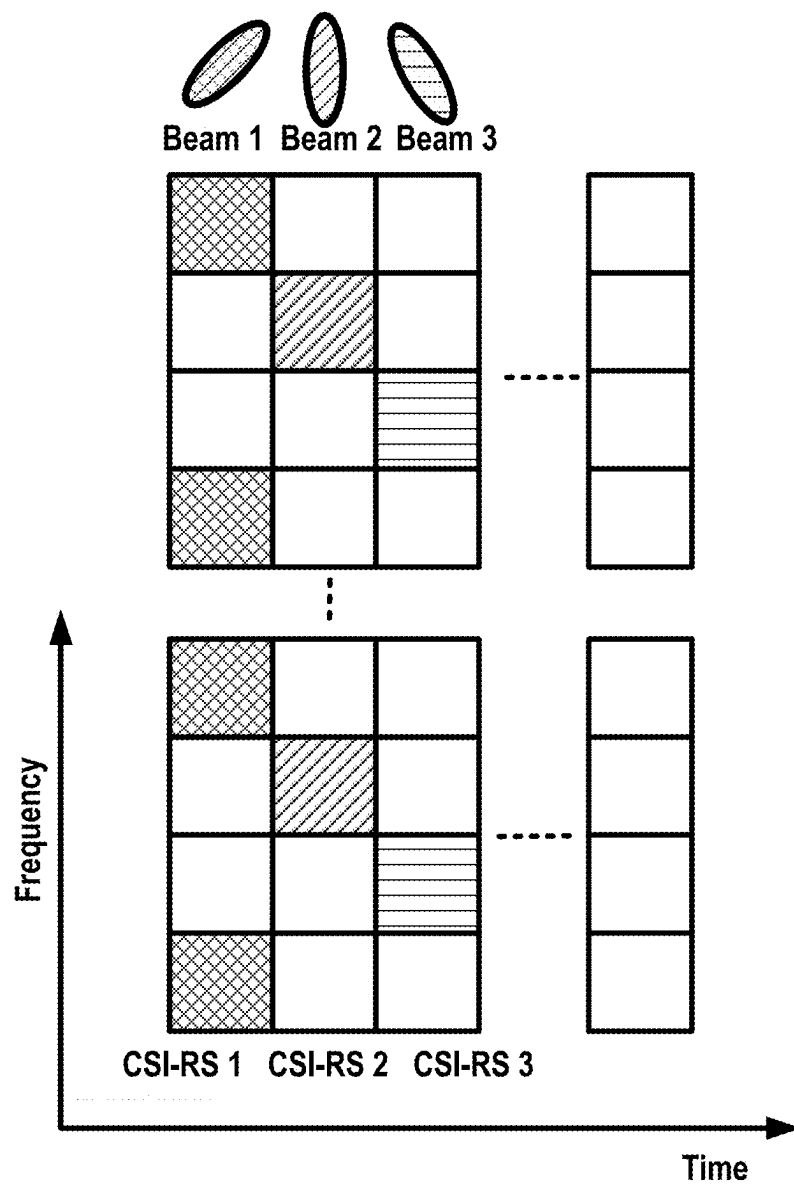


FIG. 21

```

PhysicalConfigDedicated ::= SEQUENCE {
    pdsch-ConfigDedicated PDSCH-ConfigDedicated OPTIONAL, --- Need ON
    pucch-ConfigDedicated PUCCH-ConfigDedicated OPTIONAL, --- Need ON
    pusch-ConfigDedicated PUSCH-ConfigDedicated OPTIONAL, --- Need ON
    uplinkPowerControlDedicated UplinkPowerControlDedicated OPTIONAL, --- Need ON
    tpc-PDCCH-ConfigPUCCH TPC-PDCCH-Config OPTIONAL, --- Need ON
    tpc-PDCCH-ConfigPUSCH TPC-PDCCH-Config OPTIONAL, --- Need ON
    ...
    cqi-ReportConfig-v1130 CQI-ReportConfig-v1130 OPTIONAL, --- Need ON
    ...
}
CQI-ReportConfig-v1130 ::= SEQUENCE {
    cqi-ReportPeriodic-v1130 CQI-ReportPeriodic-v1130,
    cqi-ReportBoth-r11 CQI-ReportBoth-r11
}
CQI-ReportBoth-r11 ::= SEQUENCE {
    csi-IM-ConfigToReleaseList-r11 CSI-IM-ConfigToReleaseList-r11 OPTIONAL, --- Need ON
    csi-IM-ConfigToAddModList-r11 CSI-IM-ConfigToAddModList-r11 OPTIONAL, --- Need ON
    csi-ProcessToReleaseList-r11 CSI-ProcessToReleaseList-r11 OPTIONAL, --- Need ON
    csi-ProcessToAddModList-r11 CSI-ProcessToAddModList-r11 OPTIONAL, --- Need ON
}
CSI-ProcessToAddModList-r11 ::= SEQUENCE (SIZE (1..maxCSI-Proc-r11)) OF CSI-Process-r11
CSI-Process-r11 ::= SEQUENCE {
    csi-ProcessId-r11 CSI-ProcessId-r11,
    csi-RS-ConfigNZPId-r11 CSI-RS-ConfigNZPId-r11,
    csi-IM-ConfigId-r11 CSI-IM-ConfigId-r11,
    p-C-AndCSRList-r11 F-C-AndCSRList-r11,
    cqi-ReportBothProc-r11 CQI-ReportBothProc-r11 OPTIONAL, --- Need OR
    cqi-ReportPeriodicProcId-r11 INTEGER (0..maxCQI-ProcExt-r11) OPTIONAL, --- Need OR
    cqi-ReportAperiodicProc-r11 CQI-ReportAperiodicProc-r11 OPTIONAL, --- Need OR
    ...,
    [[ alternativeCodebookEnabledFor4TXProc-r12 ENUMERATED {true} OPTIONAL, --- Need ON
    csi-IM-ConfigIdList-r12 CHOICE {
        release NULL,
        setup SEQUENCE (SIZE (1..2)) OF CSI-IM-ConfigId-r12 OPTIONAL, --- Need ON
    }
    cqi-ReportAperiodicProc2-r12 CHOICE {
        release NULL,
        setup CQI-ReportAperiodicProc-r11 OPTIONAL --- Need ON
    }
]],
    [[ cqi-ReportAperiodicProc-v1310 CHOICE {
        release NULL,
        setup CQI-ReportAperiodicProc-v1310 OPTIONAL, --- Need ON
    }
    cqi-ReportAperiodicProc2-v1310 CHOICE {
        release NULL,
        setup CQI-ReportAperiodicProc-v1310 OPTIONAL, --- Need ON
    }
    eMIMO-Type-r13 CSI-RS-ConfigEMIMO-r13 OPTIONAL --- Need ON
]],
    [[ eMIMO-Type-v1430 CSI-RS-ConfigEMIMO-v1430 OPTIONAL, --- Need ON
    eMIMO-Hybrid-r14 CSI-RS-ConfigEMIMO-Hybrid-r14 OPTIONAL, --- Need ON
    advancedCodebookEnabled-r14 BOOLEAN OPTIONAL --- Need ON
    ]]
}
CSI-RS-Config-v1430 ::= SEQUENCE {
    eMIMO-Type-v1430 CSI-RS-ConfigEMIMO-v1430 OPTIONAL, --- Need ON
    eMIMO-Hybrid-r14 CSI-RS-ConfigEMIMO-Hybrid-r14 OPTIONAL, --- Need ON
    advancedCodebookEnabled-r14 BOOLEAN OPTIONAL --- Need ON
}

```

FIG. 22A

```

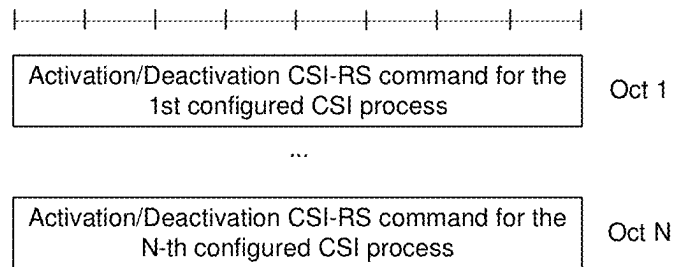
CSI-RS-ConfigEMIMO-v1430 ::= CHOICE {
    release          NULL,
    setup            CHOICE {
        nonPrecoded-v1430    CSI-RS-ConfigNonPrecoded-v1430,
        beamformed-v1430     CSI-RS-ConfigBeamformed-v1430
    }
}
CSI-RS-ConfigBeamformed-v1430 ::= SEQUENCE {
    csi-RS-ConfigNZP-ApList-r14    SEQUENCE (SIZE (1..8)) OF CSI-RS-ConfigNZP-r11
                                     OPTIONAL, -- Need OR
    nzp-ResourceConfigOriginal-v1430 CSI-RS-Config-NZP-v1430    OPTIONAL, -- Need OR
    csi-RS-NZP-Activation-r14       CSI-RS-ConfigNZP-Activation-r14 OPTIONAL -- Need OR
}

CSI-RS-ConfigNZP-Activation-r14 ::= SEQUENCE {
    csi-RS-NZP-mode-r14    ENUMERATED {semiPersistent, aperiodic},
    activatedResources-r14 INTEGER (0..4)
}

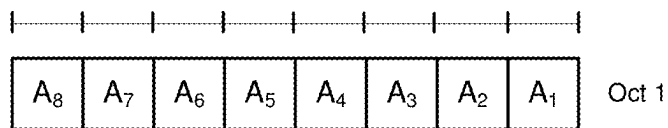
CSI-RS-ConfigNZP-r11 ::= SEQUENCE {
    csi-RS-ConfigNZPId-r11    CSI-RS-ConfigNZPId-r11,
    antennaPortsCount-r11    ENUMERATED {an1, an2, an4, an8},
    resourceConfig-r11       INTEGER (0..31),
    subframeConfig-r11       INTEGER (0..154),
    scramblingIdentity-r11    INTEGER (0..503),
    qcl-CRS-Info-r11         SEQUENCE {
        qcl-ScramblingIdentity-r11    INTEGER (0..503),
        crs-PortsCount-r11            ENUMERATED {n1, n2, n4, spare1},
        mbsfn-SubframeConfigList-r11 CHOICE {
            release          NULL,
            setup            SEQUENCE {
                subframeConfigList    MBSFN-SubframeConfigList
            }
        }
    }
}
OPTIONAL -- Need ON
OPTIONAL, -- Need OR
....
[[ csi-RS-ConfigNZPId-v1310    CSI-RS-ConfigNZPId-v1310    OPTIONAL -- Need ON
]],
[[ transmissionComb-r14       NZP-TransmissionComb-r14    OPTIONAL, -- Need OR
   frequencyDensity-r14       NZP-FrequencyDensity-r14     OPTIONAL -- Need OR
]],
[[ mbsfn-SubframeConfigList-v1430 CHOICE {
    release          NULL,
    setup            SEQUENCE {
        subframeConfigList-v1430    MBSFN-SubframeConfigList-v1430
    }
}
OPTIONAL -- Need OF
]]
}

```

FIG. 22B



A) Activation/Deactivation of CSI-RS resources MAC Control Element



B) Activation/Deactivation CSI-RS command

FIG. 23

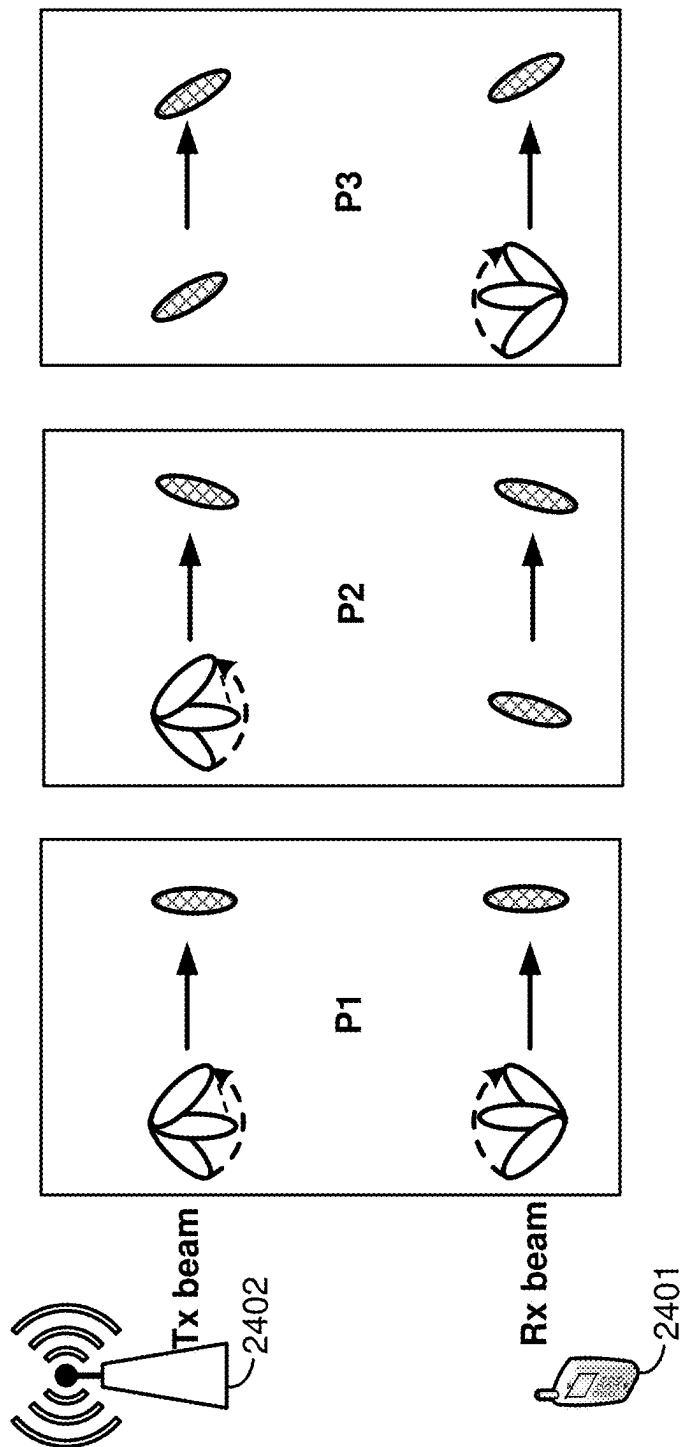


FIG. 24

```

RRCConnectionReconfiguration-v1020-IEs ::= SEQUENCE {
    sCellToReleaseList-r10          OPTIONAL,  --- Need ON
    sCellToAddModList-r10          OPTIONAL,  --- Need ON
    nonCriticalExtension            OPTIONAL
}

...
SCellToAddModList-r10 ::= SEQUENCE (SIZE (1..maxSCell-r10)) OF SCellToAddMod-r10
...
SCellToAddMod-r10 ::= SEQUENCE {
    sCellIndex-r10,
    cellIdentification-r10
    physCellId-r10
    dl-CarrierFreq-r10
}
radioResourceConfigCommonSCell-r10          OPTIONAL,  --- Cond SCellAdd
radioResourceConfigDedicatedSCell-r10      RadioResourceConfigCommonSCell-r10 OPTIONAL,  --- Cond SCellAdd
...,
[{ dl-CarrierFreq-v1090
}],
[{ antennaInfoDedicatedSCell-v10i0
}],
[{ srs-SwitchFromServCellIndex-r14
}],
}
...
SCellToReleaseList-r10 ::= SEQUENCE (SIZE (1..maxSCell-r10)) OF SCellIndex-r10

RadioResourceConfigCommonSCell-r10 ::= SEQUENCE {
    -- DL configuration as well as configuration applicable for DL and UL
    nonUL-Configuration-r10          SEQUENCE {
        -- 1: Cell characteristics
        dl-Bandwidth-r10
        -- 2: Physical configuration, general
        antennaInfoCommon-r10
        mbsfn-SubframeConfigList-r10
        -- 3: Physical configuration, control
        pchich-Config-r10
        -- 4: Physical configuration, physical channels
        pdsch-ConfigCommon-r10
        tdd-Config-r10
    },
    enumerated {n6, n15, n25, n50, n75, n100},
    antennaInfoCommon,
    mbsfn-SubframeConfigList          OPTIONAL,  --- Need OR
    pchich-Config,
    pdsch-ConfigCommon,
    tdd-Config                        OPTIONAL  --- Cond TDDSCell
},

```

FIG. 25A

FIG. 25B


```

additionalSpectrumEmissionSCell-r14      AdditionalSpectrumEmission
},
P-Max-r14                                P-Max
    soundingRS-UL-ConfigCommon-r14        SoundingRS-UL-ConfigCommon,
    ul-CyclicPrefixLength-r14             UL-CyclicPrefixLength,
    prach-ConfigSCell-r14                 PRACH-ConfigSCell-r10
    uplinkPowerControlCommonPUSCH-LessCell-v1430
LessCell-v1430 OPTIONAL -- Need OR      OPTIONAL, --- Cond TBD-OR-NoR11
                                        UplinkPowerControlCommonPUSCH-
}
    harq-ReferenceConfig-r14               OPTIONAL, --- Cond ULRS
    soundingRS-FlexibleTiming-r14          ENUMERATED {sa2,sa4,sa5} --- Need OR
},                                         OPTIONAL, --- Need OR
    ENUMERATED {true}
[[ mbsfn-SubframeConfigList-v1430        MBSFN-SubframeConfigList-v1430  OPTIONAL--- Need ON
]]
}

RadioResourceConfigDedicatedSCell-r10 ::= SEQUENCE {
    -- UE specific configuration extensions applicable for an SCell
    physicalConfigDedicatedSCell-r10      PhysicalConfigDedicatedSCell-r10  OPTIONAL, --- Need ON
    ...,
    [[ mac-MainConfigSCell-r11             MAC-MainConfigSCell-r11      OPTIONAL --- Cond SCellAdd
    ]],
    [[ naics-Info-r12                      NAICS-AssistanceInfo-r12    OPTIONAL --- Need ON
    ]],
    [[ neighCellsCRS-InfoSCell-r13         NeighCellsCRS-Info-r13     OPTIONAL --- Need ON
    ]]
}

ServCellIndex-r10 ::=
ServCellIndex-r13 ::=
    INTEGER (0..7)
    INTEGER (0..31)

PhysicalConfigDedicatedSCell-r10 ::= SEQUENCE {
    -- DL configuration as well as configuration applicable for DL and UL
    nonUL-Configuration-r10               SEQUENCE {
        antennaInfo-r10
        crossCarrierSchedulingConfig-r10  CrossCarrierSchedulingConfig-r10
        csi-RS-Config-r10                 CSI-RS-Config-r10
        pdsch-ConfigDedicated-r10         PDSCH-ConfigDedicated
    }
    }
    OPTIONAL, --- Cond SCellAdd
    OPTIONAL, --- Need ON
    OPTIONAL, --- Need ON
    OPTIONAL, --- Need ON
    OPTIONAL, --- Cond SCellAdd

```

FIG. 25C

FIG. 25E

```

[[ pucch-Cell-r13
    pucch-SCell
    release
    setup
        pucch-ConfigDedicated-r13
            schedulingRequestConfig-r13
                PUCCH-ConfigDedicated-r13 OPTIONAL, --- Need ON
            tpc-PDCCH-ConfigPUCCH-SCell-r13
                SchedulingRequestConfigSCell-r13 OPTIONAL, --- Need ON
            pusch-ConfigDedicated-r13
                TPC-PDCCH-ConfigSCell-r13 OPTIONAL, --- Need ON
            uplinkPowerControlDedicated-r13
                PUSCH-ConfigDedicated-r13 OPTIONAL, --- Cond PUSCH-SCell
                UplinkPowerControlDedicatedSCell-v1310 OPTIONAL --- Need ON
        }
    }
    crossCarrierSchedulingConfig-r13
        CrossCarrierSchedulingConfig-r13 OPTIONAL, --- Cond Cross-Carrier-Config
    pdcch-ConfigSCell-r13
        PDCCH-ConfigSCell-r13 OPTIONAL, --- Need ON
    cqi-ReportConfig-v1310
        CQI-ReportConfig-v1310 OPTIONAL, --- Need ON
    pdsch-ConfigDedicated-v1310
        PDSCH-ConfigDedicated-v1310 OPTIONAL, --- Need ON
    soundingRS-UL-ConfigDedicated-v1310
        SoundingRS-UL-ConfigDedicated-v1310 OPTIONAL, --- Need ON
    soundingRS-UL-ConfigDedicatedUpPTSExt-r13
        SoundingRS-UL-ConfigDedicatedUpPTSExt-r13 OPTIONAL, --- Need ON
    soundingRS-UL-ConfigDedicatedAperiodic-v1310
        SoundingRS-UL-ConfigDedicatedAperiodic-v1310 OPTIONAL, --- Need ON
    soundingRS-UL-ConfigDedicatedAperiodicUpPTSExt-r13
        SoundingRS-UL-ConfigDedicatedAperiodicUpPTSExt-r13 OPTIONAL, --- Need ON
    csi-RS-Config-v1310
        CSI-RS-Config-v1310 OPTIONAL, --- Need ON
    laa-SCellConfiguration-r13
        LAA-SCellConfiguration-r13 OPTIONAL, --- Need ON
    csi-RS-ConfigNZPTToAddModListExt-r13
        CSI-RS-ConfigNZPTToAddModListExt-r13 OPTIONAL, --- Need ON
    csi-RS-ConfigNZPTToReleaseListExt-r13
        CSI-RS-ConfigNZPTToReleaseListExt-r13 OPTIONAL --- Need ON
    },
    {
        cqi-ReportConfig-v1320
            CQI-ReportConfig-v1320 OPTIONAL --- Need ON
    },
    {
        laa-SCellConfiguration-v1430
            LAA-SCellConfiguration-v1430
                typeE-SRS-TPC-PDCCH-Config-r14
                    SRS-TPC-PDCCH-Config-r14 OPTIONAL, --- Need ON
                uplinkPUSCH-LessPowerControlDedicated-v1430 OPTIONAL, --- Need ON
                soundingRS-UL-PeriodicConfigDedicatedList-r14
                    SEQUENCE (SIZE (1..2)) OF SoundingRS-UL-ConfigDedicated
                    OPTIONAL, --- Cond PeriodicSRS
                soundingRS-UL-PeriodicConfigDedicatedUpPTSExtList-r14
                    SEQUENCE (SIZE (1..4)) OF SoundingRS-UL-
                    ConfigDedicatedUpPTSExt-r13
                        OPTIONAL, --- Cond PeriodicSRSExt
    }

```

```

soundingRS-UL-AperiodicConfigDedicatedList-r14
    OPTIONAL,    --- Cond AperiodicRS
    soundingRS-UL-ConfigDedicatedApPdschExtList-r14
    OPTIONAL,    --- Cond AperiodicRSExt
    must-Config-r14
        release
        setup
            k-max-r14
            p-a-must-r14
                SEQUENCE (SIZE (1..2)) OF SoundingRS-AperiodicSet-r14
                SEQUENCE (SIZE (1..4)) OF SoundingRS-AperiodicSetUpPdschExt-r14
                CHOICE {
                    NULL,
                    SEQUENCE {
                        ENUMERATED {11, 13},
                        ENUMERATED {
                            dB-6, dB-4dot77, dB-3, dB-1dot77,
                            dB0, dB1, dB2, dB3} OPTIONAL,    --- Need ON
                        }
                    }
                }
            }
        }
        pusSch-ConfigDedicated-v1430
        csi-RS-Config-v1430
        csi-RS-ConfigZP-ApList-r14
        cqi-ReportConfig-v1430
        semiOpenLoop-r14
        pdsch-ConfigDedicatedSCell-v1430
    }
}

PDCCH-ConfigSCell-r13 ::= SEQUENCE {
    skipMonitoringDCI-format0-1A-r13    ENUMERATED {true}
}

```

FIG. 25F

```

RadioResourceConfigDedicated ::=
    srb-ToAddModList          SEQUENCE {
Conn                           SRB-ToAddModList          OPTIONAL,      --- Cond HO--
    drb-ToAddModList          DRB-ToAddModList          OPTIONAL,      --- Cond HO--
toEUTRA
    drb-ToReleaseList         DRE-ToReleaseList         OPTIONAL,      --- Need ON
    mac-MainConfig            CHOICE {
        explicitValue         MAC-MainConfig,
        defaultValue          NULL
    }                        OPTIONAL,      --- Cond HO--
toEUTRA2
    sps-Config                SPS-Config                OPTIONAL,      --- Need ON
    physicalConfigDedicated    PhysicalConfigDedicated    OPTIONAL,      --- Need ON
...
}

MAC-MainConfig ::=
    ul-SCH-Config              SEQUENCE {
        maxHARQ-Tx             SEQUENCE {
            ENUMERATED {
                n1, n2, n3, n4, n5, n6, n7, n8,
                n10, n12, n16, n20, n24, n28,
                spare2, spare1} OPTIONAL,      --- Need ON
            periodicBSR-Timer    PeriodicBSR-Timer-r12    OPTIONAL,      -- Need ON
            retxBSR-Timer        RetxBSR-Timer-r12,
            ttiBundling          BOOLEAN
        }                        OPTIONAL,      --- Need ON
        drx-Config              DRX-Config              OPTIONAL,      -- Need ON
        timeAlignmentTimerDedicated TimeAlignmentTimer,
        phr-Config              CHOICE {
            release              NULL,
            setup                SEQUENCE {
                periodicPHR-Timer    ENUMERATED {sf10, sf20, sf50, sf100, sf200,
                    sf500, sf1000, infinity},
                prohibitPHR-Timer    ENUMERATED {sf0, sf10, sf20, sf50, sf100,
                    sf200, sf500, sf1000},
                dl-PathlossChange    ENUMERATED {dB1, dB3, dB6, infinity}
            }
        }                        OPTIONAL,      -- Need ON
        ...,
        [[ sr-ProhibitTimer-r9    INTEGER (0..7)          OPTIONAL      -- Need ON
        ]],
        [[ mac-MainConfig-v1020    SEQUENCE {
            sCellDeactivationTimer-r10    ENUMERATED {
                rf2, rf4, rf8, rf16, rf32, rf64, rf128,
                spare} OPTIONAL,      --- Need OF
            extendedBSR-Sizes-r10    ENUMERATED {setup}      OPTIONAL,      -- Need OR
            extendedPHR-r10          ENUMERATED {setup}      OPTIONAL      --- Need OR
        }                        OPTIONAL      -- Need ON
        ]],
        [[ stag-ToReleaseList-r11    STAG-ToReleaseList-r11    OPTIONAL,      -- Need ON
        stag-ToAddModList-r11        STAG-ToAddModList-r11    OPTIONAL,      -- Need ON
        drx-Config-v1130            DRX-Config-v1130          OPTIONAL      --- Need ON
        ]],
        [[ e-HARQ-Pattern-r12        BOOLEAN                OPTIONAL,      --- Need ON
        dualConnectivityPHR         CHOICE {
            release              NULL,
            setup                SEQUENCE {
                phr-ModeOtherCG-r12    ENUMERATED {real, virtual}
            }
        }                        OPTIONAL,      --- Need ON
        logicalChannelSR-Config-r12 CHOICE {
            release              NULL,
            setup                SEQUENCE {
                logicalChannelSR-ProhibitTimer-r12    ENUMERATED {sf20, sf40, sf64, sf128,
                    sf512, sf1024, sf2560, spare1}
            }
        }
    }
}

```

FIG. 26A

OPTIONAL -- Need ON

```

]],
  [[ drx-Config-v1310          DRX-Config-v1310          OPTIONAL,      --- Need ON
     extendedPHR2-r13          BOOLEAN          OPTIONAL,      -- Need ON
     eDRX-Config-CycleStartOffset-r13
       release
       setup
       sf5120
       sf10240
     ]
  ]],
  [[ drx-Config-r13            CHOICE {
     release                    NULL,
     setup                      DRX-Config-r13
   }
  ]],
  [[ skipUplinkTx-r14          CHOICE {
     release                    NULL,
     setup                      SEQUENCE {
       skipUplinkTxSPS-r14      ENUMERATED {true}      OPTIONAL,      --- Need
OR
       skipUplinkTxDynamic-r14  ENUMERATED {true}      OPTIONAL      --- Need
OR
     }
   }
  ]],
  dataInactivityTimerConfig-r14 CHOICE {
    release                    NULL,
    setup                      SEQUENCE {
      dataInactivityTimer-r14  DataInactivityTimer-r14
    }
  }
  ]],
  ]],
  }
}

MAC-MainConfigSCell-r11 ::= SEQUENCE {
  stag-Id-r11                STAG-Id-r11          OPTIONAL,      --- Need OP
}

DRX-Config ::= CHOICE {
  release                    NULL,
  setup                      SEQUENCE {
    onDurationTimer          ENUMERATED {
      psf1, psf2, psf3, psf4, psf5, psf6,
      psf8, psf10, psf20, psf30, psf40,
      psf50, psf60, psf80, psf100,
      psf200},
    drx-InactivityTimer      ENUMERATED {
      psf1, psf2, psf3, psf4, psf5, psf6,
      psf8, psf10, psf20, psf30, psf40,
      psf50, psf60, psf80, psf100,
      psf200, psf300, psf500, psf750,
      psf1280, psf1920, psf2560, psf0-v1020,
      spare9, spare8, spare7, spare6,
      spare5, spare4, spare3, spare2,
      spare1},
    drx-RetransmissionTimer  ENUMERATED {
      psf1, psf2, psf4, psf6, psf8, psf16,
      psf24, psf33},
    longDRX-CycleStartOffset CHOICE {
      sf10          INTEGER(0..9),
      sf20          INTEGER(0..19),
      sf32          INTEGER(0..31),
      sf40          INTEGER(0..39),
      sf64          INTEGER(0..63),
      sf80          INTEGER(0..79),
      sf128         INTEGER(0..127),
      sf160         INTEGER(0..159),

```

FIG. 26B

```

        sf256                INTEGER(0..255),
        sf320                INTEGER(0..319),
        sf512                INTEGER(0..511),
        sf640                INTEGER(0..639),
        sf1024               INTEGER(0..1023),
        sf1280               INTEGER(0..1279),
        sf2048               INTEGER(0..2047),
        sf2560               INTEGER(0..2559)
    },
    shortDRX                  SEQUENCE {
        shortDRX-Cycle        ENUMERATED {
            sf2, sf5, sf8, sf10, sf16, sf20,
            sf32, sf40, sf64, sf80, sf128, sf160,
            sf256, sf320, sf512, sf640},
            drxShortCycleTimer INTEGER (1..16)
        }
    } OPTIONAL --- Need OR
}

DRX-Config-v1130 ::=
    drx-RetransmissionTimer-v1130 SEQUENCE {
        longDRX-CycleStartOffset-v1130 ENUMERATED {psf0-v1130} OPTIONAL, ---Need OR
        CHOICE {
            sf60-v1130 INTEGER(0..59),
            sf70-v1130 INTEGER(0..69)
        }
        shortDRX-Cycle-v1130 ENUMERATED {sf4-v1130} OPTIONAL ---Need OR
    }
DRX-Config-v1310 ::= SEQUENCE {
    longDRX-CycleStartOffset-v1310 SEQUENCE {
        sf60-v1310 INTEGER(0..59)
    }
} OPTIONAL ---Need OR

DRX-Config-r13 ::= SEQUENCE {
    onDurationTimer-v1310 ENUMERATED {psf300, psf400, psf500, psf600,
        psf800, psf1000, psf1200, psf1600}
        OPTIONAL, ---Need OR
    drx-RetransmissionTimer-v1310 ENUMERATED {psf40, psf64, psf80, psf96, psf112,
        psf128, psf160, psf320}
        OPTIONAL, ---Need OR
    drx-ULRetransmissionTimer-r13 ENUMERATED {psf0, psf1, psf2, psf4, psf6, psf8,
        psf24, psf33, psf40, psf64, psf80,
        psf112, psf128, psf160, psf320}
        OPTIONAL ---Need OR
}
PeriodicBSR-Timer-r12 ::= ENUMERATED {
    sf5, sf10, sf16, sf20, sf32, sf40, sf64, sf80,
    sf128, sf160, sf320, sf640, sf1280, sf2560,
    infinity, spare1}
RetxBSR-Timer-r12 ::= ENUMERATED {
    sf320, sf640, sf1280, sf2560, sf5120,
    sf10240, spare2, spare1}
STAG-ToReleaseList-r11 ::= SEQUENCE (SIZE (1..maxSTAG-r11)) OF STAG-Id-r11
STAG-ToAddModList-r11 ::= SEQUENCE (SIZE (1..maxSTAG-r11)) OF STAG-ToAddMod-r11
STAG-ToAddMod-r11 ::= SEQUENCE {
    stag-Id-r11 STAG-Id-r11,
    timeAlignmentTimerSTAG-r11 TimeAlignmentTimer,
    ...
}
STAG-Id-r11 ::= INTEGER (1..maxSTAG-r11)

```

FIG. 26C

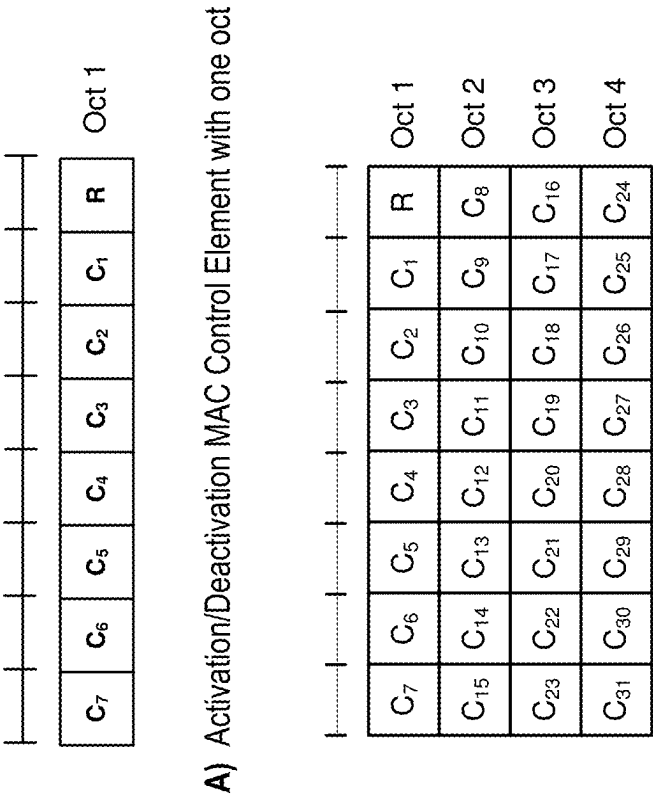


FIG. 27

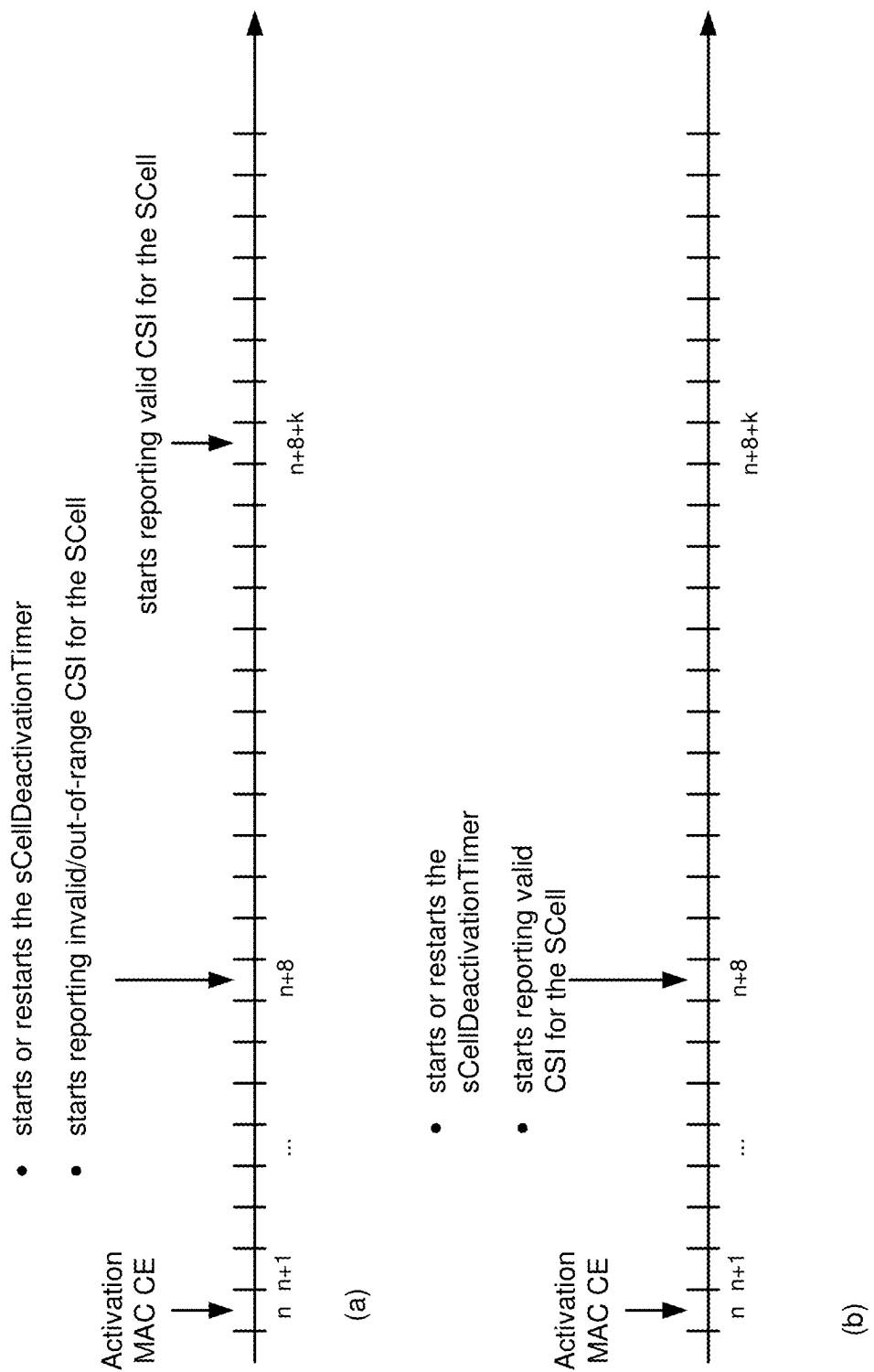


FIG. 28

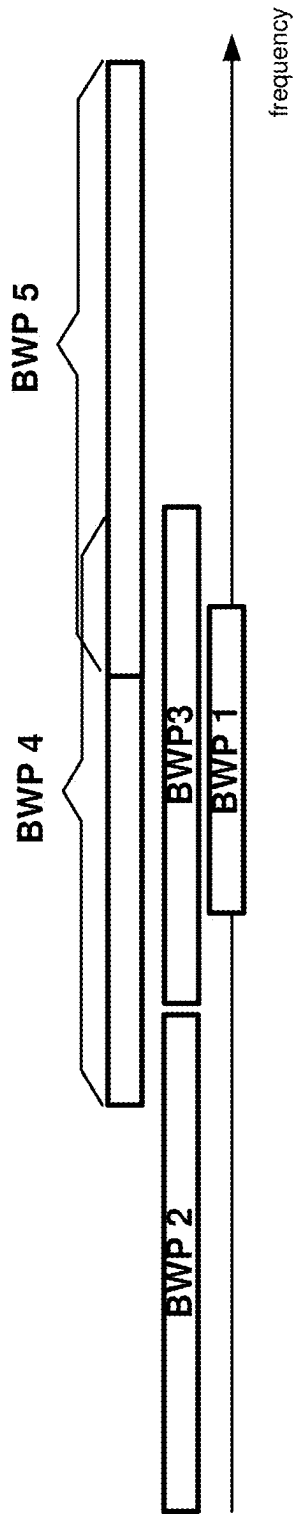


FIG. 29

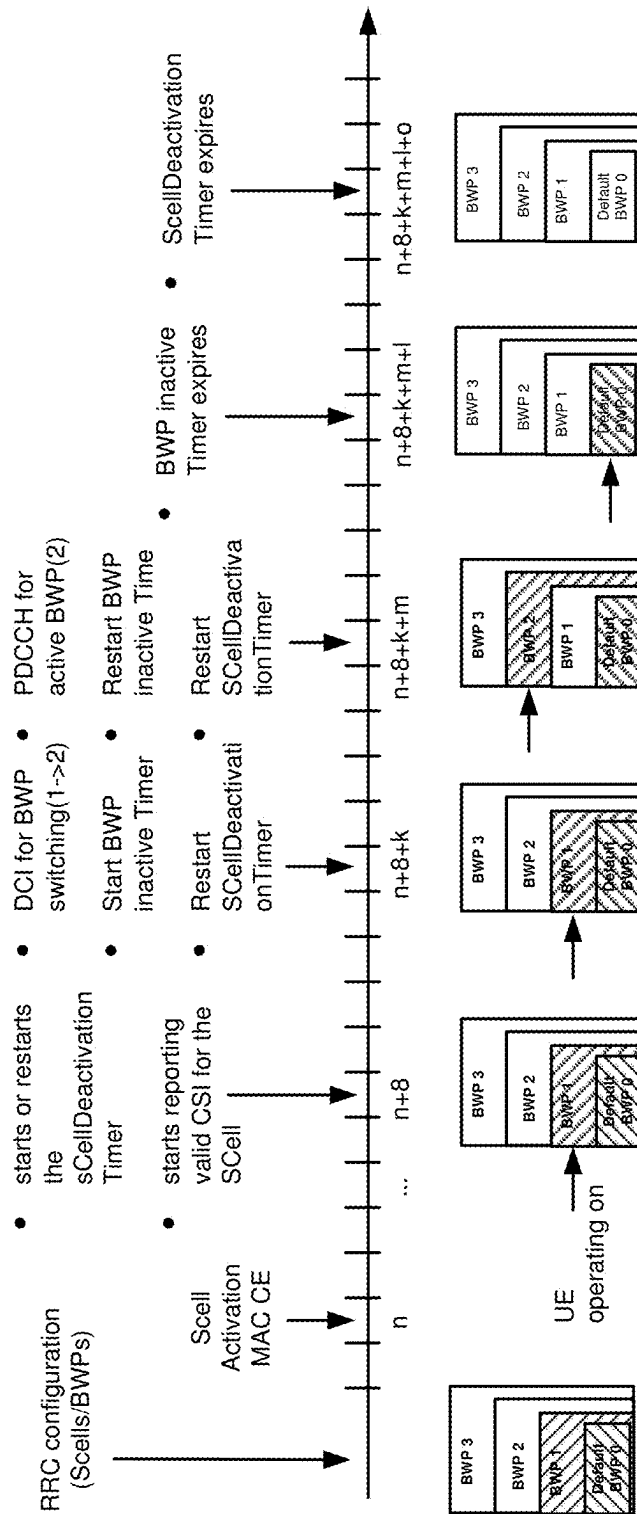


FIG. 30

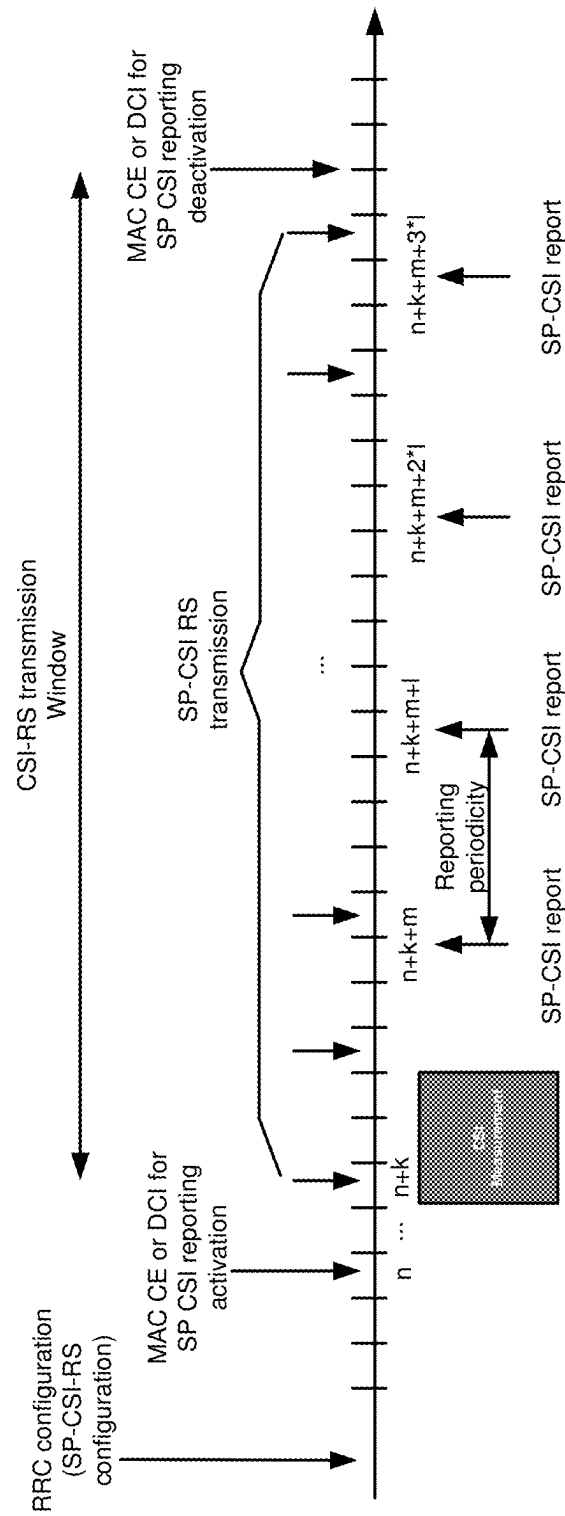


FIG. 31

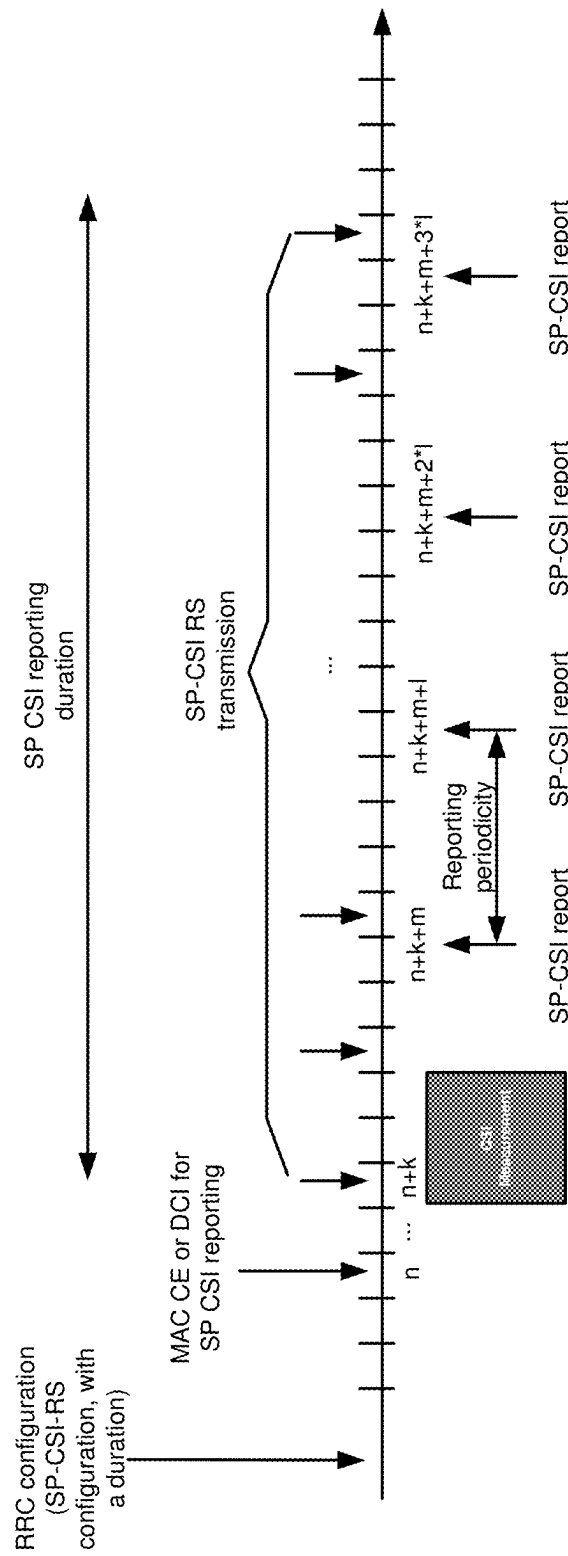


FIG. 32

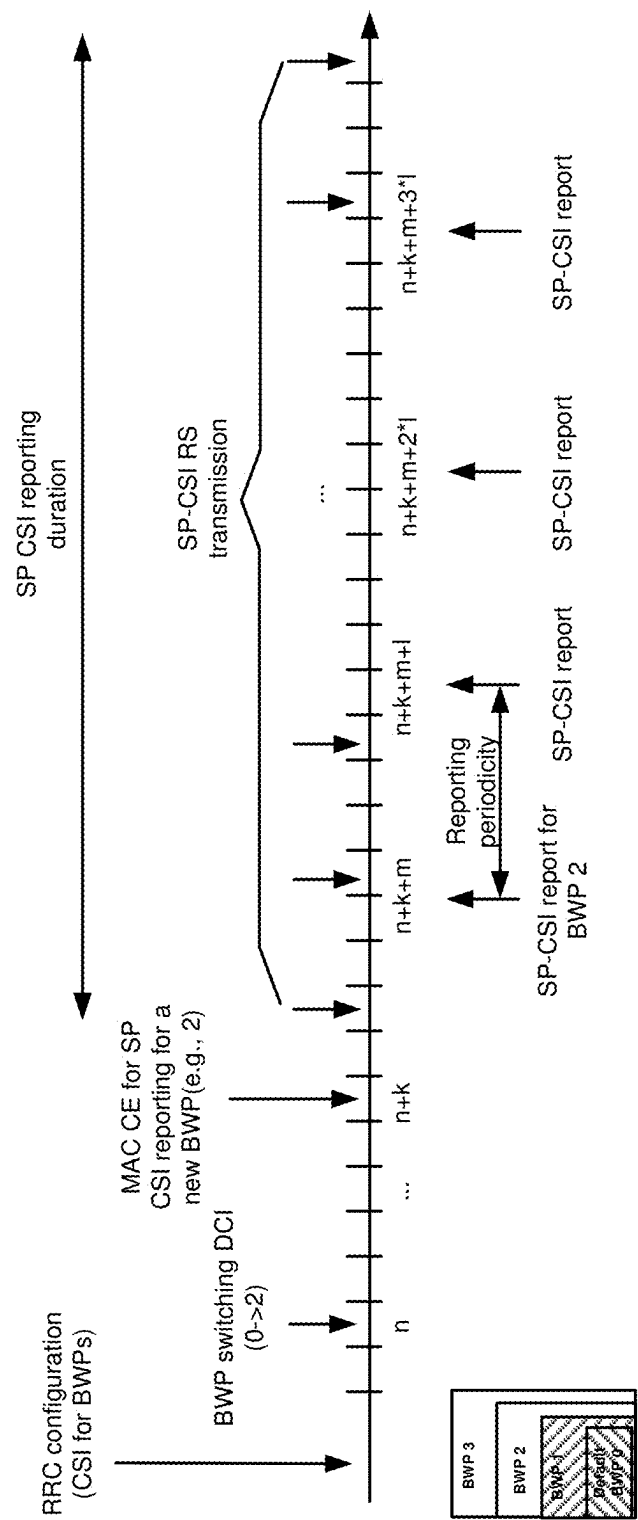


FIG. 33

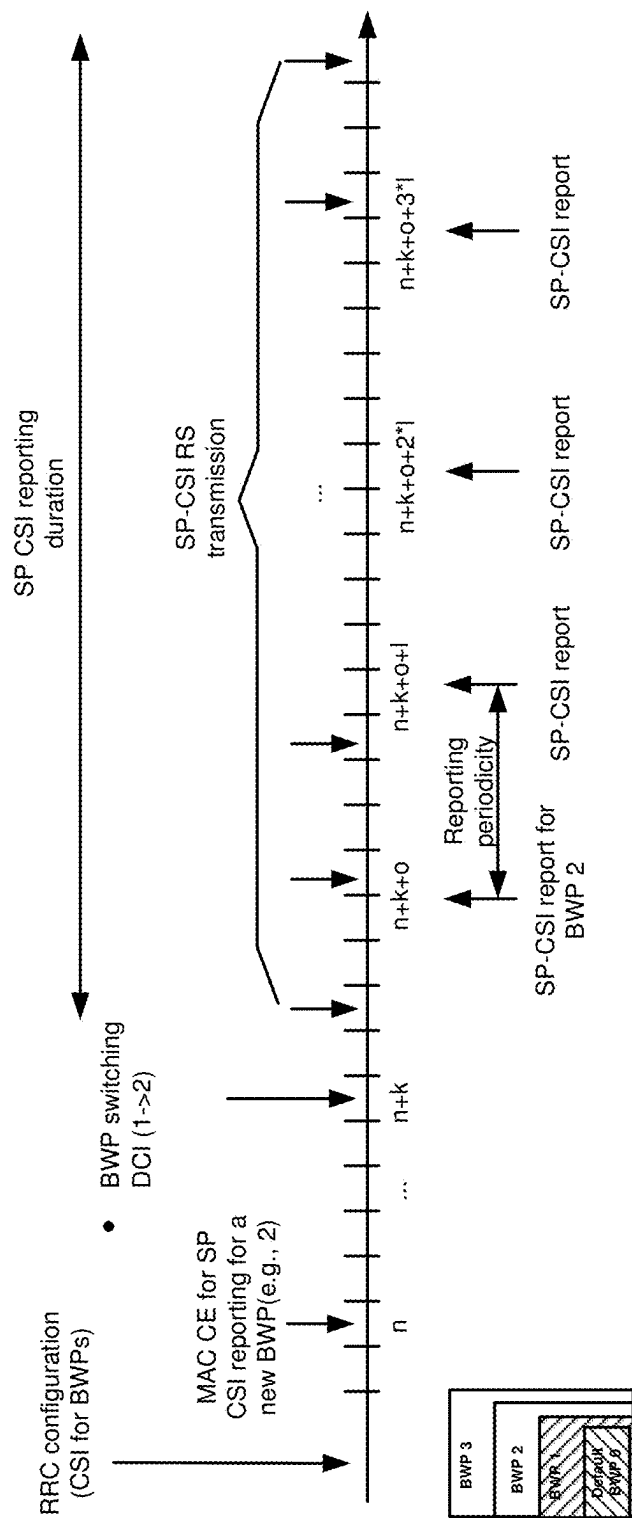
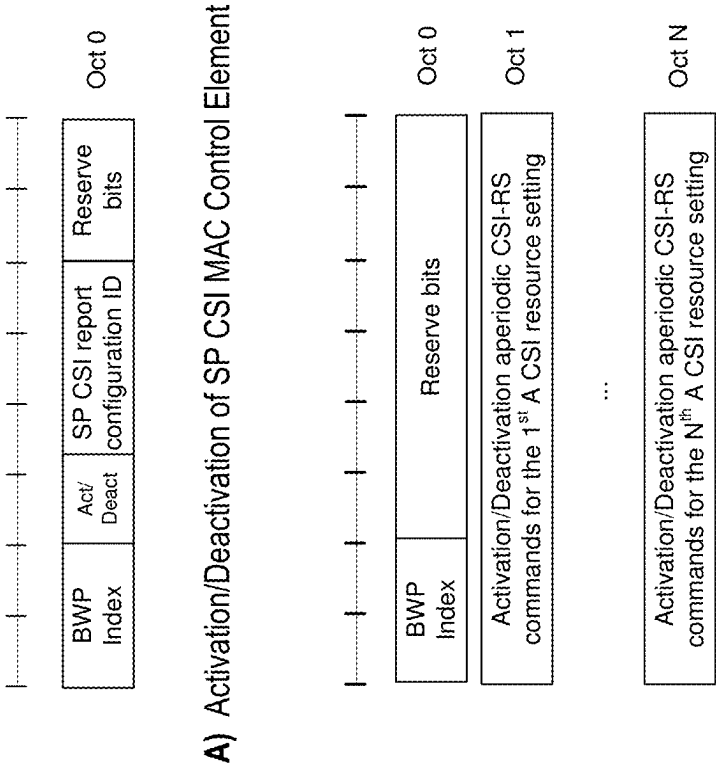


FIG. 34



B) Activation/Deactivation A-CSI MAC CE

FIG. 35

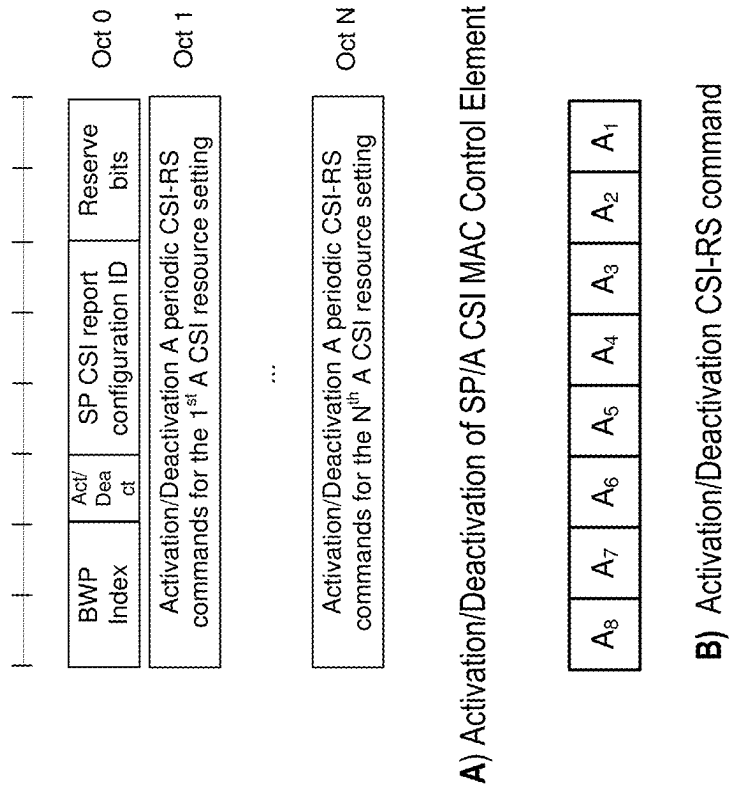


FIG. 36

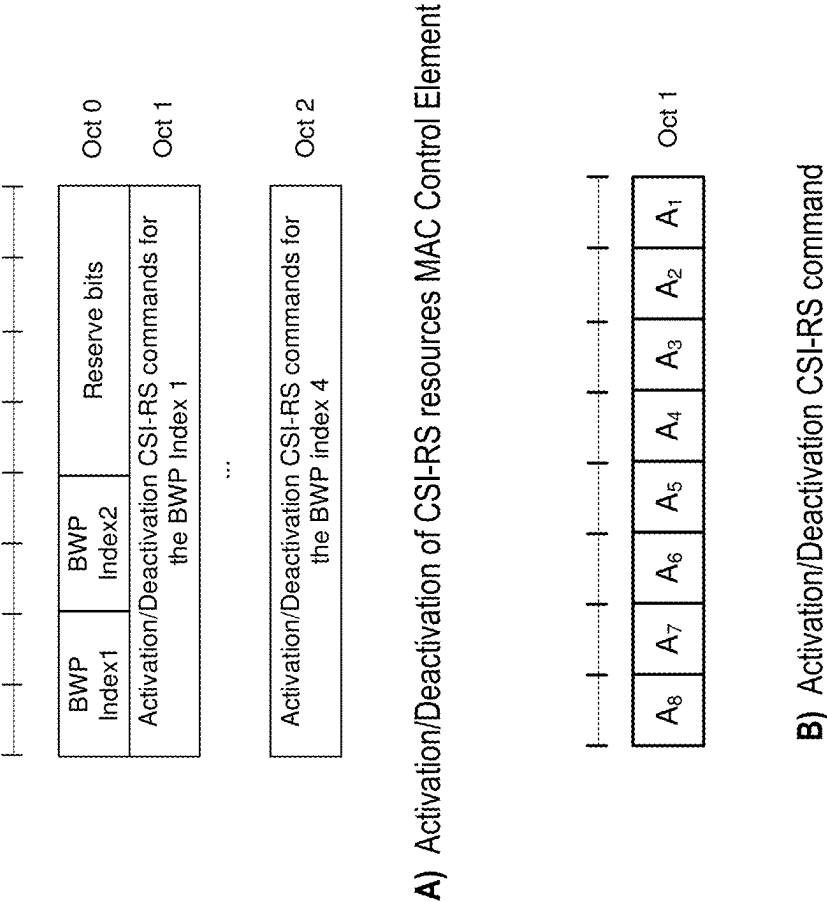


FIG. 37

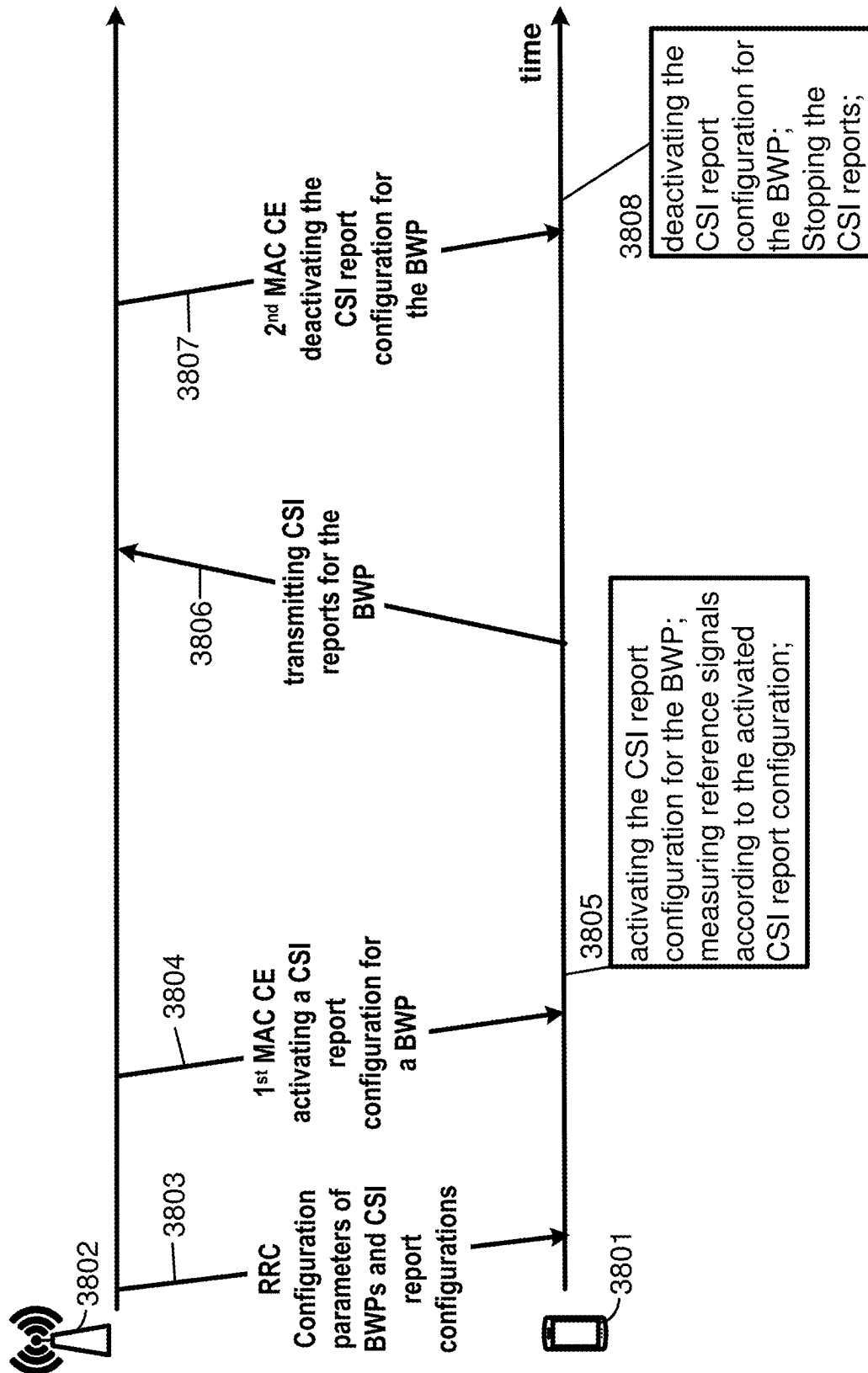
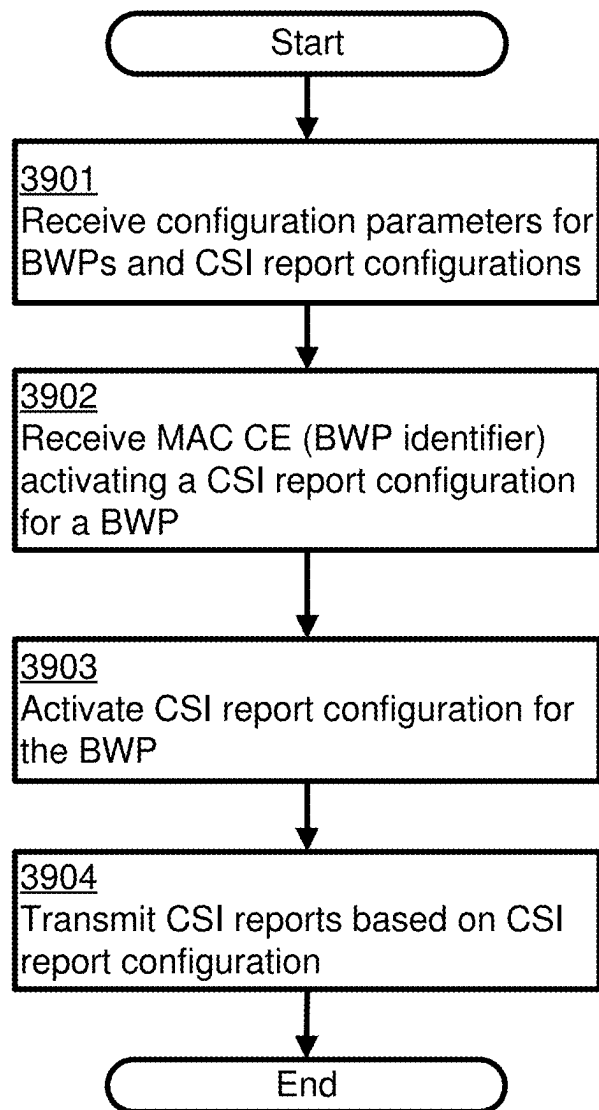


FIG. 38

**FIG. 39**

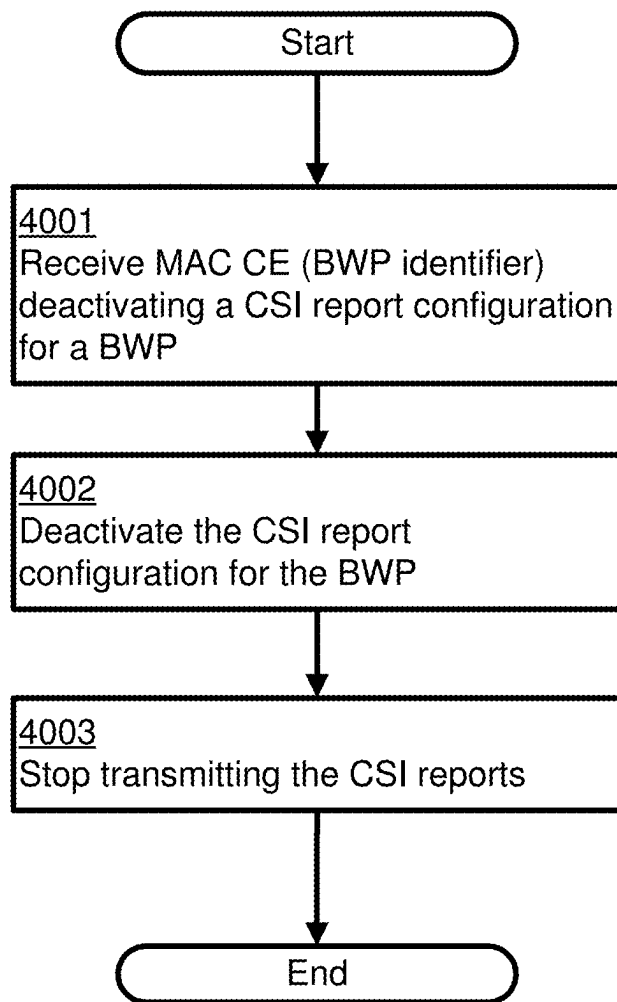
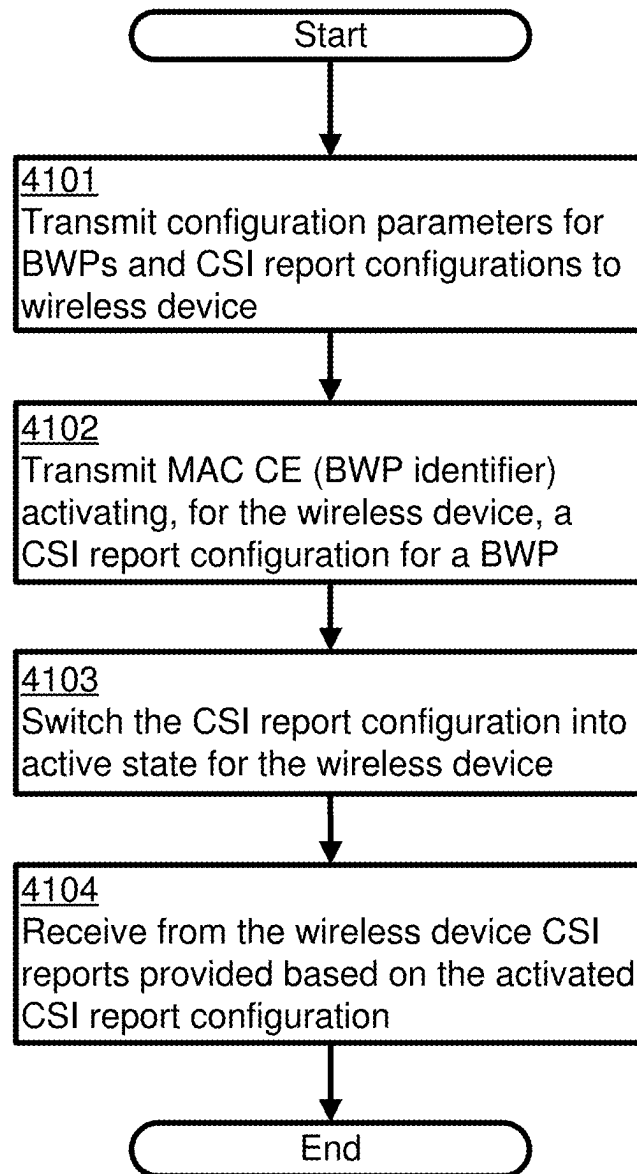


FIG. 40

**FIG. 41**

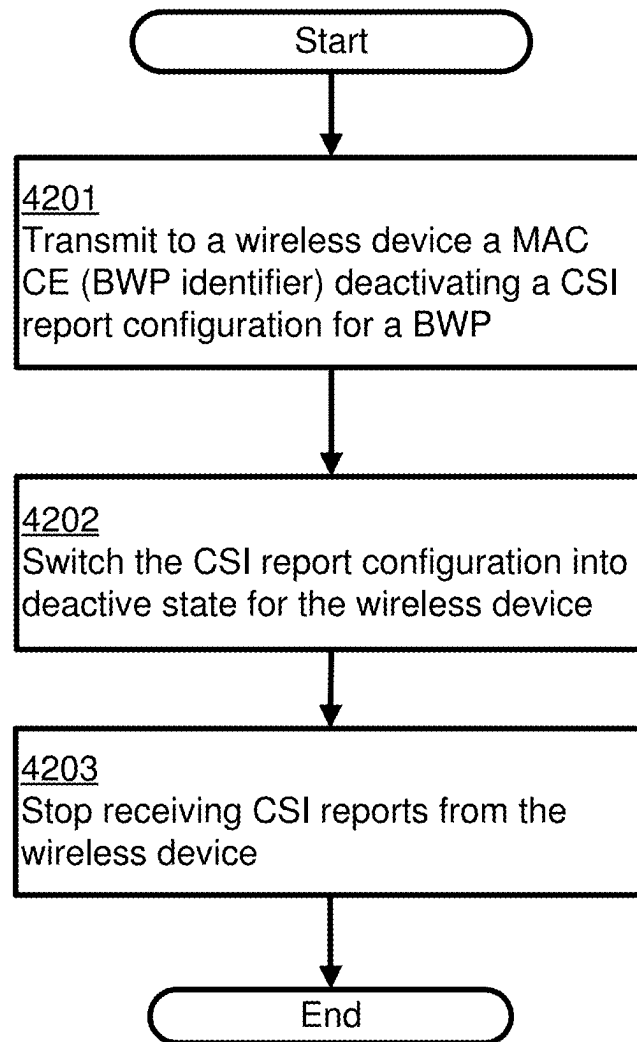


FIG. 42

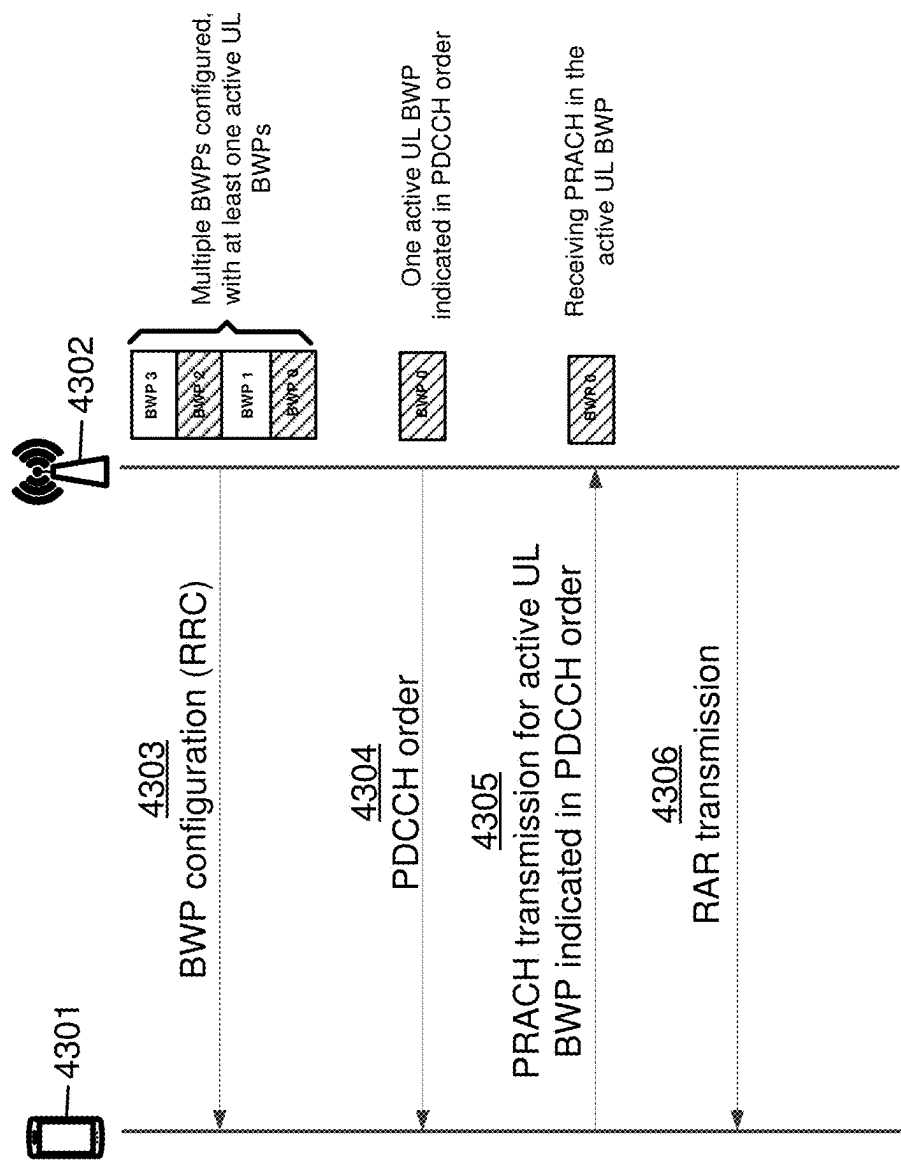


FIG. 43

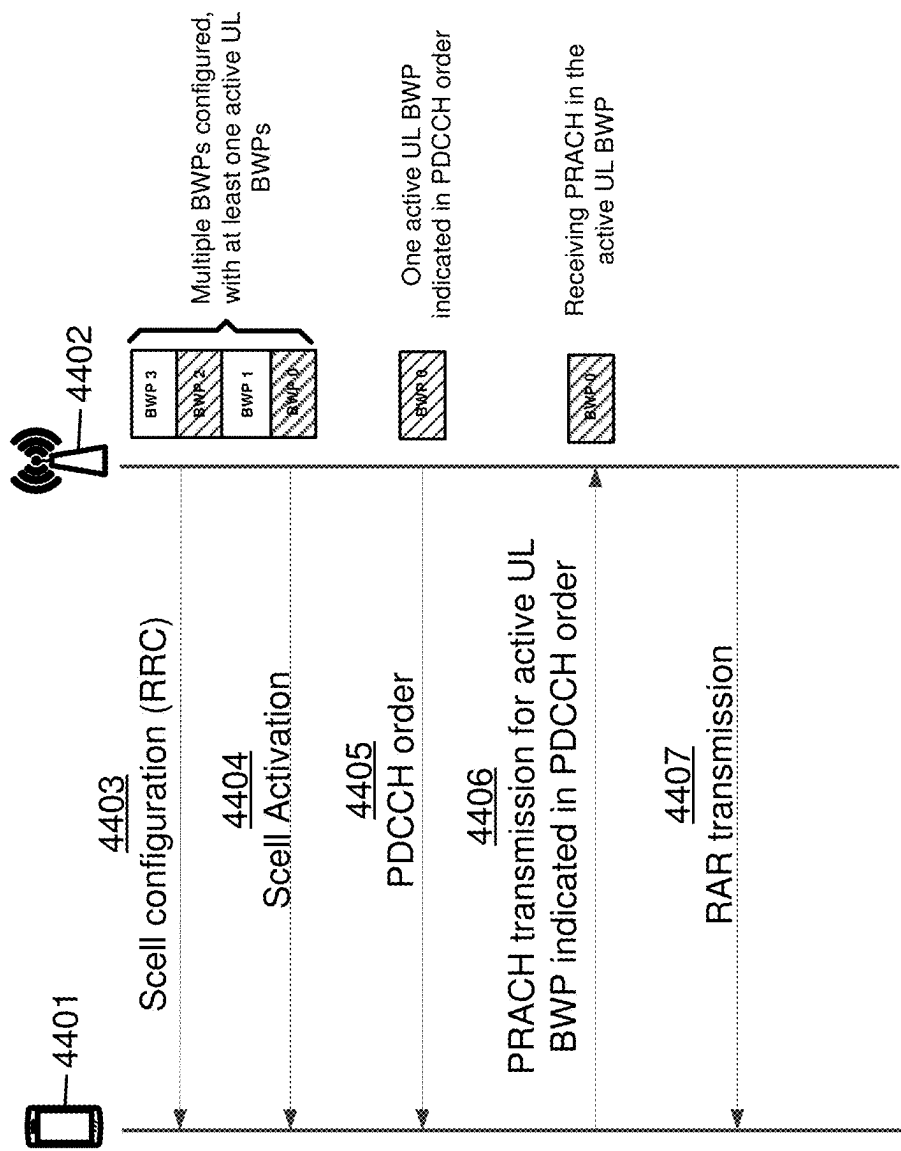


FIG. 44

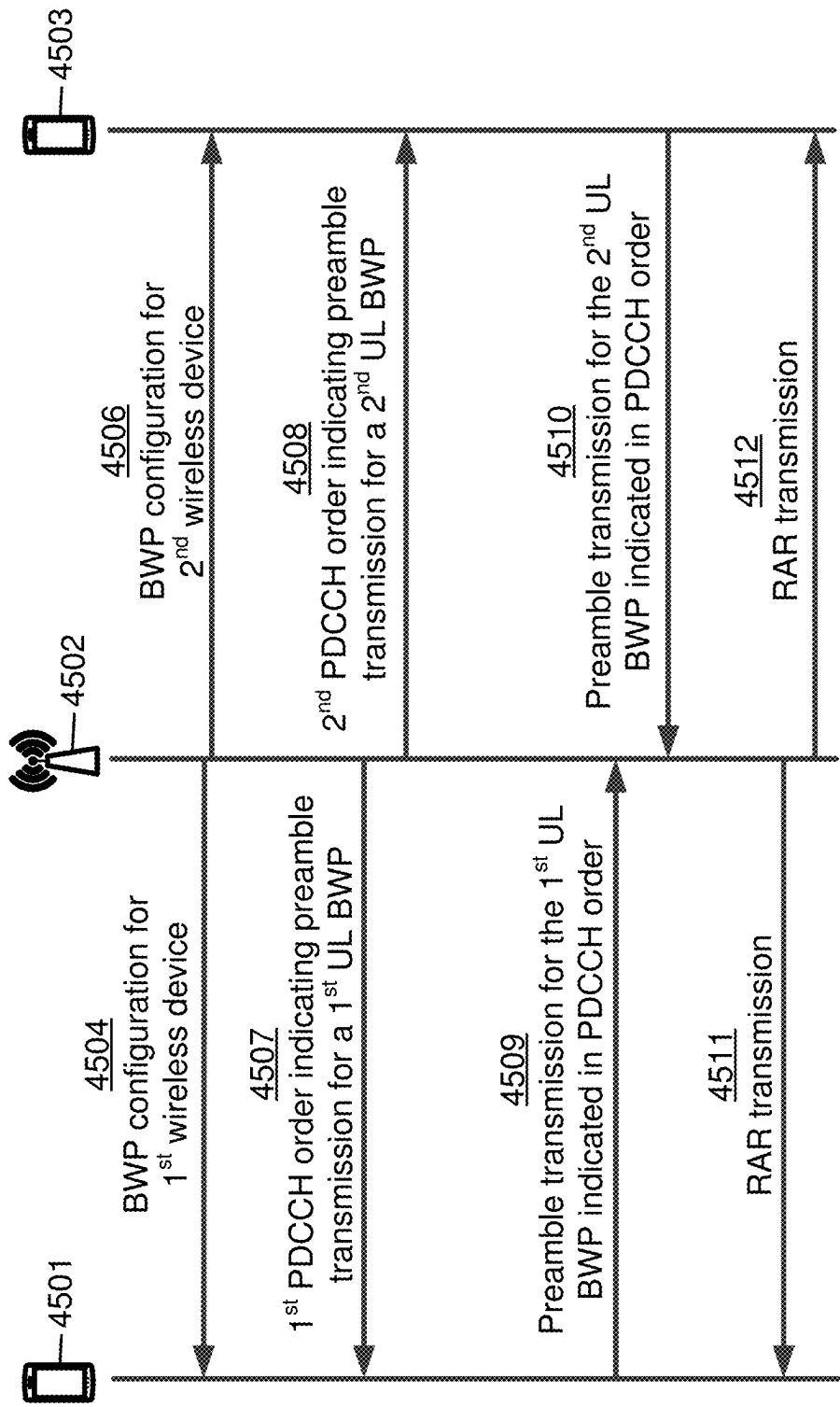


FIG. 45

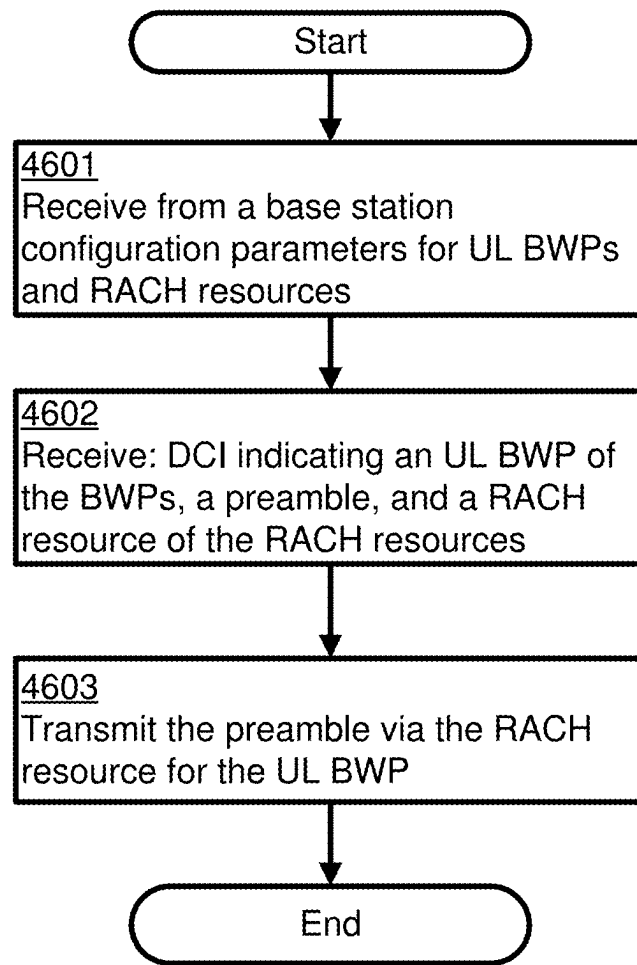


FIG. 46

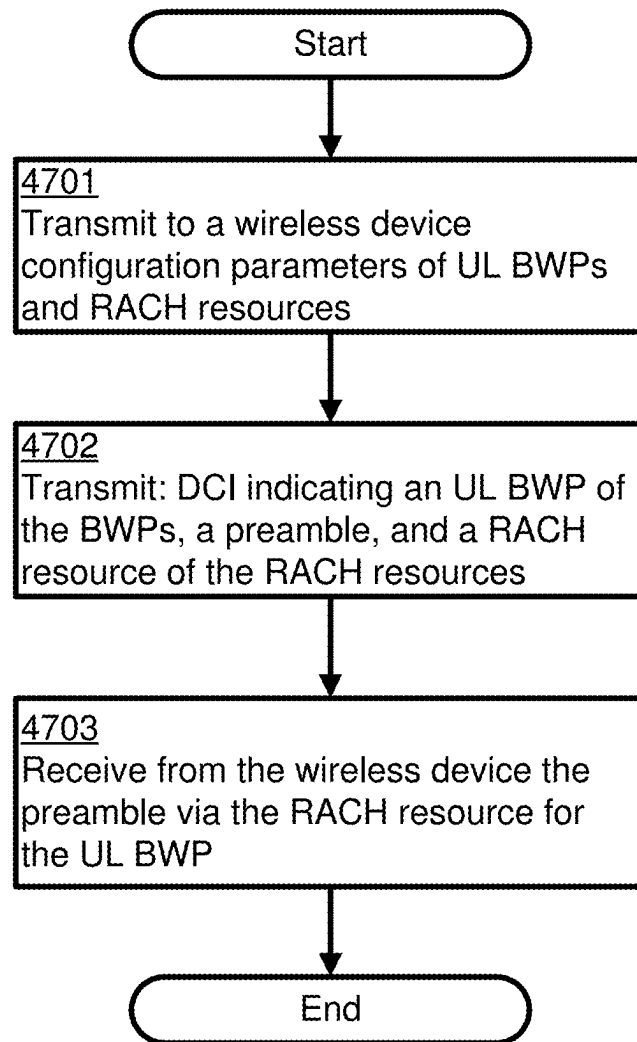


FIG. 47

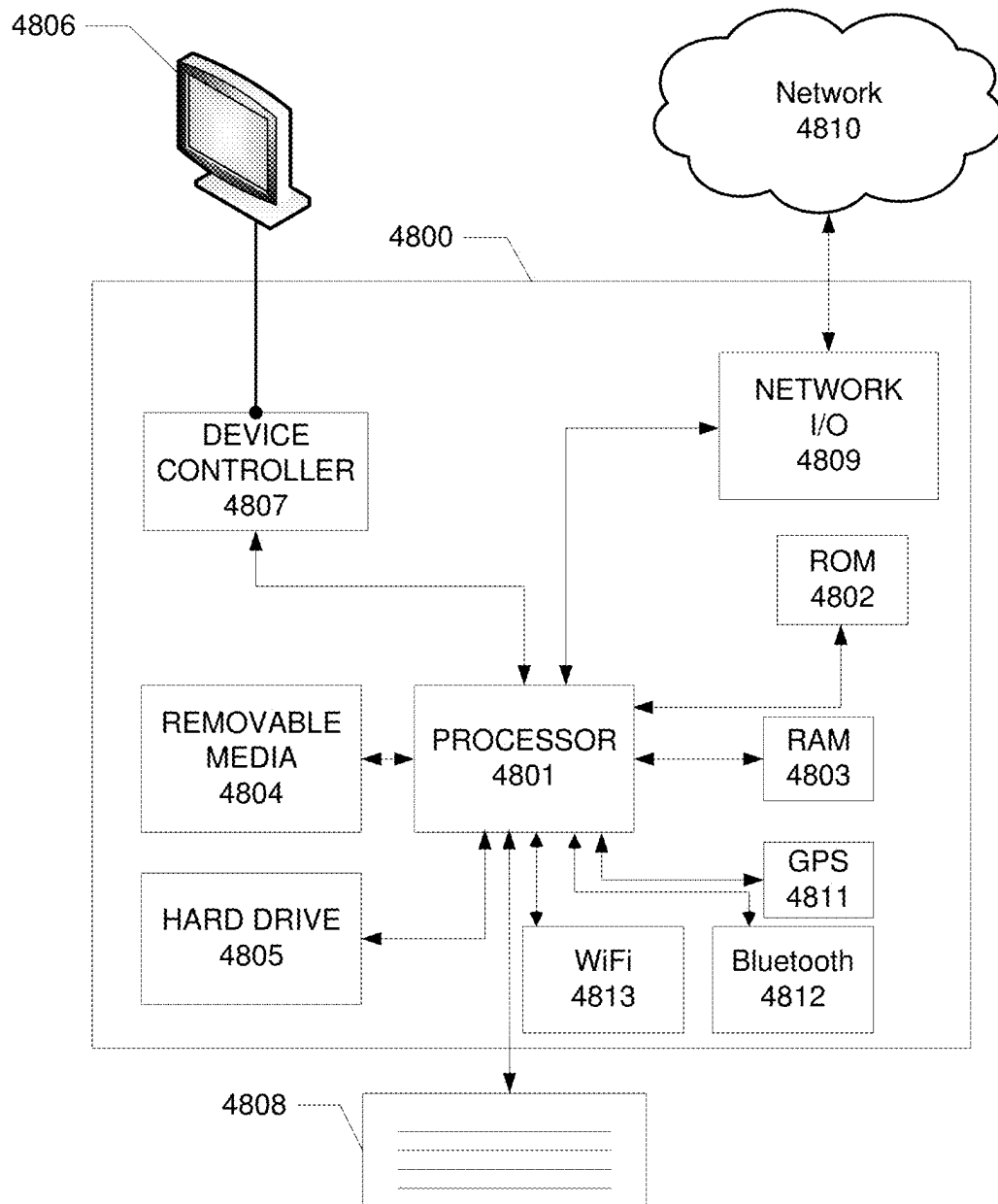


FIG. 48

CSI TRANSMISSION WITH MULTIPLE BANDWIDTH PARTS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 18/487,354, filed Oct. 16, 2023, which is a continuation of U.S. patent application Ser. No. 17/476,991, filed Sep. 16, 2021, which is a continuation of U.S. patent application Ser. No. 16/185,821, filed on Nov. 9, 2018, which claims the benefit of U.S. Provisional Application No. 62/583,650, filed on Nov. 9, 2017, and U.S. Provisional Application No. 62/587,058, filed on Nov. 16, 2017, all of which are hereby incorporated by reference in their entireties.

BACKGROUND

In wireless communications, bandwidth parts and other wireless resources may be used by wireless devices. A base station may determine that one or more wireless devices should use or switch to one or more bandwidth parts or other wireless resources. It is desired to improve wireless communications without adversely increasing signaling overhead and/or decreasing spectral efficiency.

SUMMARY

The following summary presents a simplified summary of certain features. The summary is not an extensive overview and is not intended to identify key or critical elements.

Systems, apparatuses, and methods are described for activating and deactivating bandwidth parts and/or other wireless resources. A base station may transmit, to a wireless device, a medium access control (MAC) control element (CE) comprising an indication of a bandwidth part or other wireless resource. The MAC CE may comprise an indication for activation and/or deactivation of a channel state information (CSI) report configuration. After or in response to receiving the MAC CE, the wireless device may activate the CSI report configuration for the indicated bandwidth part. By including an indication of a bandwidth part in the MAC CE, the downlink data scheduling delay may be reduced and/or spectrum efficiency of downlink data transmission may be increased.

A base station may transmit, to a wireless device, BWP switching and/or activation information such as an uplink bandwidth part index indicating an uplink bandwidth part; and random access procedure information such as random access channel parameters indicating one or more random access resources and/or preamble indices indicating one or more preambles. The base station may transmit downlink control information (DCI) comprising the BWP switching information and random access procedure information. The wireless device may receive the DCI. The wireless device may transmit, to the base station via the random access channel resource for the uplink bandwidth part indicated by the uplink bandwidth part index, the preamble. By including the BWP switching, and/or BWP activation information, and random access information in a DCI, BWP switching delay may be reduced and/or spectrum efficiency may be increased.

These and other features and advantages are described in greater detail below.

BRIEF DESCRIPTION OF THE DRAWINGS

Some features are shown by way of example, and not by limitation, in the accompanying drawings. In the drawings, like numerals reference similar elements.

FIG. 1 shows example sets of orthogonal frequency division multiplexing (OFDM) subcarriers.

FIG. 2 shows example transmission time and reception time for two carriers in a carrier group.

FIG. 3 shows example OFDM radio resources.

FIG. 4 shows hardware elements of a base station and a wireless device.

FIG. 5A, FIG. 5B, FIG. 5C and FIG. 5D show examples for uplink and downlink signal transmission.

FIG. 6 shows an example protocol structure with multi-connectivity.

FIG. 7 shows an example protocol structure with carrier aggregation (CA) and dual connectivity (DC).

FIG. 8 shows example timing advance group (TAG) configurations.

FIG. 9 shows example message flow in a random access process in a secondary TAG.

FIG. 10A and FIG. 10B show examples for interfaces between a 5G core network and base stations.

FIG. 11A, FIG. 11B, FIG. 11C, FIG. 11D, FIG. 11E, and FIG. 11F show examples for architectures of tight interworking between a 5G RAN and a long term evolution (LTE) radio access network (RAN).

FIG. 12A, FIG. 12B, and FIG. 12C show examples for radio protocol structures of tight interworking bearers.

FIG. 13A and FIG. 13B show examples for gNodeB (gNB) deployment.

FIG. 14 shows functional split option examples of a centralized gNB deployment.

FIG. 15 is an example configuration of a synchronization signal (SS) burst set.

FIG. 16 shows an example of random access (RA) procedures.

FIG. 17 shows an example media access control (MAC) packet data unit (PDU) format of an example of MAC PDU comprising a MAC header and MAC random access responses (RARs) for a four-step RA procedure.

FIG. 18 shows an example MAC RAR format of an example of MAC RAR comprising a timing advance command, uplink (UL) grant, and temporary cell-radio network temporary identifier for a four-step RA procedure.

FIG. 19 shows an example of a random access procedure for an SS burst.

FIG. 20 shows an example of transmitting channel state information reference signals for a beam.

FIG. 21 shows an example of channel state information reference signal mapping.

FIG. 22A and FIG. 22B show examples for radio resource control signaling.

FIG. 23 shows an example of activation/deactivation of a CSI-RS resources MAC control element and a CSI-RS command.

FIG. 24 shows an example of downlink beam management.

FIGS. 25A to 25F show examples for radio resource control signaling.

FIGS. 26A to 26C show examples for radio resource control signaling.

FIG. 27 shows an example of activation/deactivation MAC control elements.

FIG. 28 shows an example of a sCellDeactivationTimer starting and CSI reporting for an SCell.

FIG. 29 shows an example of multiple bandwidth parts (BWPs) configurations in a frequency domain.

FIG. 30 shows an example of a BWP inactive timer and a sCellDeactivationTimer relation for an activated SCell.

FIG. 31 shows an example of a semi-persistent (SP) CSI configuration with a CSI activation MAC CE or DCI and a CSI deactivation MAC CE or DCI.

FIG. 32 shows an example of SP CSI configuration with a CSI activation MAC CE or DCI and a CSI reporting duration.

FIG. 33 shows an example of SP CSI activation MAC CE transmission after a DCI for BWP switching.

FIG. 34 shows an example of SP CSI activation MAC CE transmission before a DCI for BWP switching.

FIG. 35 shows an example of separate SP/aperiodic CSI activation/deactivation MAC CEs with a BWP index.

FIG. 36 shows an example of joint SP/aperiodic CSI activation/deactivation MAC CE with a BWP index.

FIG. 37 shows an example of aperiodic CSI activation MAC CE with multiple BWPs.

FIG. 38 shows an example of base station and wireless device transmissions for activating and/or deactivating CSI reporting for a BWP.

FIG. 39 shows an example of wireless device processes for activating CSI reporting for a BWP.

FIG. 40 shows an example of wireless device processes for deactivating CSI reporting for a BWP.

FIG. 41 shows an example of base station processes for activating CSI reporting for a BWP.

FIG. 42 shows an example of base station processes for deactivating CSI reporting for a BWP.

FIG. 43 shows an example of a random access procedure on an active UL BWP for a wireless device.

FIG. 44 shows an example of a random access procedure for an active UL BWP in an SCell.

FIG. 45 shows an example of random access procedures with multiple wireless devices for multiple UL BWPs.

FIG. 46 shows an example of wireless device processes for a random access procedure for an active UL BWP.

FIG. 47 shows an example of base station processes for a random access procedure for an active UL BWP.

FIG. 48 shows example elements of a computing device that may be used to implement any of the various devices described herein.

DETAILED DESCRIPTION

The accompanying drawings, which form a part hereof, show examples of the disclosure. It is to be understood that the examples shown in the drawings and/or discussed herein are non-exclusive and that there are other examples of how the disclosure may be practiced.

Examples may enable operation of carrier aggregation and may be used in the technical field of multicarrier communication systems. Examples may relate to transmissions with bandwidth parts in multicarrier communication systems.

The following acronyms are used throughout the present disclosure, provided below for convenience although other acronyms may be introduced in the detailed description:

3GPP 3rd Generation Partnership Project

5G 5th generation wireless systems

5GC 5G Core Network

ACK Acknowledgement

AMF Access and Mobility Management Function

ASIC application-specific integrated circuit

BPSK binary phase shift keying

CA carrier aggregation

CC component carrier

CDMA code division multiple access

CP cyclic prefix

CPLD complex programmable logic devices

CSI channel state information

CSS common search space

CU central unit

DC dual connectivity

DCI downlink control information

DFTS-OFDM discrete Fourier transform spreading OFDM

DL downlink

DU distributed unit

eLTE enhanced LTE

eMBB enhanced mobile broadband

eNB evolved Node B

EPC evolved packet core

E-UTRAN evolved-universal terrestrial radio access network

FDD frequency division multiplexing

FPGA field programmable gate arrays

Fs-C Fs-control plane

Fs-U Fs-user plane

gNB next generation node B

HARQ hybrid automatic repeat request

HDL hardware description languages

ID identifier

IE information element

LTE long term evolution

MAC media access control

MCG master cell group

MeNB master evolved node B

MIB master information block

MME mobility management entity

mMTC massive machine type communications

NACK Negative Acknowledgement

NAS non-access stratum

NG CP next generation control plane core

NGC next generation core

NG-C NG-control plane

NG-U NG-user plane

NR MAC new radio MAC

NR PDCP new radio PDCP

NR PHY new radio physical

NR RLC new radio RLC

NR RRC new radio RRC

NR new radio

NSSAI network slice selection assistance information

OFDM orthogonal frequency division multiplexing

PCC primary component carrier

PCell primary cell

PDCCH physical downlink control channel

PDCC packet data convergence protocol

PDU packet data unit

PHICH physical HARQ indicator channel

PHY physical

PLMN public land mobile network

PSCell primary secondary cell

pTAG primary timing advance group

PUCCH physical uplink control channel

PUSCH physical uplink shared channel

QAM quadrature amplitude modulation

QPSK quadrature phase shift keying

RA random access

RACH random access channel

RAN radio access network

5

RAP random access preamble
 RAR random access response
 RB resource blocks
 RBG resource block groups
 RLC radio link control
 RRC radio resource control
 RRM radio resource management
 RV redundancy version
 SCC secondary component carrier
 SCell secondary cell
 SCG secondary cell group
 SC-OFDM single carrier-OFDM
 SDU service data unit
 SeNB secondary evolved node B
 SFN system frame number
 S-GW serving gateway
 SIB system information block
 SC-OFDM single carrier orthogonal frequency division
 multiplexing
 SRB signaling radio bearer
 sTAG(s) secondary timing advance group(s)
 TA timing advance
 TAG timing advance group
 TAI tracking area identifier
 TAT time alignment timer
 TDD time division duplexing
 TDMA time division multiple access
 TTI transmission time interval
 TB transport block
 UE user equipment
 UL uplink
 UPGW user plane gateway
 URLLC ultra-reliable low-latency communications
 VHDL VHIC hardware description language
 Xn-C Xn-control plane
 Xn-U Xn-user plane
 Xx-C Xx-control plane
 Xx-U Xx-user plane

Examples may be implemented using various physical layer modulation and transmission mechanisms. Example transmission mechanisms may include, but are not limited to: CDMA, OFDM, TDMA, Wavelet technologies, and/or the like. Hybrid transmission mechanisms such as TDMA/CDMA, and OFDM/CDMA may also be employed. Various modulation schemes may be used for signal transmission in the physical layer. Examples of modulation schemes include, but are not limited to: phase, amplitude, code, a combination of these, and/or the like. An example radio transmission method may implement QAM using BPSK, QPSK, 16-QAM, 64-QAM, 256-QAM, and/or the like. Physical radio transmission may be enhanced by dynamically or semi-dynamically changing the modulation and coding scheme depending on transmission requirements and radio conditions.

FIG. 1 shows example sets of OFDM subcarriers. As shown in this example, arrow(s) in the diagram may depict a subcarrier in a multicarrier OFDM system. The OFDM system may use technology such as OFDM technology, DFTS-OFDM, SC-OFDM technology, or the like. For example, arrow 101 shows a subcarrier transmitting information symbols. FIG. 1 is shown as an example, and a typical multicarrier OFDM system may include more subcarriers in a carrier. For example, the number of subcarriers in a carrier may be in the range of 10 to 10,000 subcarriers. FIG. 1 shows two guard bands 106 and 107 in a transmission band. As shown in FIG. 1, guard band 106 is between subcarriers 103 and subcarriers 104. The example set of

6

subcarriers A 102 includes subcarriers 103 and subcarriers 104. FIG. 1 also shows an example set of subcarriers B 105. As shown, there is no guard band between any two subcarriers in the example set of subcarriers B 105. Carriers in a multicarrier OFDM communication system may be contiguous carriers, non-contiguous carriers, or a combination of both contiguous and non-contiguous carriers.

FIG. 2 shows an example timing arrangement with transmission time and reception time for two carriers. A multicarrier OFDM communication system may include one or more carriers, for example, ranging from 1 to 10 carriers. Carrier A 204 and carrier B 205 may have the same or different timing structures. Although FIG. 2 shows two synchronized carriers, carrier A 204 and carrier B 205 may or may not be synchronized with each other. Different radio frame structures may be supported for FDD and TDD duplex mechanisms. FIG. 2 shows an example FDD frame timing. Downlink and uplink transmissions may be organized into radio frames 201. In this example, radio frame duration is 10 milliseconds (msec). Other frame durations, for example, in the range of 1 to 100 msec may also be supported. In this example, each 10 msec radio frame 201 may be divided into ten equally sized subframes 202. Other subframe durations such as including 0.5 msec, 1 msec, 2 msec, and 5 msec may also be supported. Subframe(s) may consist of two or more slots (e.g., slots 206 and 207). For the example of FDD, 10 subframes may be available for downlink transmission and 10 subframes may be available for uplink transmissions in each 10 msec interval. Uplink and downlink transmissions may be separated in the frequency domain. A slot may be 7 or 14 OFDM symbols for the same subcarrier spacing of up to 60 kHz with normal CP. A slot may be 14 OFDM symbols for the same subcarrier spacing higher than 60 kHz with normal CP. A slot may include all downlink, all uplink, or a downlink part and an uplink part, and/or alike. Slot aggregation may be supported, for example, data transmission may be scheduled to span one or multiple slots. For example, a mini-slot may start at an OFDM symbol in a subframe. A mini-slot may have a duration of one or more OFDM symbols. Slot(s) may include a plurality of OFDM symbols 203. The number of OFDM symbols 203 in a slot 206 may depend on the cyclic prefix length and subcarrier spacing.

FIG. 3 shows an example of OFDM radio resources. The resource grid structure in time 304 and frequency 305 is shown in FIG. 3. The quantity of downlink subcarriers or RBs may depend, at least in part, on the downlink transmission bandwidth 306 configured in the cell. The smallest radio resource unit may be called a resource element (e.g., 301). Resource elements may be grouped into resource blocks (e.g., 302). Resource blocks may be grouped into larger radio resources called Resource Block Groups (RBG) (e.g., 303). The transmitted signal in slot 206 may be described by one or several resource grids of a plurality of subcarriers and a plurality of OFDM symbols. Resource blocks may be used to describe the mapping of certain physical channels to resource elements. Other pre-defined groupings of physical resource elements may be implemented in the system depending on the radio technology. For example, 24 subcarriers may be grouped as a radio block for a duration of 5 msec. A resource block may correspond to one slot in the time domain and 180 kHz in the frequency domain (for 15 kHz subcarrier bandwidth and 12 subcarriers).

Multiple numerologies may be supported. A numerology may be derived by scaling a basic subcarrier spacing by an integer N. Scalable numerology may allow at least from 15 kHz to 480 kHz subcarrier spacing. The numerology with 15

kHz and scaled numerology with different subcarrier spacing with the same CP overhead may align at a symbol boundary every 1 msec in a NR carrier.

FIG. 4 shows hardware elements of a base station **401** and a wireless device **406**. A communication network **400** may include at least one base station **401** and at least one wireless device **406**. The base station **401** may include at least one communication interface **402**, one or more processors **403**, and at least one set of program code instructions **405** stored in non-transitory memory **404** and executable by the one or more processors **403**. The wireless device **406** may include at least one communication interface **407**, one or more processors **408**, and at least one set of program code instructions **410** stored in non-transitory memory **409** and executable by the one or more processors **408**. A communication interface **402** in the base station **401** may be configured to engage in communication with a communication interface **407** in the wireless device **406**, such as via a communication path that includes at least one wireless link **411**. The wireless link **411** may be a bi-directional link. The communication interface **407** in the wireless device **406** may also be configured to engage in communication with the communication interface **402** in the base station **401**. The base station **401** and the wireless device **406** may be configured to send and receive data over the wireless link **411** using multiple frequency carriers. Base stations, wireless devices, and other communication devices may include structure and operations of transceiver(s). A transceiver is a device that includes both a transmitter and receiver. Transceivers may be employed in devices such as wireless devices, base stations, relay nodes, and/or the like. Examples for radio technology implemented in the communication interfaces **402**, **407** and the wireless link **411** are shown in FIG. 1, FIG. 2, FIG. 3, FIG. 5, and associated text. The communication network **400** may comprise any number and/or type of devices, such as, for example, computing devices, wireless devices, mobile devices, handsets, tablets, laptops, internet of things (IoT) devices, hotspots, cellular repeaters, computing devices, and/or, more generally, user equipment (e.g., UE). Although one or more of the above types of devices may be referenced herein (e.g., UE, wireless device, computing device, etc.), it should be understood that any device herein may comprise any one or more of the above types of devices or similar devices. The communication network **400**, and any other network referenced herein, may comprise an LTE network, a 5G network, or any other network for wireless communications. Apparatuses, systems, and/or methods described herein may generally be described as implemented on one or more devices (e.g., wireless device, base station, eNB, gNB, computing device, etc.), in one or more networks, but it will be understood that one or more features and steps may be implemented on any device and/or in any network. As used throughout, the term “base station” may comprise one or more of: a base station, a node, a Node B, a gNB, an eNB, an ng-eNB, a relay node (e.g., an integrated access and backhaul (IAB) node), a donor node (e.g., a donor eNB, a donor gNB, etc.), an access point (e.g., a WiFi access point), a computing device, a device capable of wirelessly communicating, or any other device capable of sending and/or receiving signals. As used throughout, the term “wireless device” may comprise one or more of: a UE, a handset, a mobile device, a computing device, a node, a device capable of wirelessly communicating, or any other device capable of sending and/or receiving signals. Any reference to one or more of these terms/devices also considers use of any other term/device mentioned above.

The communications network **400** may comprise Radio Access Network (RAN) architecture. The RAN architecture may comprise one or more RAN nodes that may be a next generation Node B (gNB) (e.g., **401**) providing New Radio (NR) user plane and control plane protocol terminations towards a first wireless device (e.g. **406**). A RAN node may be a next generation evolved Node B (ng-eNB), providing Evolved UMTS Terrestrial Radio Access (E-UTRA) user plane and control plane protocol terminations towards a second wireless device. The first wireless device may communicate with a gNB over a Uu interface. The second wireless device may communicate with a ng-eNB over a Uu interface. Base station **401** may comprise one or more of a gNB, ng-eNB, and/or the like.

A gNB or an ng-eNB may host functions such as: radio resource management and scheduling, IP header compression, encryption and integrity protection of data, selection of Access and Mobility Management Function (AMF) at User Equipment (UE) attachment, routing of user plane and control plane data, connection setup and release, scheduling and transmission of paging messages (originated from the AMF), scheduling and transmission of system broadcast information (originated from the AMF or Operation and Maintenance (O&M)), measurement and measurement reporting configuration, transport level packet marking in the uplink, session management, support of network slicing, Quality of Service (QoS) flow management and mapping to data radio bearers, support of wireless devices in RRC_IDLE/INACTIVE state, distribution function for Non-Access Stratum (NAS) messages, RAN sharing, and dual connectivity or tight interworking between NR and E-UTRA.

One or more gNBs and/or one or more ng-eNBs may be interconnected with each other by means of Xn interface. A gNB or an ng-eNB may be connected by means of NG interfaces to 5G Core Network (5GC). 5GC may comprise one or more AMF/User Plane Function (UPF) functions. A gNB or an ng-eNB may be connected to a UPF by means of an NG-User plane (NG-U) interface. The NG-U interface may provide delivery (e.g., non-guaranteed delivery) of user plane Protocol Data Units (PDUs) between a RAN node and the UPF. A gNB or an ng-eNB may be connected to an AMF by means of an NG-Control plane (e.g., NG-C) interface. The NG-C interface may provide functions such as NG interface management, UE context management, UE mobility management, transport of NAS messages, paging, PDU session management, configuration transfer or warning message transmission.

A UPF may host functions such as anchor point for intra-/inter-Radio Access Technology (RAT) mobility (if applicable), external PDU session point of interconnect to data network, packet routing and forwarding, packet inspection and user plane part of policy rule enforcement, traffic usage reporting, uplink classifier to support routing traffic flows to a data network, branching point to support multi-homed PDU session, QoS handling for user plane, for example, packet filtering, gating, Uplink (UL)/Downlink (DL) rate enforcement, uplink traffic verification (e.g. Service Data Flow (SDF) to QoS flow mapping), downlink packet buffering and/or downlink data notification triggering.

An AMF may host functions such as NAS signaling termination, NAS signaling security, Access Stratum (AS) security control, inter Core Network (CN) node signaling for mobility between 3rd Generation Partnership Project (3GPP) access networks, idle mode UE reachability (e.g., control and execution of paging retransmission), registration area management, support of intra-system and inter-system

mobility, access authentication, access authorization including check of roaming rights, mobility management control (subscription and policies), support of network slicing and/or Session Management Function (SMF) selection.

An interface may be a hardware interface, a firmware interface, a software interface, and/or a combination thereof. The hardware interface may include connectors, wires, electronic devices such as drivers, amplifiers, and/or the like. A software interface may include code stored in a memory device to implement protocol(s), protocol layers, communication drivers, device drivers, combinations thereof, and/or the like. A firmware interface may include a combination of embedded hardware and code stored in and/or in communication with a memory device to implement connections, electronic device operations, protocol(s), protocol layers, communication drivers, device drivers, hardware operations, combinations thereof, and/or the like.

The term configured may relate to the capacity of a device whether the device is in an operational or a non-operational state. Configured may also refer to specific settings in a device that effect the operational characteristics of the device whether the device is in an operational or a non-operational state. In other words, the hardware, software, firmware, registers, memory values, and/or the like may be “configured” within a device, whether the device is in an operational or a nonoperational state, to provide the device with specific characteristics. Terms such as “a control message to cause in a device” may mean that a control message has parameters that may be used to configure specific characteristics in the device, whether the device is in an operational or a non-operational state.

A network may include a multitude of base stations, providing a user plane NR PDCP/NR RLC/NR MAC/NR PHY and control plane (e.g., NR RRC) protocol terminations towards the wireless device. The base station(s) may be interconnected with other base station(s) (e.g., employing an Xn interface). The base stations may also be connected employing, for example, an NG interface to an NGC. FIG. 10A and FIG. 10B show examples for interfaces between a 5G core network (e.g., NGC) and base stations (e.g., gNB and eLTE eNB). For example, the base stations may be interconnected to the NGC control plane (e.g., NG CP) employing the NG-C interface and to the NGC user plane (e.g., UPGW) employing the NG-U interface. The NG interface may support a many-to-many relation between 5G core networks and base stations.

A base station may include many sectors, for example: 1, 2, 3, 4, or 6 sectors. A base station may include many cells, for example, ranging from 1 to 50 cells or more. A cell may be categorized, for example, as a primary cell or secondary cell. At RRC connection establishment/re-establishment/handover, one serving cell may provide the NAS (non-access stratum) mobility information (e.g., TAI), and at RRC connection re-establishment/handover, one serving cell may provide the security input. This cell may be referred to as the Primary Cell (PCell). In the downlink, the carrier corresponding to the PCell may be the Downlink Primary Component Carrier (DL PCC); in the uplink, the carrier corresponding to the PCell may be the Uplink Primary Component Carrier (UL PCC). Depending on wireless device capabilities, Secondary Cells (SCells) may be configured to form together with the PCell a set of serving cells. In the downlink, the carrier corresponding to an SCell may be a Downlink Secondary Component Carrier (DL SCC); in the uplink, the carrier corresponding to an SCell may be an Uplink Secondary Component Carrier (UL SCC). An SCell may or may not have an uplink carrier.

A cell, comprising a downlink carrier and optionally an uplink carrier, may be assigned a physical cell ID and a cell index. A carrier (downlink or uplink) may belong to only one cell. The cell ID or cell index may also identify the downlink carrier or uplink carrier of the cell (depending on the context in which it is used). The cell ID may be equally referred to a carrier ID, and cell index may be referred to carrier index. In implementation, the physical cell ID or cell index may be assigned to a cell. A cell ID may be determined using a synchronization signal transmitted on a downlink carrier. A cell index may be determined using RRC messages. For example, reference to a first physical cell ID for a first downlink carrier may indicate that the first physical cell ID is for a cell comprising the first downlink carrier. The same concept may apply to, for example, carrier activation. Reference to a first carrier that is activated may indicate that the cell comprising the first carrier is activated.

A device may be configured to operate as needed by freely combining any of the examples. The disclosed mechanisms may be performed if certain criteria are met, for example, in a wireless device, a base station, a radio environment, a network, a combination of the above, and/or the like. Example criteria may be based, at least in part, on for example, traffic load, initial system set up, packet sizes, traffic characteristics, a combination of the above, and/or the like. One or more criteria may be satisfied. It may be possible to implement examples that selectively implement disclosed protocols.

A base station may communicate with a variety of wireless devices. Wireless devices may support multiple technologies, and/or multiple releases of the same technology. Wireless devices may have some specific capability(ies) depending on its wireless device category and/or capability(ies). A base station may comprise multiple sectors. Reference to a base station communicating with a plurality of wireless devices may indicate that a base station may communicate with a subset of the total wireless devices in a coverage area. A plurality of wireless devices of a given LTE or 5G release, with a given capability and in a given sector of the base station, may be used. The plurality of wireless devices may refer to a selected plurality of wireless devices, and/or a subset of total wireless devices in a coverage area which perform according to disclosed methods, and/or the like. There may be a plurality of wireless devices in a coverage area that may not comply with the disclosed methods, for example, because those wireless devices perform based on older releases of LTE or 5G technology.

A base station may transmit (e.g., to a wireless device) one or more messages (e.g. RRC messages) that may comprise a plurality of configuration parameters for one or more cells. One or more cells may comprise at least one primary cell and at least one secondary cell. An RRC message may be broadcasted or unicasted to the wireless device. Configuration parameters may comprise common parameters and dedicated parameters.

Services and/or functions of an RRC sublayer may comprise at least one of: broadcast of system information related to AS and NAS; paging initiated by 5GC and/or NG-RAN; establishment, maintenance, and/or release of an RRC connection between a wireless device and NG-RAN, which may comprise at least one of addition, modification and release of carrier aggregation; or addition, modification, and/or release of dual connectivity in NR or between E-UTRA and NR. Services and/or functions of an RRC sublayer may further comprise at least one of security functions comprising key management; establishment, configuration, maintenance, and/or release of Signaling Radio Bearers (SRBs) and/or

11

Data Radio Bearers (DRBs); mobility functions which may comprise at least one of a handover (e.g. intra NR mobility or inter-RAT mobility) and a context transfer; or a wireless device cell selection and reselection and control of cell selection and reselection. Services and/or functions of an RRC sublayer may further comprise at least one of QoS management functions; a wireless device measurement configuration/reporting; detection of and/or recovery from radio link failure; or NAS message transfer to/from a core network entity (e.g. AMF, Mobility Management Entity (MME)) from/to the wireless device.

An RRC sublayer may support an RRC_Idle state, an RRC_Inactive state and/or an RRC_Connected state for a wireless device. In an RRC_Idle state, a wireless device may perform at least one of: Public Land Mobile Network (PLMN) selection; receiving broadcasted system information; cell selection/re-selection; monitoring/receiving a paging for mobile terminated data initiated by 5GC; paging for mobile terminated data area managed by 5GC; or DRX for CN paging configured via NAS. In an RRC_Inactive state, a wireless device may perform at least one of: receiving broadcasted system information; cell selection/re-selection; monitoring/receiving a RAN/CN paging initiated by NG-RAN/5GC; RAN-based notification area (RNA) managed by NG-RAN; or DRX for RAN/CN paging configured by NG-RAN/NAS. In an RRC_Idle state of a wireless device, a base station (e.g. NG-RAN) may keep a 5GC-NG-RAN connection (both C/U-planes) for the wireless device; and/or store a UE AS context for the wireless device. In an RRC_Connected state of a wireless device, a base station (e.g. NG-RAN) may perform at least one of: establishment of 5GC-NG-RAN connection (both C/U-planes) for the wireless device; storing a UE AS context for the wireless device; transmit/receive of unicast data to/from the wireless device; or network-controlled mobility based on measurement results received from the wireless device. In an RRC_Connected state of a wireless device, an NG-RAN may know a cell that the wireless device belongs to.

System information (SI) may be divided into minimum SI and other SI. The minimum SI may be periodically broadcast. The minimum SI may comprise basic information required for initial access and information for acquiring any other SI broadcast periodically or provisioned on-demand, i.e. scheduling information. The other SI may either be broadcast, or be provisioned in a dedicated manner, either triggered by a network or upon request from a wireless device. A minimum SI may be transmitted via two different downlink channels using different messages (e.g. MasterInformationBlock and SystemInformationBlockType1). The other SI may be transmitted via SystemInformationBlockType2. For a wireless device in an RRC_Connected state, dedicated RRC signaling may be employed for the request and delivery of the other SI. For the wireless device in the RRC_Idle state and/or the RRC_Inactive state, the request may trigger a random-access procedure.

A wireless device may send its radio access capability information which may be static. A base station may request what capabilities for a wireless device to report based on band information. If allowed by a network, a temporary capability restriction request may be sent by the wireless device to signal the limited availability of some capabilities (e.g. due to hardware sharing, interference or overheating) to the base station. The base station may confirm or reject the request. The temporary capability restriction may be transparent to 5GC (e.g., static capabilities may be stored in 5GC).

12

If CA is configured, a wireless device may have an RRC connection with a network. At RRC connection establishment/re-establishment/handover procedure, one serving cell may provide NAS mobility information, and at RRC connection re-establishment/handover, one serving cell may provide a security input. This cell may be referred to as the PCell. Depending on the capabilities of the wireless device, SCells may be configured to form together with the PCell a set of serving cells. The configured set of serving cells for the wireless device may comprise one PCell and one or more SCells.

The reconfiguration, addition and removal of SCells may be performed by RRC. At intra-NR handover, RRC may also add, remove, or reconfigure SCells for usage with the target PCell. If adding a new SCell, dedicated RRC signaling may be employed to send all required system information of the SCell. In connected mode, wireless devices may not need to acquire broadcasted system information directly from the SCells.

An RRC connection reconfiguration procedure may be used to modify an RRC connection, (e.g. to establish, modify and/or release RBs, to perform handover, to setup, modify, and/or release measurements, to add, modify, and/or release SCells and cell groups). As part of the RRC connection reconfiguration procedure, NAS dedicated information may be transferred from the network to the wireless device. The RRCConnectionReconfiguration message may be a command to modify an RRC connection. It may convey information for measurement configuration, mobility control, radio resource configuration (e.g. RBs, MAC main configuration and physical channel configuration) comprising any associated dedicated NAS information and security configuration. If the received RRC Connection Reconfiguration message includes the sCellToReleaseList, the wireless device may perform an SCell release. If the received RRC Connection Reconfiguration message includes the sCellToAddModList, the wireless device may perform SCell additions or modification.

An RRC connection establishment (or reestablishment, resume) procedure may be used to establish (or reestablish, resume) an RRC connection. An RRC connection establishment procedure may comprise SRB1 establishment. The RRC connection establishment procedure may be used to transfer the initial NAS dedicated information message from a wireless device to E-UTRAN. The RRCConnectionReestablishment message may be used to re-establish SRB1.

A measurement report procedure may be to transfer measurement results from a wireless device to NG-RAN. The wireless device may initiate a measurement report procedure, for example, after successful security activation. A measurement report message may be employed to transmit measurement results.

FIG. 5A, FIG. 5B, FIG. 5C, and FIG. 5D show examples of architecture for uplink and downlink signal transmission. FIG. 5A shows an example for an uplink physical channel. The baseband signal representing the physical uplink shared channel may be processed according to the following processes, which may be performed by structures described below. These structures and corresponding functions are shown as examples, however, it is anticipated that other structures and/or functions may be implemented in various examples. The structures and corresponding functions may comprise, for example, one or more scrambling devices 501A and 501B configured to perform scrambling of coded bits in each of the codewords to be transmitted on a physical channel; one or more modulation mappers 502A and 502B configured to perform modulation of scrambled bits to

13

generate complex-valued symbols; a layer mapper **503** configured to perform mapping of the complex-valued modulation symbols onto one or several transmission layers; one or more transform precoders **504A** and **504B** to generate complex-valued symbols; a precoding device **505** configured to perform precoding of the complex-valued symbols; one or more resource element mappers **506A** and **506B** configured to perform mapping of precoded complex-valued symbols to resource elements; one or more signal generators **507A** and **507B** configured to perform the generation of a complex-valued time-domain DFTS-OFDM/SC-FDMA signal for each antenna port; and/or the like.

FIG. **5B** shows an example for performing modulation and up-conversion to the carrier frequency of the complex-valued DFTS-OFDM/SC-FDMA baseband signal, for example, for each antenna port and/or for the complex-valued physical random access channel (PRACH) baseband signal. For example, the baseband signal, represented as $s_1(t)$, may be split, by a signal splitter **510**, into real and imaginary components, $\text{Re}\{s_1(t)\}$ and $\text{Im}\{s_1(t)\}$, respectively. The real component may be modulated by a modulator **511A**, and the imaginary component may be modulated by a modulator **511B**. The output signal of the modulator **511A** and the output signal of the modulator **511B** may be mixed by a mixer **512**. The output signal of the mixer **512** may be input to a filtering device **513**, and filtering may be employed by the filtering device **513** prior to transmission.

FIG. **5C** shows an example structure for downlink transmissions. The baseband signal representing a downlink physical channel may be processed by the following processes, which may be performed by structures described below. These structures and corresponding functions are shown as examples, however, it is anticipated that other structures and/or functions may be implemented in various examples. The structures and corresponding functions may comprise, for example, one or more scrambling devices **531A** and **531B** configured to perform scrambling of coded bits in each of the codewords to be transmitted on a physical channel; one or more modulation mappers **532A** and **532B** configured to perform modulation of scrambled bits to generate complex-valued modulation symbols; a layer mapper **533** configured to perform mapping of the complex-valued modulation symbols onto one or several transmission layers; a precoding device **535** configured to perform precoding of the complex-valued modulation symbols on each layer for transmission on the antenna ports; one or more resource element mappers **536A** and **536B** configured to perform mapping of complex-valued modulation symbols for each antenna port to resource elements; one or more OFDM signal generators **537A** and **537B** configured to perform the generation of complex-valued time-domain OFDM signal for each antenna port; and/or the like.

FIG. **5D** shows an example structure for modulation and up-conversion to the carrier frequency of the complex-valued OFDM baseband signal for each antenna port. For example, the baseband signal, represented as $s_1^{(p)}(t)$, may be split, by a signal splitter **520**, into real and imaginary components, $\text{Re}\{s_1^{(p)}(t)\}$ and $\text{Im}\{s_1^{(p)}(t)\}$, respectively. The real component may be modulated by a modulator **521A**, and the imaginary component may be modulated by a modulator **521B**. The output signal of the modulator **521A** and the output signal of the modulator **521B** may be mixed by a mixer **522**. The output signal of the mixer **522** may be input to a filtering device **523**, and filtering may be employed by the filtering device **523** prior to transmission.

FIG. **6** and FIG. **7** show examples for protocol structures with CA and multi-connectivity. NR may support multi-

14

connectivity operation, whereby a multiple receiver/transmitter (RX/TX) wireless device in RRC_CONNECTED may be configured to utilize radio resources provided by multiple schedulers located in multiple gNBs connected via a non-ideal or ideal backhaul over the Xn interface. gNBs involved in multi-connectivity for a certain wireless device may assume two different roles: a gNB may either act as a master gNB (e.g., **600**) or as a secondary gNB (e.g., **610** or **620**). In multi-connectivity, a wireless device may be connected to one master gNB (e.g., **600**) and one or more secondary gNBs (e.g., **610** and/or **620**). Any one or more of the Master gNB **600** and/or the secondary gNBs **610** and **620** may be a Next Generation (NG) NodeB. The master gNB **600** may comprise protocol layers NR MAC **601**, NR RLC **602** and **603**, and NR PDCP **604** and **605**. The secondary gNB may comprise protocol layers NR MAC **611**, NR RLC **612** and **613**, and NR PDCP **614**. The secondary gNB may comprise protocol layers NR MAC **621**, NR RLC **622** and **623**, and NR PDCP **624**. The master gNB **600** may communicate via an interface **606** and/or via an interface **607**, the secondary gNB **610** may communicate via an interface **615**, and the secondary gNB **620** may communicate via an interface **625**. The master gNB **600** may also communicate with the secondary gNB **610** and the secondary gNB **620** via interfaces **608** and **609**, respectively, which may include Xn interfaces. For example, the master gNB **600** may communicate via the interface **608**, at layer NR PDCP **605**, and with the secondary gNB **610** at layer NR RLC **612**. The master gNB **600** may communicate via the interface **609**, at layer NR PDCP **605**, and with the secondary gNB **620** at layer NR RLC **622**.

FIG. **7** shows an example structure for the UE side MAC entities, for example, if a Master Cell Group (MCG) and a Secondary Cell Group (SCG) are configured. Media Broadcast Multicast Service (MBMS) reception may be included but is not shown in this figure for simplicity.

In multi-connectivity, the radio protocol architecture that a particular bearer uses may depend on how the bearer is set up. As an example, three alternatives may exist, an MCG bearer, an SCG bearer, and a split bearer, such as shown in FIG. **6**. NR RRC may be located in a master gNB and SRBs may be configured as a MCG bearer type and may use the radio resources of the master gNB. Multi-connectivity may have at least one bearer configured to use radio resources provided by the secondary gNB. Multi-connectivity may or may not be configured or implemented.

For multi-connectivity, the wireless device may be configured with multiple NR MAC entities: e.g., one NR MAC entity for a master gNB, and other NR MAC entities for secondary gNBs. In multi-connectivity, the configured set of serving cells for a wireless device may comprise two subsets: e.g., the Master Cell Group (MCG) including the serving cells of the master gNB, and the Secondary Cell Groups (SCGs) including the serving cells of the secondary gNBs.

At least one cell in a SCG may have a configured UL component carrier (CC) and one of the UL CCs, for example, named PSCell (or PCell of SCG, or sometimes called PCell), may be configured with PUCCH resources. If the SCG is configured, there may be at least one SCG bearer or one split bearer. If a physical layer problem or a random access problem on a PSCell occurs or is detected, if the maximum number of NR RLC retransmissions has been reached associated with the SCG, or if an access problem on a PSCell during a SCG addition or a SCG change occurs or is detected, then an RRC connection re-establishment procedure may not be triggered, UL transmissions towards cells

of the SCG may be stopped, a master gNB may be informed by the wireless device of a SCG failure type, and for a split bearer the DL data transfer over the master gNB may be maintained. The NR RLC Acknowledge Mode (AM) bearer may be configured for the split bearer. Like the PCell, a PSCell may not be de-activated. The PSCell may be changed with an SCG change (e.g., with a security key change and a RACH procedure). A direct bearer type may change between a split bearer and an SCG bearer, or a simultaneous configuration of an SCG and a split bearer may or may not be supported.

A master gNB and secondary gNBs may interact for multi-connectivity. The master gNB may maintain the RRM measurement configuration of the wireless device, and the master gNB may, (e.g., based on received measurement reports, and/or based on traffic conditions and/or bearer types), decide to ask a secondary gNB to provide additional resources (e.g., serving cells) for a wireless device. If a request from the master gNB is received, a secondary gNB may create a container that may result in the configuration of additional serving cells for the wireless device (or the secondary gNB decide that it has no resource available to do so). For wireless device capability coordination, the master gNB may provide some or all of the Active Set (AS) configuration and the wireless device capabilities to the secondary gNB. The master gNB and the secondary gNB may exchange information about a wireless device configuration, such as by employing NR RRC containers (e.g., inter-node messages) carried in Xn messages. The secondary gNB may initiate a reconfiguration of its existing serving cells (e.g., PUCCH towards the secondary gNB). The secondary gNB may decide which cell is the PSCell within the SCG. The master gNB may or may not change the content of the NR RRC configuration provided by the secondary gNB. In an SCG addition and an SCG SCell addition, the master gNB may provide the latest measurement results for the SCG cell(s). Both a master gNB and a secondary gNBs may know the system frame number (SFN) and subframe offset of each other by operations, administration, and maintenance (OAM) (e.g., for the purpose of discontinuous reception (DRX) alignment and identification of a measurement gap). If adding a new SCG SCell, dedicated NR RRC signaling may be used for sending required system information of the cell for CA, except, for example, for the SFN acquired from an MIB of the PSCell of an SCG.

FIG. 7 shows an example of dual-connectivity (DC) for two MAC entities at a wireless device side. A first MAC entity may comprise a lower layer of an MCG 700, an upper layer of an MCG 718, and one or more intermediate layers of an MCG 719. The lower layer of the MCG 700 may comprise, for example, a paging channel (PCH) 701, a broadcast channel (BCH) 702, a downlink shared channel (DL-SCH) 703, an uplink shared channel (UL-SCH) 704, and a random access channel (RACH) 705. The one or more intermediate layers of the MCG 719 may comprise, for example, one or more hybrid automatic repeat request (HARQ) processes 706, one or more random access control processes 707, multiplexing and/or de-multiplexing processes 709, logical channel prioritization on the uplink processes 710, and a control processes 708 providing control for the above processes in the one or more intermediate layers of the MCG 719. The upper layer of the MCG 718 may comprise, for example, a paging control channel (PCCH) 711, a broadcast control channel (BCCH) 712, a common control channel (CCCH) 713, a dedicated control channel (DCCH) 714, a dedicated traffic channel (DTCH) 715, and a MAC control 716.

A second MAC entity may comprise a lower layer of an SCG 720, an upper layer of an SCG 738, and one or more intermediate layers of an SCG 739. The lower layer of the SCG 720 may comprise, for example, a BCH 722, a DL-SCH 723, an UL-SCH 724, and a RACH 725. The one or more intermediate layers of the SCG 739 may comprise, for example, one or more HARQ processes 726, one or more random access control processes 727, multiplexing and/or de-multiplexing processes 729, logical channel prioritization on the uplink processes 730, and a control processes 728 providing control for the above processes in the one or more intermediate layers of the SCG 739. The upper layer of the SCG 738 may comprise, for example, a BCCH 732, a DCCH 714, a DTCH 735, and a MAC control 736.

Serving cells may be grouped in a TA group (TAG). Serving cells in one TAG may use the same timing reference. For a given TAG, a wireless device may use at least one downlink carrier as a timing reference. For a given TAG, a wireless device may synchronize uplink subframe and frame transmission timing of uplink carriers belonging to the same TAG. Serving cells having an uplink to which the same TA applies may correspond to serving cells hosted by the same receiver. A wireless device supporting multiple TAs may support two or more TA groups. One TA group may include the PCell and may be called a primary TAG (pTAG). In a multiple TAG configuration, at least one TA group may not include the PCell and may be called a secondary TAG (sTAG). Carriers within the same TA group may use the same TA value and/or the same timing reference. If DC is configured, cells belonging to a cell group (e.g., MCG or SCG) may be grouped into multiple TAGs including a pTAG and one or more sTAGs.

FIG. 8 shows example TAG configurations. In Example 1, a pTAG comprises a PCell, and an sTAG comprises an SCell1. In Example 2, a pTAG comprises a PCell and an SCell1, and an sTAG comprises an SCell2 and an SCell3. In Example 3, a pTAG comprises a PCell and an SCell1, and an sTAG1 comprises an SCell2 and an SCell3, and an sTAG2 comprises a SCell4. Up to four TAGs may be supported in a cell group (MCG or SCG), and other example TAG configurations may also be provided. In various examples, structures and operations are described for use with a pTAG and an sTAG. Some of the examples may be used for configurations with multiple sTAGs.

An eNB may initiate an RA procedure, via a PDCCH order, for an activated SCell. The PDCCH order may be sent on a scheduling cell of this SCell. If cross carrier scheduling is configured for a cell, the scheduling cell may be different than the cell that is employed for preamble transmission, and the PDCCH order may include an SCell index. At least a non-contention based RA procedure may be supported for SCell(s) assigned to sTAG(s).

FIG. 9 shows an example of random access processes, and a corresponding message flow, in a secondary TAG. A base station, such as an eNB, may transmit an activation command 900 to a wireless device, such as a UE. The activation command 900 may be transmitted to activate an SCell. The base station may also transmit a PDCCH order 901 to the wireless device, which may be transmitted, for example, after the activation command 900. The wireless device may begin to perform a RACH process for the SCell, which may be initiated, for example, after receiving the PDCCH order 901. A wireless device may transmit to the base station (e.g., as part of a RACH process) a preamble 902 (e.g., Msg1), such as a random access preamble (RAP). The preamble 902 may be transmitted after or in response to the PDCCH order 901. The wireless device may transmit the preamble 902 via

an SCell belonging to an sTAG. Preamble transmission for SCells may be controlled by a network using PDCCH format 1A. The base station may send a random access response (RAR) 903 (e.g., Msg2 message) to the wireless device. The RAR 903 may be after or in response to the preamble 902 transmission via the SCell. The RAR 903 may be addressed to a random access radio network temporary identifier (RA-RNTI) in a PCell common search space (CSS). If the wireless device receives the RAR 903, the RACH process may conclude. The RACH process may conclude, for example, after or in response to the wireless device receiving the RAR 903 from the base station. After the RACH process, the wireless device may transmit an uplink transmission 904. The uplink transmission 904 may comprise uplink packets transmitted via the same SCell used for the preamble 902 transmission.

Timing alignment (e.g., initial timing alignment) for communications between the wireless device and the base station may be performed through a random access procedure, such as described above regarding FIG. 9. The random access procedure may involve a wireless device, such as a UE, transmitting a random access preamble and a base station, such as an eNB, responding with an initial TA command N_{TA} (amount of timing advance) within a random access response window. The start of the random access preamble may be aligned with the start of a corresponding uplink subframe at the wireless device assuming $N_{TA}=0$. The eNB may estimate the uplink timing from the random access preamble transmitted by the wireless device. The TA command may be derived by the eNB based on the estimation of the difference between the desired UL timing and the actual UL timing. The wireless device may determine the initial uplink transmission timing relative to the corresponding downlink of the sTAG on which the preamble is transmitted.

The mapping of a serving cell to a TAG may be configured by a serving eNB with RRC signaling. The mechanism for TAG configuration and reconfiguration may be based on RRC signaling. If an eNB performs an SCell addition configuration, the related TAG configuration may be configured for the SCell. An eNB may modify the TAG configuration of an SCell by removing (e.g., releasing) the SCell and adding (e.g., configuring) a new SCell (with the same physical cell ID and frequency) with an updated TAG ID. The new SCell with the updated TAG ID may initially be inactive subsequent to being assigned the updated TAG ID. The eNB may activate the updated new SCell and start scheduling packets on the activated SCell. In some examples, it may not be possible to change the TAG associated with an SCell, but rather, the SCell may need to be removed and a new SCell may need to be added with another TAG. For example, if there is a need to move an SCell from an sTAG to a pTAG, at least one RRC message, such as at least one RRC reconfiguration message, may be sent to the wireless device. The at least one RRC message may be sent to the wireless device to reconfigure TAG configurations, for example, by releasing the SCell and configuring the SCell as a part of the pTAG. If, for example, an SCell is added or configured without a TAG index, the SCell may be explicitly assigned to the pTAG. The PCell may not change its TAG group and may be a member of the pTAG.

In LTE Release-10 and Release-11 CA, a PUCCH transmission is only transmitted on a PCell (e.g., a PSCell) to an eNB. In LTE-Release 12 and earlier, a wireless device may transmit PUCCH information on one cell (e.g., a PCell or a PSCell) to a given eNB. As the number of CA capable wireless devices increase, and as the number of aggregated carriers increase, the number of PUCCHs and the PUCCH

payload size may increase. Accommodating the PUCCH transmissions on the PCell may lead to a high PUCCH load on the PCell. A PUCCH on an SCell may be used to offload the PUCCH resource from the PCell. More than one PUCCH may be configured. For example, a PUCCH on a PCell may be configured and another PUCCH on an SCell may be configured. One, two, or more cells may be configured with PUCCH resources for transmitting CSI, acknowledgment (ACK), and/or non-acknowledgment (NACK) to a base station. Cells may be grouped into multiple PUCCH groups, and one or more cell within a group may be configured with a PUCCH. In some examples, one SCell may belong to one PUCCH group. SCells with a configured PUCCH transmitted to a base station may be called a PUCCH SCell, and a cell group with a common PUCCH resource transmitted to the same base station may be called a PUCCH group.

A MAC entity may have a configurable timer, for example, timeAlignmentTimer, per TAG. The timeAlignmentTimer may be used to control how long the MAC entity considers the serving cells belonging to the associated TAG to be uplink time aligned. If a Timing Advance Command MAC control element is received, the MAC entity may apply the Timing Advance Command for the indicated TAG; and/or the MAC entity may start or restart the timeAlignmentTimer associated with a TAG that may be indicated by the Timing Advance Command MAC control element. If a Timing Advance Command is received in a Random Access Response message for a serving cell belonging to a TAG, the MAC entity may apply the Timing Advance Command for this TAG and/or start or restart the timeAlignmentTimer associated with this TAG. Additionally or alternatively, if the Random Access Preamble is not selected by the MAC entity, the MAC entity may apply the Timing Advance Command for this TAG and/or start or restart the timeAlignmentTimer associated with this TAG. If the timeAlignmentTimer associated with this TAG is not running, the Timing Advance Command for this TAG may be applied, and the timeAlignmentTimer associated with this TAG may be started. If the contention resolution is not successful, a timeAlignmentTimer associated with this TAG may be stopped. If the contention resolution is successful, the MAC entity may ignore the received Timing Advance Command. The MAC entity may determine whether the contention resolution is successful or whether the contention resolution is not successful.

FIG. 10A and FIG. 10B show examples for interfaces between a 5G core network (e.g., NGC) and base stations (e.g., gNB and eLTE eNB). A base station, such as a gNB 1020, may be interconnected to an NGC 1010 control plane employing an NG-C interface. The base station, for example, the gNB 1020, may also be interconnected to an NGC 1010 user plane (e.g., UPGW) employing an NG-U interface. As another example, a base station, such as an eLTE eNB 1040, may be interconnected to an NGC 1030 control plane employing an NG-C interface. The base station, for example, the eLTE eNB 1040, may also be interconnected to an NGC 1030 user plane (e.g., UPGW) employing an NG-U interface. An NG interface may support a many-to-many relation between 5G core networks and base stations.

FIG. 11A, FIG. 11B, FIG. 11C, FIG. 11D, FIG. 11E, and FIG. 11F are examples for architectures of tight interworking between a 5G RAN and an LTE RAN. The tight interworking may enable a multiple receiver/transmitter (RX/TX) wireless device in an RRC_CONNECTED state to be configured to utilize radio resources provided by two

schedulers located in two base stations (e.g., an eLTE eNB and a gNB). The two base stations may be connected via a non-ideal or ideal backhaul over the Xx interface between an LTE eNB and a gNB, or over the Xn interface between an eLTE eNB and a gNB. Base stations involved in tight interworking for a certain wireless device may assume different roles. For example, a base station may act as a master base station or a base station may act as a secondary base station. In tight interworking, a wireless device may be connected to both a master base station and a secondary base station. Mechanisms implemented in tight interworking may be extended to cover more than two base stations.

A master base station may be an LTE eNB 1102A or an LTE eNB 1102B, which may be connected to EPC nodes 1101A or 1101B, respectively. This connection to EPC nodes may be, for example, to an MME via the S1-C interface and/or to an S-GW via the S1-U interface. A secondary base station may be a gNB 1103A or a gNB 1103B, either or both of which may be a non-standalone node having a control plane connection via an Xx-C interface to an LTE eNB (e.g., the LTE eNB 1102A or the LTE eNB 1102B). In the tight interworking architecture of FIG. 11A, a user plane for a gNB (e.g., the gNB 1103A) may be connected to an S-GW (e.g., the EPC 1101A) through an LTE eNB (e.g., the LTE eNB 1102A), via an Xx-U interface between the LTE eNB and the gNB, and via an S1-U interface between the LTE eNB and the S-GW. In the architecture of FIG. 11B, a user plane for a gNB (e.g., the gNB 1103B) may be connected directly to an S-GW (e.g., the EPC 1101B) via an S1-U interface between the gNB and the S-GW.

A master base station may be a gNB 1103C or a gNB 1103D, which may be connected to NGC nodes 1101C or 1101D, respectively. This connection to NGC nodes may be, for example, to a control plane core node via the NG-C interface and/or to a user plane core node via the NG-U interface. A secondary base station may be an eLTE eNB 1102C or an eLTE eNB 1102D, either or both of which may be a non-standalone node having a control plane connection via an Xn-C interface to a gNB (e.g., the gNB 1103C or the gNB 1103D). In the tight interworking architecture of FIG. 11C, a user plane for an eLTE eNB (e.g., the eLTE eNB 1102C) may be connected to a user plane core node (e.g., the NGC 1101C) through a gNB (e.g., the gNB 1103C), via an Xn-U interface between the eLTE eNB and the gNB, and via an NG-U interface between the gNB and the user plane core node. In the architecture of FIG. 11D, a user plane for an eLTE eNB (e.g., the eLTE eNB 1102D) may be connected directly to a user plane core node (e.g., the NGC 1101D) via an NG-U interface between the eLTE eNB and the user plane core node.

A master base station may be an eLTE eNB 1102E or an eLTE eNB 1102F, which may be connected to NGC nodes 1101E or 1101F, respectively. This connection to NGC nodes may be, for example, to a control plane core node via the NG-C interface and/or to a user plane core node via the NG-U interface. A secondary base station may be a gNB 1103E or a gNB 1103F, either or both of which may be a non-standalone node having a control plane connection via an Xn-C interface to an eLTE eNB (e.g., the eLTE eNB 1102E or the eLTE eNB 1102F). In the tight interworking architecture of FIG. 11E, a user plane for a gNB (e.g., the gNB 1103E) may be connected to a user plane core node (e.g., the NGC 1101E) through an eLTE eNB (e.g., the eLTE eNB 1102E), via an Xn-U interface between the eLTE eNB and the gNB, and via an NG-U interface between the eLTE eNB and the user plane core node. In the architecture of FIG.

11F, a user plane for a gNB (e.g., the gNB 1103F) may be connected directly to a user plane core node (e.g., the NGC 1101F) via an NG-U interface between the gNB and the user plane core node.

FIG. 12A, FIG. 12B, and FIG. 12C are examples for radio protocol structures of tight interworking bearers.

An LTE eNB 1201A may be an S1 master base station, and a gNB 1210A may be an S1 secondary base station. An example for a radio protocol architecture for a split bearer and an SCG bearer is shown. The LTE eNB 1201A may be connected to an EPC with a non-standalone gNB 1210A, via an Xx interface between the PDCP 1206A and an NR RLC 1212A. The LTE eNB 1201A may include protocol layers MAC 1202A, RLC 1203A and RLC 1204A, and PDCP 1205A and PDCP 1206A. An MCG bearer type may interface with the PDCP 1205A, and a split bearer type may interface with the PDCP 1206A. The gNB 1210A may include protocol layers NR MAC 1211A, NR RLC 1212A and NR RLC 1213A, and NR PDCP 1214A. An SCG bearer type may interface with the NR PDCP 1214A.

A gNB 1201B may be an NG master base station, and an eLTE eNB 1210B may be an NG secondary base station. An example for a radio protocol architecture for a split bearer and an SCG bearer is shown. The gNB 1201B may be connected to an NGC with a non-standalone eLTE eNB 1210B, via an Xn interface between the NR PDCP 1206B and an RLC 1212B. The gNB 1201B may include protocol layers NR MAC 1202B, NR RLC 1203B and NR RLC 1204B, and NR PDCP 1205B and NR PDCP 1206B. An MCG bearer type may interface with the NR PDCP 1205B, and a split bearer type may interface with the NR PDCP 1206B. The eLTE eNB 1210B may include protocol layers MAC 1211B, RLC 1212B and RLC 1213B, and PDCP 1214B. An SCG bearer type may interface with the PDCP 1214B.

An eLTE eNB 1201C may be an NG master base station, and a gNB 1210C may be an NG secondary base station. An example for a radio protocol architecture for a split bearer and an SCG bearer is shown. The eLTE eNB 1201C may be connected to an NGC with a non-standalone gNB 1210C, via an Xn interface between the PDCP 1206C and an NR RLC 1212C. The eLTE eNB 1201C may include protocol layers MAC 1202C, RLC 1203C and RLC 1204C, and PDCP 1205C and PDCP 1206C. An MCG bearer type may interface with the PDCP 1205C, and a split bearer type may interface with the PDCP 1206C. The gNB 1210C may include protocol layers NR MAC 1211C, NR RLC 1212C and NR RLC 1213C, and NR PDCP 1214C. An SCG bearer type may interface with the NR PDCP 1214C.

In a 5G network, the radio protocol architecture that a particular bearer uses may depend on how the bearer is setup. At least three alternatives may exist, for example, an MCG bearer, an SCG bearer, and a split bearer, such as shown in FIG. 12A, FIG. 12B, and FIG. 12C. The NR RRC may be located in a master base station, and the SRBs may be configured as an MCG bearer type and may use the radio resources of the master base station. Tight interworking may have at least one bearer configured to use radio resources provided by the secondary base station. Tight interworking may or may not be configured or implemented.

The wireless device may be configured with two MAC entities: e.g., one MAC entity for a master base station, and one MAC entity for a secondary base station. In tight interworking, the configured set of serving cells for a wireless device may comprise of two subsets: e.g., the Master Cell Group (MCG) including the serving cells of the

master base station, and the Secondary Cell Group (SCG) including the serving cells of the secondary base station.

At least one cell in a SCG may have a configured UL CC and one of them, for example, a PSCell (or the PCell of the SCG, which may also be called a PCell), is configured with PUCCH resources. If the SCG is configured, there may be at least one SCG bearer or one split bearer. If one or more of a physical layer problem or a random access problem is detected on a PSCell, if the maximum number of (NR) RLC retransmissions associated with the SCG has been reached, and/or if an access problem on a PSCell during an SCG addition or during an SCG change is detected, then: an RRC connection re-establishment procedure may not be triggered, UL transmissions towards cells of the SCG may be stopped, a master base station may be informed by the wireless device of a SCG failure type, and/or for a split bearer the DL data transfer over the master base station may be maintained. The RLC AM bearer may be configured for the split bearer. Like the PCell, a PSCell may not be de-activated. A PSCell may be changed with an SCG change, for example, with security key change and a RACH procedure. A direct bearer type change, between a split bearer and an SCG bearer, may not be supported. Simultaneous configuration of an SCG and a split bearer may not be supported.

A master base station and a secondary base station may interact. The master base station may maintain the RRM measurement configuration of the wireless device. The master base station may determine to ask a secondary base station to provide additional resources (e.g., serving cells) for a wireless device. This determination may be based on, for example, received measurement reports, traffic conditions, and/or bearer types. If a request from the master base station is received, a secondary base station may create a container that may result in the configuration of additional serving cells for the wireless device, or the secondary base station may determine that it has no resource available to do so. The master base station may provide at least part of the AS configuration and the wireless device capabilities to the secondary base station, for example, for wireless device capability coordination. The master base station and the secondary base station may exchange information about a wireless device configuration such as by using RRC containers (e.g., inter-node messages) carried in Xn or Xx messages. The secondary base station may initiate a reconfiguration of its existing serving cells (e.g., PUCCH towards the secondary base station). The secondary base station may determine which cell is the PSCell within the SCG. The master base station may not change the content of the RRC configuration provided by the secondary base station. If an SCG is added and/or an SCG SCell is added, the master base station may provide the latest measurement results for the SCG cell(s). Either or both of a master base station and a secondary base station may know the SFN and subframe offset of each other by OAM, (e.g., for the purpose of DRX alignment and identification of a measurement gap). If a new SCG SCell is added, dedicated RRC signaling may be used for sending required system information of the cell, such as for CA, except, for example, for the SFN acquired from an MIB of the PSCell of an SCG.

FIG. 13A and FIG. 13B show examples for gNB deployment. A core 1301 and a core 1310 may interface with other nodes via RAN-CN interfaces. In a non-centralized deployment example, the full protocol stack (e.g., NR RRC, NR PDPCP, NR RLC, NR MAC, and NR PHY) may be supported at one node, such as a gNB 1302, a gNB 1303, and/or an eLTE eNB or LTE eNB 1304. These nodes (e.g., the gNB 1302, the gNB 1303, and the eLTE eNB or LTE eNB 1304)

may interface with one of more of each other via a respective inter-BS interface. In a centralized deployment example, upper layers of a gNB may be located in a Central Unit (CU) 1311, and lower layers of the gNB may be located in Distributed Units (DU) 1312, 1313, and 1314. The CU-DU interface (e.g., Fs interface) connecting CU 1311 and DUs 1312, 1313, and 1314 may be ideal or non-ideal. The Fs-C may provide a control plane connection over the Fs interface, and the Fs-U may provide a user plane connection over the Fs interface. In the centralized deployment, different functional split options between the CU 1311 and the DUs 1312, 1313, and 1314 may be possible by locating different protocol layers (e.g., RAN functions) in the CU 1311 and in the DU 1312, 1313, and 1314. The functional split may support flexibility to move the RAN functions between the CU 1311 and the DUs 1312, 1313, and 1314 depending on service requirements and/or network environments. The functional split option may change during operation (e.g., after the Fs interface setup procedure), or the functional split option may change only in the Fs setup procedure (e.g., the functional split option may be static during operation after Fs setup procedure).

FIG. 14 shows examples for different functional split options of a centralized gNB deployment. Element numerals that are followed by "A" or "B" designations in FIG. 14 may represent the same elements in different traffic flows, for example, either receiving data (e.g., data 1402A) or sending data (e.g., 1402B). In the split option example 1, an NR RRC 1401 may be in a CU, and an NR PDPCP 1403, an NR RLC (e.g., comprising a High NR RLC 1404 and/or a Low NR RLC 1405), an NR MAC (e.g., comprising a High NR MAC 1406 and/or a Low NR MAC 1407), an NR PHY (e.g., comprising a High NR PHY 1408 and/or a Low NR PHY 1409), and an RF 1410 may be in a DU. In the split option example 2, the NR RRC 1401 and the NR PDPCP 1403 may be in a CU, and the NR RLC, the NR MAC, the NR PHY, and the RF 1410 may be in a DU. In the split option example 3, the NR RRC 1401, the NR PDPCP 1403, and a partial function of the NR RLC (e.g., the High NR RLC 1404) may be in a CU, and the other partial function of the NR RLC (e.g., the Low NR RLC 1405), the NR MAC, the NR PHY, and the RF 1410 may be in a DU. In the split option example 4, the NR RRC 1401, the NR PDPCP 1403, and the NR RLC may be in a CU, and the NR MAC, the NR PHY, and the RF 1410 may be in a DU. In the split option example 5, the NR RRC 1401, the NR PDPCP 1403, the NR RLC, and a partial function of the NR MAC (e.g., the High NR MAC 1406) may be in a CU, and the other partial function of the NR MAC (e.g., the Low NR MAC 1407), the NR PHY, and the RF 1410 may be in a DU. In the split option example 6, the NR RRC 1401, the NR PDPCP 1403, the NR RLC, and the NR MAC may be in CU, and the NR PHY and the RF 1410 may be in a DU. In the split option example 7, the NR RRC 1401, the NR PDPCP 1403, the NR RLC, the NR MAC, and a partial function of the NR PHY (e.g., the High NR PHY 1408) may be in a CU, and the other partial function of the NR PHY (e.g., the Low NR PHY 1409) and the RF 1410 may be in a DU. In the split option example 8, the NR RRC 1401, the NR PDPCP 1403, the NR RLC, the NR MAC, and the NR PHY may be in a CU, and the RF 1410 may be in a DU.

The functional split may be configured per CU, per DU, per wireless device, per bearer, per slice, and/or with other granularities. In a per CU split, a CU may have a fixed split, and DUs may be configured to match the split option of the CU. In a per DU split, each DU may be configured with a different split, and a CU may provide different split options

for different DUs. In a per wireless device split, a gNB (e.g., a CU and a DU) may provide different split options for different wireless devices. In a per bearer split, different split options may be utilized for different bearer types. In a per slice split, different split options may be applied for different slices.

A new radio access network (new RAN) may support different network slices, which may allow differentiated treatment customized to support different service requirements with end to end scope. The new RAN may provide a differentiated handling of traffic for different network slices that may be pre-configured, and the new RAN may allow a single RAN node to support multiple slices. The new RAN may support selection of a RAN part for a given network slice, for example, by one or more slice ID(s) or NSSAI(s) provided by a wireless device or provided by an NGC (e.g., an NG CP). The slice ID(s) or NSSAI(s) may identify one or more of pre-configured network slices in a PLMN. For an initial attach, a wireless device may provide a slice ID and/or an NSSAI, and a RAN node (e.g., a gNB) may use the slice ID or the NSSAI for routing an initial NAS signaling to an NGC control plane function (e.g., an NG CP). If a wireless device does not provide any slice ID or NSSAI, a RAN node may send a NAS signaling to a default NGC control plane function. For subsequent accesses, the wireless device may provide a temporary ID for a slice identification, which may be assigned by the NGC control plane function, to enable a RAN node to route the NAS message to a relevant NGC control plane function. The new RAN may support resource isolation between slices. If the RAN resource isolation is implemented, shortage of shared resources in one slice does not cause a break in a service level agreement for another slice.

The amount of data traffic carried over networks is expected to increase for many years to come. The number of users and/or devices is increasing, and each user/device accesses an increasing number and variety of services, for example, video delivery, large files, and images. This requires not only high capacity in the network, but also provisioning very high data rates to meet customers' expectations on interactivity and responsiveness. More spectrum may be required for network operators to meet the increasing demand. Considering user expectations of high data rates along with seamless mobility, it is beneficial that more spectrum be made available for deploying macro cells as well as small cells for communication systems.

Striving to meet the market demands, there has been increasing interest from operators in deploying some complementary access utilizing unlicensed spectrum to meet the traffic growth. This is exemplified by the large number of operator-deployed Wi-Fi networks and the 3GPP standardization of LTE/WLAN interworking solutions. This interest indicates that unlicensed spectrum, if present, may be an effective complement to licensed spectrum for network operators, for example, to help address the traffic explosion in some examples, such as hotspot areas. Licensed Assisted Access (LAA) offers an alternative for operators to make use of unlicensed spectrum, for example, if managing one radio network, offering new possibilities for optimizing the network's efficiency.

Listen-before-talk (clear channel assessment) may be implemented for transmission in an LAA cell. In a listen-before-talk (LBT) procedure, equipment may apply a clear channel assessment (CCA) check before using the channel. For example, the CCA may utilize at least energy detection to determine the presence or absence of other signals on a channel to determine if a channel is occupied or clear,

respectively. For example, European and Japanese regulations mandate the usage of LBT in the unlicensed bands. Apart from regulatory requirements, carrier sensing via LBT may be one way for fair sharing of the unlicensed spectrum.

Discontinuous transmission on an unlicensed carrier with limited maximum transmission duration may be enabled. Some of these functions may be supported by one or more signals to be transmitted from the beginning of a discontinuous LAA downlink transmission. Channel reservation may be enabled by the transmission of signals, by an LAA node, after gaining channel access, for example, via a successful LBT operation, so that other nodes that receive the transmitted signal with energy above a certain threshold sense the channel to be occupied. Functions that may need to be supported by one or more signals for LAA operation with discontinuous downlink transmission may include one or more of the following: detection of the LAA downlink transmission (including cell identification) by wireless devices, time synchronization of wireless devices, and frequency synchronization of wireless devices.

DL LAA design may employ subframe boundary alignment according to LTE-A carrier aggregation timing relationships across serving cells aggregated by CA. This may not indicate that the eNB transmissions may start only at the subframe boundary. LAA may support transmitting PDSCH if not all OFDM symbols are available for transmission in a subframe according to LBT. Delivery of necessary control information for the PDSCH may also be supported.

LBT procedures may be employed for fair and friendly coexistence of LAA with other operators and technologies operating in unlicensed spectrum. LBT procedures on a node attempting to transmit on a carrier in unlicensed spectrum may require the node to perform a clear channel assessment to determine if the channel is free for use. An LBT procedure may involve at least energy detection to determine if the channel is being used. For example, regulatory requirements in some regions, for example, in Europe, specify an energy detection threshold such that if a node receives energy greater than this threshold, the node assumes that the channel is not free. Nodes may follow such regulatory requirements. A node may optionally use a lower threshold for energy detection than that specified by regulatory requirements. LAA may employ a mechanism to adaptively change the energy detection threshold, for example, LAA may employ a mechanism to adaptively lower the energy detection threshold from an upper bound. Adaptation mechanism may not preclude static or semi-static setting of the threshold. A Category 4 LBT mechanism or other type of LBT mechanisms may be implemented.

Various example LBT mechanisms may be implemented. For some signals, in some implementation scenarios, in some situations, and/or in some frequencies, no LBT procedure may be performed by the transmitting entity. For example, Category 2 (e.g., LBT without random back-off) may be implemented. The duration of time that the channel is sensed to be idle before the transmitting entity transmits may be deterministic. For example, Category 3 (e.g., LBT with random back-off with a contention window of fixed size) may be implemented. The LBT procedure may have the following procedure as one of its components. The transmitting entity may draw a random number N within a contention window. The size of the contention window may be specified by the minimum and maximum value of N. The size of the contention window may be fixed. The random number N may be employed in the LBT procedure to determine the duration of time that the channel is sensed to be idle, for example, before the transmitting entity transmits

on the channel. For example, Category 4 (e.g., LBT with random back-off with a contention window of variable size) may be implemented. The transmitting entity may draw a random number N within a contention window. The size of contention window may be specified by the minimum and maximum value of N . The transmitting entity may vary the size of the contention window if drawing the random number N . The random number N may be used in the LBT procedure to determine the duration of time that the channel is sensed to be idle, for example, before the transmitting entity transmits on the channel.

LAA may employ uplink LBT at the wireless device. The UL LBT scheme may be different from the DL LBT scheme, for example, by using different LBT mechanisms or parameters. These differences in schemes may be due to the LAA UL being based on scheduled access, which may affect a wireless device's channel contention opportunities. Other considerations motivating a different UL LBT scheme may include, but are not limited to, multiplexing of multiple wireless devices in a single subframe.

LAA may use uplink LBT at the wireless device. The UL LBT scheme may be different from the DL LBT scheme, for example, by using different LBT mechanisms or parameters. These differences in schemes may be due to the LAA UL being based on scheduled access, which may affect a wireless device's channel contention opportunities. Other considerations motivating a different UL LBT scheme may include, but are not limited to, multiplexing of multiple wireless devices in a single subframe.

A DL transmission burst may be a continuous transmission from a DL transmitting node, for example, with no transmission immediately before or after from the same node on the same CC. An UL transmission burst from a wireless device perspective may be a continuous transmission from a wireless device, for example, with no transmission immediately before or after from the same wireless device on the same CC. A UL transmission burst may be defined from a wireless device perspective or from an eNB perspective. If an eNB is operating DL and UL LAA over the same unlicensed carrier, DL transmission burst(s) and UL transmission burst(s) on LAA may be scheduled in a TDM manner over the same unlicensed carrier. An instant in time may be part of a DL transmission burst or part of an UL transmission burst.

A wireless device configured for operation with wireless resources (e.g., bandwidth parts (BWPs)) of a serving cell may be configured by higher layers for the serving cell. The wireless device may be configured for a set of BWPs for receptions by the wireless device (e.g., DL BWP set) and/or or a set of BWPs for transmissions by the wireless device (e.g., UL BWP set). For a DL BWP, an UL BWP in a set of DL BWPs, or an UL BWPs, the wireless device may be configured with at least one of following for the serving cell: a subcarrier spacing (SCS) for DL BWP and/or UL BWP, a cyclic prefix (CP) for DL BWP and/or UL BWP, a number of contiguous PRBs for DL BWP and/or UL BWP, an offset of the first PRB of DL BWP and/or UL BWP in the number of contiguous PRBs relative to the first PRB of a reference location, and/or Q control resource sets (e.g., if the BWP is a DL BWP). Higher layer signaling may configure a wireless device with Q control resource sets, for example, for each serving cell. For a control resource set q , such that $0 \leq q < Q$, the configuration may comprise one or more of following: a first OFDM symbol, a number of consecutive OFDM symbols, a set of resource blocks, a CCE-to-REG mapping, a REG bundle size (e.g., for interleaved CCE-to-REG mapping), and/or antenna port quasi-collocation.

A control resource set may comprise a set of CCEs numbered from 0 to $N_{CCE,q}-1$, where $N_{CCE,q}$ may be the number of CCEs in control resource set q . Sets of PDCCH candidates that a wireless device monitors may be defined in terms of PDCCH wireless device-specific search spaces. A PDCCH wireless device-specific search space at CCE aggregation level $L \in \{1, 2, 4, 8\}$ may be defined by a set of PDCCH candidates for CCE aggregation level L . A wireless device may be configured (e.g., for a DCI format), per serving cell by one or more higher layer parameters, for a number of PDCCH candidates per CCE aggregation level L .

A wireless device may monitor (e.g., in non-DRX mode operation) one or more PDCCH candidate in control resource set q according to a periodicity of $W_{PDCCH,q}$ symbols. The symbols may be configured by one or more higher layer parameters for control resource set q . The carrier indicator field value may correspond to $cif-InSchedulingCell$, for example, if a wireless device is configured with a higher layer parameter (e.g., $cif-InSchedulingCell$). For the serving cell on which a wireless device may monitor one or more PDCCH candidate in a wireless device-specific search space, the wireless device may monitor the one or more PDCCH candidates without carrier indicator field (e.g., if the wireless device is not configured with a carrier indicator field). For the serving cell on which a wireless device may monitor one or more PDCCH candidates in a wireless device-specific search space, the wireless device may monitor the one or more PDCCH candidates with carrier indicator field (e.g., if a wireless device is configured with a carrier indicator field). A wireless device may not monitor one or more PDCCH candidates on a secondary cell, for example, if the wireless device is configured to monitor one or more PDCCH candidates with carrier indicator field corresponding to that secondary cell in another serving cell. For the serving cell on which the wireless device may monitor one or more PDCCH candidates, the wireless device may monitor the one or more PDCCH candidates at least for the same serving cell.

A wireless device may receive PDCCH and/or PDSCH in a DL BWP according to a configured SCS and CP length for the DL BWP. A wireless device may transmit PUCCH and/or PUSCH in an UL BWP according to a configured SCS and CP length for the UL BWP.

A wireless device may be configured, by one or more higher layer parameters, for a DL BWP from a configured DL BWP set for DL receptions. A wireless device may be configured, by one or more higher layer parameters, for an UL BWP from a configured UL BWP set for UL transmissions. A DL BWP index field value may indicate a DL BWP (such as from the configured DL BWP set) for DL receptions, for example, if the DL BWP index field is configured in a DCI format scheduling PDSCH reception to a wireless device. An UL-BWP index field value may indicate the UL BWP (such as from the configured UL BWP set) for UL transmissions, for example, if the UL-BWP index field is configured in a DCI format scheduling PUSCH transmission from a wireless device.

A wireless device may determine that the center frequency for the DL BWP is or should be the same as the center frequency for the UL BWP, such as for TDD. The wireless device may not monitor PDCCH, for example, if the wireless device performs measurements over a bandwidth that is not within the DL BWP for the wireless device.

A wireless device may identify the bandwidth and/or frequency of an initial active DL BWP, such as for an initial active DL BWP. The wireless device may identify the bandwidth and/or frequency after or in response to receiving

the NR-PBCH. A bandwidth of an initial active DL BWP may be confined within the wireless device minimum bandwidth for the given frequency band. The bandwidth may be indicated in PBCH, such as for flexible DL information scheduling. Some bandwidth candidates may be predefined. A number of bits (e.g., x bits) may be used for a bandwidth indication.

A frequency location of an initial active DL BWP may be derived from the bandwidth and SS block (e.g., a center frequency of the initial active DL BWP). The edge of the SS block PRB and data PRB boundary may not be aligned. An SS block may have a frequency offset, for example, if the edge of the SS block PRB and data PRB are not aligned. Predefining the frequency location of an SS block and an initial active DL BWP may reduce the PBCH payload size such that additional bits may not be needed for an indication of a frequency location of an initial active DL BWP. The bandwidth and frequency location may be informed in RMSI, for example, for the paired UL BWP.

A base station may configure a set of BWPs for a wireless device by RRC signaling. The wireless device may transmit or receive in an active BWP from the configured BWPs in a given time instance. Activation and/or a deactivation of DL bandwidth part may be based on a timer for a wireless device. The wireless device may switch its active DL bandwidth part to a default DL bandwidth part, for example, if a timer expires. If the wireless device has not received scheduling DCI for a time period (e.g., X ms, or after expiry of a timer), the wireless device may switch to the default DL BWP.

A new timer (e.g., BWPDeactivationTimer) may be defined to deactivate the original BWP and/or switch to the default BWP. The new timer (e.g., BWPDeactivationTimer) may be started if the original BWP is activated by the activation and/or deactivation DCI. If PDCCH on the original BWP is received, a wireless device may restart the timer (e.g., BWPDeactivationTimer) associated with the original BWP. If the timer (e.g., BWPDeactivationTimer) expires, a wireless device may deactivate the original BWP, switch to the default BWP, stop the timer for the original BWP, and/or flush (or not flush) all HARQ buffers associated with the original BWP.

A base station and a wireless device may have a different understanding of the starting of the timer, for example, if the wireless device misses one or more scheduling grants. The wireless device may be triggered to switch to the default BWP, but the base station may schedule the wireless device in the previous active BWP. The base station may restrict the location of the CORESET of BWP2 to be within BWP1 (e.g., the narrow band BWP1 may be the default BWP), for example, if the default BWP is nested within other BWPs. The wireless device may receive an indication (e.g., CORESET) and switch back to BWP2, for example, if the wireless device previously mistakenly switched to the default BWP.

Restricting the location of the indication (e.g., CORESET) may not solve a miss switching problem, for example, if the default BWP and the other BWPs are not overlapped in frequency domain. The base station may maintain a timer for a wireless device. If the timer expires (e.g., if there is no data scheduled for the wireless device for a time period such as Y ms), and/or if the base station has not received feedback from the wireless device for a time period (such as Y' ms), the wireless device may switch to the default BWP. The wireless device may switch to the default BWP to send a paging signal and/or to re-schedule the wireless device in the default BWP.

A base station may not fix the default BWP to be the same as an initial active BWP. The initial active DL BWP may be the SS block bandwidth which is common to wireless devices in the cell. The traffic load may be very heavy, for example, if many wireless devices fall back to a small bandwidth for data transmission. Configuring the wireless devices with different default BWPs may help to balance the load in the system bandwidth.

There may be no initial active BWP on an SCell, for example, if the initial access is performed on the PCell. An DL BWP and/or UL BWP that is initially activated based on the SCell being activated may be configured or reconfigured by RRC signaling. The default BWP of the SCell may also be configured and/or reconfigured by RRC signaling. The default BWP may be configured or reconfigured by the RRC signaling, and/or the default BWP may be one of the configured BWPs of the wireless device, which may provide a unified design for both PCell and SCell.

The base station may configure a wireless device-specific default DL BWP other than an initial active BWP. The base station may configure the wireless device-specific default DL BWP, for example, after RRC connection, which may be performed for the purpose of load balancing. The default BWP may support connected mode operations other than operations supported by initial active BWP. Other connected mode operations may comprise, for example, fall back and/or connected mode paging. The default BWP may comprise a common search space, such as at least the search space needed for monitoring the pre-emption indications. The default DL and UL BWPs may be independently configured to the wireless device, such as for FDD.

The initial active DL BWP and/or UL BWP may be set as default DL BWP and/or UL BWP, respectively. A wireless device may return to default DL BWP and/or UL BWP. For example, if a wireless device does not receive control for a long time (e.g., based on a timer expiration or a time duration reaching a threshold), the wireless device may fall back to a default BWP (e.g., default DL BWP and/or default UL BWP).

A base station may configure a wireless device with multiple BWPs. The multiple BWPs may share at least one CORESET including a default BWP. CORESET for RMSI may be shared for all configured BWPs. The wireless device may receive control information via the common CORESET, for example, without going back to another BWP or a default BWP. The common CORESET may schedule data within only a default BWP, which may minimize the ambiguity of resource allocation, for example, if a frequency region of a default BWP may belong to all or more than one of the configured BWPs.

A semi-static pattern of BWP switching to default BWP may be performed, for example, if the configured BWP is associated with a different numerology from a default BWP. Switching to a default BWP may be performed, for example, to check RMSI at least periodically. Switching to a default BWP may be necessary particularly if BWPs use different numerologies.

Reconfiguration of a default BWP from an initial BWP may be performed, such as for RRC connected wireless devices. A default BWP may be the same as an initial BWP, such as for RRC IDLE wireless devices. Additionally or alternatively, a wireless device (e.g., RRC IDLE wireless device) may fall back to an initial BWP regardless of a default BWP. If a wireless device performs a measurement based on SS block, reconfiguration of a default BWP outside of an initial BWP may become very inefficient, for example, due to frequent measurement gaps. If a default BWP is

reconfigured to outside of an initial BWP, the following conditions may be satisfied: a wireless device may be in a CONNECTED mode, and/or a wireless device may not be configured with an SS block based measurement for both serving cell and neighbor cells.

A DL BWP other than the initial active DL BWP may be configured as the default DL BWP for a wireless device. Reconfiguring the default DL BWP may be performed based on load balancing and/or different numerologies used for an active DL BWP and an initial active DL BWP. A default BWP on a PCell may be an initial active DL BWP for a transmission of RMSI. The transmission of RMSI may comprise one or more of an RMSI CORESET with a CSS, and/or a wireless device-specific search space (e.g., USS). The initial active BWP and/or default BWP may remain an active BWP for a user after a wireless device becomes RRC connected.

Downlink and uplink BWPs may be independently activated, such as for a paired spectrum. Downlink and uplink bandwidth parts may be jointly activated, such as for an unpaired spectrum. In bandwidth adaptation (e.g., where the bandwidth of the active downlink BWP may be changed), a joint activation of a new downlink BWP and a new uplink BWP may be performed (e.g., for an unpaired spectrum). A new DL/UL BWP pair may be activated such that the bandwidth of the uplink BWPs may be the same (e.g., there may not be a change of an uplink BWP).

There may be an association of DL BWP and UL BWP in RRC configuration. For example, a wireless device may not retune the center frequency of a channel bandwidth (BW) between DL and UL, such as for TDD. If the RF is shared between DL and UL (e.g., in TDD), a wireless device may not retune the RF BW for every alternating DL-to-UL and UL-to-DL switching.

Applying an association between a DL BWP and an UL BWP may enable an activation and/or deactivation command to switch both DL and UL BWPs. Such switching may comprise switching a DL BWP together with switching an UL BWP. If an association is not applied between a DL BWP and an UL BWP, separate BWP switching commands may be necessary.

ADL BWP and an UL BWP may be configured separately for the wireless device. Pairing of the DL BWP and the UL BWP may impose constraints on the configured BWPs (e.g., the paired DL BWP and UL BWP may be activated simultaneously or near simultaneously such as within a threshold time period). A base station may indicate a DL BWP and an UL BWP to a wireless device for activation, for example, in an FDD system. A base station may indicate to a wireless device a DL BWP and an UL BWP with the same center frequency for activation, for example, in a TDD system. No pairing and/or association of the DL BWP and UL BWP may be mandatory, even for TDD system, for example, if the activation and/or deactivation of the BWP for the wireless device is instructed by the base station. Pairing and/or association of the DL BWP and UL BWP may be determined by a base station.

An association between a DL carrier and an UL carrier within a serving cell may be performed by carrier association. A wireless device may not be expected to retune the center frequency of a channel BW between DL and UL, such as for a TDD system. An association between a DL BWP and an UL BWP may be required for a wireless device. An association may be performed by grouping DL BWP configurations with same center frequency as one set of DL BWPs and grouping UL BWP configurations with same center frequency as one set of UL BWPs. The set of DL

BWPs may be associated with the set of UL BWPs sharing the same center frequency. There may be no association between a DL BWP and an UL BWP, for example, if the association between a DL carrier and an UL carrier within a serving cell may be performed by carrier association, such as for an FDD serving cell.

A wireless device may identify a BWP identity from a DCI, which may simplify an indication process. The total number of bits for a BWP identity may depend on the number of bits that may be used within a scheduling DCI (and/or a switching DCI), and/or the wireless device minimum BW. The number of BWPs may be determined based on the wireless device supported minimum BW and/or the network maximum BW. The maximum number of BWPs may be determined based on the network maximum BW and/or the wireless device minimum BW. For example, if 400 MHz is the network maximum BW and 50 MHz is the wireless device minimum BW, 8 BWPs may be configured to the wireless device such that 3 bits may be required within the DCI to indicate the BWP. Such a split of the network BW (e.g., depending on the wireless device minimum BW) may be useful for creating one or more default BWPs from the network side by distributing wireless devices across the entire network BW (e.g., for load balancing purposes).

At least two DL and two UL BWPs may be supported by a wireless device for a BWP adaption. The total number of BWPs supported by a wireless device may be given by 2 number of DL/UL BWP floor (network maximum BW/wireless device minimum DL/UL BW), where floor(x) may be a floor function that returns the greatest integer being less than or equal to x. For example, a maximum number of configured BWPs may be four for DL and UL, respectively, or a maximum number of configured BWPs for UL may be two. Any other number of BWPs, for example, greater than or equal to 2 and less than or equal to a floor, may be supported by a wireless device.

Different sets of BWPs may be configured for different DCI formats and/or scheduling types, respectively. BWPs may be configured for non-slot-based scheduling (e.g., for larger BWPs) or for slot-based scheduling (e.g., for smaller BWPs). If different DCI formats are defined for slot-based scheduling and non-slot-based scheduling, different BWPs may be configured for different DCI formats. Different BWP configurations may provide flexibility between different scheduling types without increasing DCI overhead. A 2-bit field may be used to indicate a BWP among four BWPs for a DCI format. For example, four DL BWPs or two or four UL BWPs may be configured for each DCI format. The same or different BWPs may be configured for different DCI formats.

A required maximum number of configured BWPs (which may exclude the initial BWP) may depend on the flexibility needed for a BWP functionality. It may be sufficient to be able to configure one DL BWP and one UL BWP (or a single DL/UL BWP pair for an unpaired spectrum), which may correspond to minimal support of bandlimited devices. There may be a need to configure at least two DL BWPs and at least a single uplink BWP for a paired spectrum (or two DL/UL BWP pairs for an unpaired spectrum), such as to support bandwidth adaptation. There may be a need to configure one or more DL (or UL) BWPs that jointly cover different parts of the downlink (or uplink) carrier, such as to support dynamic load balancing between different parts of the spectrum. Two BWPs may be sufficient, for example, for dynamic load balancing. In addition to the two BWPs, two other BWPs may be needed, such as for bandwidth adaptation. A maximum number of configured BWPs may be four

DL BWPs and two UL BWPs for a paired spectrum. A maximum number of configured BWPs may be four DL/UL BWP pairs for an unpaired spectrum.

A wireless device may monitor for RMSI and broadcasted OSI, which may be transmitted by a base station within a common search space (CSS) on the PCell. RACH response and paging control monitoring on the PCell may be transmitted within the CSS. A wireless device may not monitor the common search space, for example, if the wireless device is allowed to be on an active BWP configured with a wireless device-specific search space (USSS or USS).

At least one of configured DL bandwidth parts may comprise at least one CORESET with a CSS type, such as for a PCell. To monitor RMSI and broadcast OSI, the wireless device may periodically switch to the BWP containing the CSS. The wireless device may periodically switch to the BWP containing the CSS for RACH response and paging control monitoring on the PCell.

BWP switching to monitor the CSS may result in increasing overhead, for example, if the BWP switching occurs frequently. The overhead due to the CSS monitoring may depend on an overlapping in frequency between any two BWPs. In a nested BWP configuration (e.g., where one BWP may be a subset of another BWP), the same CORESET configuration may be used across the BWPs. A default BWP may comprise the CSS, and another BWP may comprise the CSS, for example, if the default BWP is a subset of another BWP. The BWPs may be partially overlapping. A CSS may be across a first BWP and a second BWP, for example, if the overlapping region is sufficient. Two non-overlapping BWP configurations may exist.

There may be one or more benefits from configuring the same CORESET containing the CSS across BWPs. For example, the RMSI and broadcast OSI monitoring may be performed without necessitating BWP switching, RACH response and paging control monitoring on the PCell may be performed without switching, and/or robustness for BWP switching may improve. A base station and a wireless device may be out-of-sync as to which BWP is currently active and the DL control channel may still work, for example, if CORESET configuration is the same across BWPs. One or more constraints on BWP configuration may be acceptable. A BWP may provide power saving, such that various configurations, including a nested configuration, may be very versatile for different applications.

Group-common search space (GCSS) may be supported (e.g., in NR). The GCSS may be used in addition to or as an alternative to CSS for certain information. A base station may configure GCSS within a BWP for a wireless device. Information such as RACH response and paging control may be transmitted on GCSS. The wireless device may monitor GCSS, for example, instead of switching to the BWP containing the CSS for such information. A base station may transmit information on GCSS, for example, for a pre-emption indication and other group-based commands on a serving cell. A wireless device may monitor the GCSS for the information (e.g., for the SCell which may not have CSS).

A CORESET may be configured without using a BWP. The CORESET may be configured based on a BWP, which may reduce signaling overhead. A first CORESET for a wireless device during an initial access may be configured based on a default BWP. A CORESET for monitoring PDCCH for RAR and paging may be configured based on a DL BWP. The CORESET for monitoring group common (GC)-PDCCH for SFI may be configured based on a DL BWP. The CORESET for monitoring GC-DCI for a pre-

emption indication may be configured based on a DL BWP. A BWP index may be indicated in the CORESET configuration. A default BWP index may not be indicated in the CORESET configuration.

A contention-based random access (CBRA) RACH procedure may be supported via an initial active DL BWP and/or an initial active UL BWP, for example, if the wireless device identity is unknown to the base station. The contention-free random access (CFRA) RACH procedure may be supported via the USS configured in an active DL BWP for the wireless device. An additional CSS for RACH purposes may not need to be configured per BWP, such as for the CFRA RACH procedure supported via the USS configured in an active DL BWP for the wireless device. Idle mode paging may be supported via an initial active DL BWP. Connected mode paging may be supported via a default BWP. No additional configurations for the BWP for paging purposes may be needed for paging. A configured BWP (e.g., on a serving cell) may have the CSS configured for monitoring pre-emption indications for a pre-emption.

A group-common search space may be associated with at least one CORESET configured for the same DL BWP (e.g., for a configured DL BWP). The wireless device may or may not autonomously switch to a default BWP (e.g., where a group-common search space may be available) to monitor for a DCI, for example, depending on the monitoring periodicity of different group-common control information types. A group-common search space may be configured in the same CORESET, for example, if there is at least one CORESET configured on a DL BWP.

A center frequency of an activated DL BWP may or may not be changed. If the center frequency of the activated DL BWP and the deactivated DL BWP is not aligned, the active UL BWP may be switched implicitly (e.g., for TDD).

BWPs with different numerologies may be overlapped. Rate matching for CSI-RS and/or SRS of another BWP in the overlapped region may be performed, which may achieve dynamic resource allocation of different numerologies in a FDM and/or a TDM manner. For a CSI measurement within one BWP, if the CSI-RS and/or SRS collides with data and/or an RS in another BWP, the collision region in another BWP may be rate matched. CSI information over the two or more BWPs may be determined by a base station based on wireless device reporting. Dynamic resource allocation with different numerologies in a FDM manner may be achieved by base station scheduling.

PUCCH resources may be configured in a configured UL BWP, in a default UL BWP, and/or in both a configured UL BWP and a default UL BWP. If the PUCCH resources are configured in the default UL BWP, a wireless device may retune to the default UL BWP for transmitting an SR. The PUCCH resources may be configured per a default BWP or per a BWP other than the default BWP. The wireless device may transmit an SR in the current active BWP without retuning. If a configured SCell is activated for a wireless device, a DL BWP may be associated with an UL BWP at least for the purpose of PUCCH transmission, and/or a default DL BWP may be activated. If the wireless device is configured for UL transmission in the same serving cell, a default UL BWP may be activated.

At least one of configured DL BWPs may comprise one CORESET with common search space (CSS), for example, at least in a primary component carrier. The CSS may be needed at least for RACH response (e.g., a msg2) and/or a pre-emption indication. One or more of configured DL bandwidth parts for a PCell may comprise a CORESET with the CSS type for RMSI and/or OSI, for example, if there is

no periodic gap for RACH response monitoring on the PCell. A configured DL BWP for a PCell may comprise one CORESET with the CSS type for RACH response and paging control for a system information update. A configured DL BWP for a serving cell may comprise a CORESET with the CSS type for a pre-emption indication and/or other group-based commands.

BWPs may be configured with respect to common reference point (e.g., PRB 0) on a component carrier. The BWPs may be configured using TYPE1 RA as a set of contiguous PRBs, with PRB granularity for the START and LENGTH. The minimum length may be determined by the minimum supported size of a CORESET. A CSS may be configured on a non-initial BWP, such as for RAR and paging.

To monitor common channel or group common channel for a connected wireless device (e.g., RRC CONNECTED UE), an initial DL BWP may comprise a control channel for RMSI, OSI, and/or paging. The wireless device may switch a BWP to monitor such a control channel. A configured DL BWP may comprise a control channel (e.g., for a Msg2). A configured DL BWP may comprise a control channel for a SFH. A configured DL BWP may comprise a pre-emption indication and/or other group common indicators such as for power control.

A DCI may explicitly indicate activation and/or deactivation of a BWP. A DCI without data assignment may comprise an indication to activate and/or deactivate BWP. A wireless device may receive a first indication via a first DCI to activate and/or deactivate a BWP. A second DCI with a data assignment may be transmitted by the base station, for example, for a wireless device to start receiving data. The wireless device may receive the first DCI in a target CORESET within a target BWP. A base station scheduler may make conservative scheduling decisions, for example, until the base station receives CSI feedback.

A DCI without scheduling for active BWP switching may be transmitted, for example, to measure the CSI before scheduling. A DCI with scheduling for active BWP switching may comprise setting the resource allocation field to zero, such that no data may be scheduled. Other fields in the DCI may comprise one or more CSI and/or SRS request fields.

Single scheduling a DCI to trigger active BWP switching may provide dynamic BWP adaptation for wireless device power saving during active state. Wireless device power saving during active state may occur for an ON duration, and/or if an inactivity timer is running and/or if C-DRX is configured. A wireless device may consume a significant amount of power monitoring PDCCH, without decoding any grant, for example if a C-DRX is enabled. To reduce the power consumption during PDCCH monitoring, two BWPs may be configured: a narrower BWP for PDCCH monitoring, and a wider BWP for scheduled data. The wireless device may switch back-and-forth between the narrower BWP and the wider BWP, depending on the burstiness of the traffic. The wireless device may revisit a BWP that it has previously used. Combining a BWP switching indication and a scheduling grant may provide an advantage of low latency and/or reduced signaling overhead for BWP switching.

An SCell activation and/or deactivation may or may not trigger a corresponding action for its configured BWP. A dedicated BWP activation and/or deactivation DCI may impact a DCI format. A scheduling DCI with a dummy grant may be used. The dummy grant may be constructed by invalidating one or some of the fields, such as the resource allocation field. A fallback scheduling DCI format (which

may contain a smaller payload) may be used, which may improve the robustness for BWP DCI signaling without incurring extra work by introducing a new DCI format.

A DCI with data assignment may comprise an indication to activate and/or deactivate a BWP along with a data assignment. A wireless device may receive a combined data allocation and BWP activation and/or deactivation message. A DCI format may comprise a field to indicate BWP activation and/or deactivation and/or a field indicating an UL grant and/or a DL grant. The wireless device may start receiving data with a single DCI, such as the DCI format described above. The DCI may indicate one or more target resources of a target BWP. A base station scheduler may have insufficient information about the CSI in the target BW and may make conservative scheduling decisions.

The DCI may be transmitted on a current active BWP, and scheduling information may be for a new BWP, for example, for the DCI with data assignment. There may be a single active BWP. There may be one DCI in a slot for scheduling the current BWP or scheduling another BWP. The same CORESET may be used for the DCI scheduling of the current BWP and the DCI scheduling of another BWP. The DCI payload size for the DCI scheduling current BWP and the scheduling DCI for BWP switching may be the same, which may reduce the number of blind decoding attempts.

A BWP group may be configured by a base station, in which a numerology in one group may be the same, which may support using the scheduling DCI for BWP switching. The BWP switching for the BWP group may be configured, such that BIF may be present in the CORESETs for one or more BWPs in the group. Scheduling DCI for BWP switching may be configured per BWP group, in which an active BWP in the group may be switched to any other BWP in the group.

A DCI comprising a scheduling assignment and/or grant may not comprise an active-BWP indicator. A scheduling DCI may switch a wireless device's active BWP to the transmission direction for which the scheduling is valid (e.g., for a paired spectrum). A scheduling DCI may switch the wireless device's active DL/UL BWP pair regardless of the transmission direction for which the scheduling is valid (e.g., for an unpaired spectrum). A downlink scheduling assignment and/or grant with no assignment may occur, which may allow for a switching of an active BWP without scheduling downlink and/or uplink transmissions.

A timer-based activation and/or deactivation BWP may be supported. A timer for activation and/or deactivation of DL BWP may reduce signaling overhead and may allow wireless device power savings. The activation and/or deactivation of a DL BWP may be based on an inactivity timer, which may be referred to as a BWP inactive (or inactivity) timer. A wireless device may start and/or reset a timer upon reception of a DCI. The timer may expire, for example, if the wireless device is not scheduled for the duration of the timer. The wireless device may activate and/or deactivate the appropriate BWP based on the expiry of the timer. The wireless device may, for example, activate the default BWP and/or deactivate the active BWP.

A BWP inactive timer may be beneficial for power saving for a wireless device. A wireless device may reduce power, for example, by switching to a default BWP with a smaller bandwidth. A wireless device may use a BWP inactive timer, for example, for a fallback if missing a DCI based activation and/or deactivation signaling, such as by switching from one BWP to another BWP. Triggering conditions of the BWP inactive timer may follow triggering conditions for the DRX timer in LTE or any other system. An on-duration of the

35

BWP inactive timer may be configured and/or the timer may start, for example, if a wireless device-specific PDCCH is successfully decoded indicating a new transmission during the on-duration. The timer may restart, for example, if a wireless device-specific PDCCH is successfully decoded indicating a new transmission. The timer may stop, for example, if the wireless device is scheduled to switch to the default DL BWP. The BWP inactive timer may start, for example, if the wireless device switches to a new DL BWP. The timer may restart, for example, if a wireless device-specific PDCCH is successfully decoded, wherein the wireless device-specific PDCCH may be associated with a new transmission, a retransmission, SPS activation and/or deactivation, or another purpose.

A wireless device may switch to a default BWP, for example, if the wireless device does not receive any control and/or data from the network during the running of the BWP inactive timer. The timer may be reset, for example, upon reception of any control and/or data. The timer may be triggered, for example, if wireless device receives a DCI to switch its active DL BWP from the default BWP to another BWP. The timer may be reset, for example, if a wireless device receives a DCI to schedule PDSCH(s) in the BWP other than the default BWP.

A DL BWP inactive timer may be defined separately from a UL BWP inactive timer. Timers for the DL BWP and UL BWP may be set independently and/or jointly. For the separate timers (e.g., if there is DL data and UL timer expires), UL BWP may not be deactivated since PUCCH configuration may be affected if both DL BWP and UL BWP are activated. For the uplink, if there is UL feedback signal related to DL transmission, the timer may be reset. The UL timer may not be set if there is DL data. If there is UL data and the DL timer expires, there may be no issue if the DL BWP is deactivated since UL grant is transmitted in the default DL BWP. A BWP inactivity-timer may allow fallback to default BWP on a PCell and/or SCell.

A timer-based activation and/or deactivation of BWP may be similar to a wireless device DRX timer. There may not be a separate inactivity timer for BWP activation and/or deactivation for the wireless device DRX timer. A wireless device DRX inactivity timer may trigger BWP activation and/or deactivation. There may be separate inactivity timers for BWP activation and/or deactivation for the wireless device DRX timer. For example, the DRX timers may be defined in a MAC layer, and the BWP timer may be defined in a physical layer. A wireless device may stay in a wider BWP for as long as the inactivity timer is running, for example, if the same DRX inactivity timer is used for BWP activation and/or deactivation. The DRX inactivity timer may be set to a large value of 100-200 milliseconds for a C-DRX cycle of 320 milliseconds, which may be larger than the ON duration (e.g., 10 milliseconds). Setting the DRX inactivity timer in the above manner may provide power savings, for example, based on a narrower BWP not being achievable. To realize wireless device power saving promised by BWP switching, a new timer may be defined and it may be configured to be smaller than the DRX inactivity timer. From the point of view of DRX operation, BWP switching may allow wireless device to operate at different power levels during the active state, effectively providing intermediate operating points between the ON and OFF states.

With a DCI explicit activation and/or deactivation of BWP, a wireless device and a base station may not be synchronized with respect to which BWP is activated and/or deactivated. The base station scheduler may not have CSI

36

information related to a target BWP for channel-sensitive scheduling. The base station may be limited to conservative scheduling for one or more first several scheduling occasions. The base station may rely on periodic or aperiodic CSI-RS and associated CQI report(s) to perform channel-sensitive scheduling. Relying on periodic or aperiodic CSI-RS and associated CQI report(s) may delay channel-sensitive scheduling and/or lead to signaling overhead, such as if aperiodic CQI is requested (e.g., by a base station). To mitigate a delay in acquiring synchronization and channel state information, a wireless device may transmit an acknowledgement upon receiving an activation and/or deactivation of a BWP. A CSI report based on the provided CSI-RS resource may be transmitted after activation of a BWP and may be used as acknowledgment of activation and/or deactivation.

A base station may provide a sounding reference signal for a target BWP after a wireless device tunes to a new BWP. The wireless device may report the CSI, which may be used as an acknowledgement by the base station to confirm that the wireless device receives an explicit DCI command and activates and/or deactivates the appropriate BWPs. For an explicit activation and/or deactivation via DCI with data assignment, a first data assignment may be carried out without a CSI for the target BWP.

A guard period may be defined to take RF retuning and related operations into account. A wireless device may neither transmit nor receive signals in the guard period. A base station may need to know the length of the guard period. For example, the length of the guard period may be reported to the base station as a wireless device capability. The length of the guard period may be based on the numerologies of the BWPs and the length of the slot. The length of the guard period for RF retuning may be reported as a wireless device capability. The wireless device may report the guard period as an absolute time and/or in symbols.

The base station may maintain the time domain position of guard period in alignment between the base station and the wireless device, for example, if the base station knows the length of the guard period. The guard period for RF retuning may be predefined for time pattern triggered BWP switching. The BWP switching and/or guard period may be triggered by DCI and/or a timer. For BWP switching following a time pattern, the position of the guard period may be defined. The guard period may not affect the symbols carrying CSS, for example, if the wireless device is configured to switch periodically to a default BWP for CSS monitoring.

A single DCI may switch the wireless device's active BWP from one to another within a given serving cell. The active BWP may be switched to a second BWP of the same link direction, for example an UL BWP or a DL BWP. A separate field may be used in the scheduling DCI to indicate the index of the BWP for activation such that wireless device may determine the current DL/UL BWP according to a detected DL/UL grant without requiring any other control information. The multiple scheduling DCIs transmitted in this duration may comprise the indication to the same BWP, for example, if the BWP change does not happen during a certain time duration. During the transit time wherein potential ambiguity may happen, base station may send scheduling grants in the current BWP or together in the other BWPs containing the same target BWP index, such that wireless device may obtain the target BWP index by detecting the scheduling DCI in either one of the BWPs. The duplicated scheduling DCI may be transmitted an arbitrary number

(e.g., K) times. A wireless device may switch to the target BWP and start to receive or transmit (UL) in the target BWP according to the BWP indication field, for example, if the wireless device receives one of the K times transmissions.

Switching between BWPs may introduce time gaps, for example, if wireless device is unable to receive one or more messages due to re-tuning. Breaks of several time slots may severely affect the TCP ramp up as the wireless device may not be able to transmit and receive during those slots, affecting obtained RTT and data rate. A break in reception may make wireless device out of reach from network point of view reducing network interest to utilize short inactivity timer. If BWP switching takes significant time and a wireless device requires new reference symbols to update AGC, channel estimation, etc., active BWP switching may not be adopted in the wireless device. In some configurations, BWP switching may be performed where the BWP center frequency remains the same if switching between BWPs.

A frequency location of a wireless device's RF bandwidth may be indicated by base station. The RF bandwidth of the wireless device may be smaller than the carrier bandwidth for considering the wireless device RF bandwidth capability. The supported RF bandwidth for a wireless device is usually a set of discrete values (e.g., 10 MHz, 20 MHz, 50 MHz, etc.). For energy saving purpose, the wireless device RF bandwidth may be determined as the minimum available bandwidth supporting the bandwidth of the BWP. The granularity of BWP bandwidth may be PRB level, which may be decoupled with wireless device RF bandwidth. As a result, the wireless device RF bandwidth may be larger than the BWP bandwidth. The wireless device may receive signals outside the carrier bandwidth, especially if the BWP is configured near the edge of the carrier bandwidth. Inter-system interference or the interference from an adjacent cell outside the carrier bandwidth may affect the receiving performance of the BWP. To keep the wireless device RF bandwidth in the carrier bandwidth, the frequency location of the wireless device RF bandwidth may be indicated by the base station.

A gap duration may be determined based on a measurement duration and a retuning gap. The retuning gap may vary. If a wireless device does not need to switch its center, the retuning may be relatively short, such as 20 μ s. A wireless device may indicate the necessary retuning gap for a measurement configuration, for example, if the network may not know whether the wireless device needs to switch its center or not to perform measurement. The retuning gap may depend on the current active BWP that may be dynamically switched via a switching mechanism. Wireless devices may need to indicate the retuning gap dynamically.

The measurement gap may be indirectly created, for example, if the network may configure a certain measurement gap. The measurement gap may comprise the smallest retuning latency. The smallest returning latency may be determined, for example, if a small retuning gap may be utilized and/or if both measurement bandwidth and active BWP is included within the wireless device maximum RF capability and the center frequency of the current active BWP may be not changed. The wireless device may skip receiving and/or transmitting, for example, if a wireless device needs more gap than the configured.

A different measurement gap and retuning gap may be utilized for RRM and CSI. For CSI measurement, if periodic CSI measurement outside of active BWP may be configured, a wireless device may need to perform its measurement periodically per measurement configuration. For RRM, it may be up to wireless device implementation where to

perform the measurement as long as it satisfies the measurement requirements. The worst case retuning latency for a measurement may be used. As the retuning latency may be different between intra-band and inter-band retuning, separate measurement gap configurations between intra-band and inter-band measurement may be considered.

A respective DCI format may comprise an explicit identifier to distinguish them, for example, for multiple DCI formats with the same DCI size of a same RNTI. The same DCI size may come from zero-padding bits in at least a wireless device-specific search space.

In BWP switching, a DCI in the current BWP may need to indicate resource allocation in the next BWP that the wireless device may be expected to switch. The resource allocation may be based on the wireless device-specific PRB indexing, which may be per BWP. A range of the PRB indices may change as the BWP changes. The DCI to be transmitted in the current BWP may be based on the PRB indexing for the current BWP. The DCI may need to indicate the RA in the new BWP, which may cause a resource conflict. To resolve the conflict without significantly increasing wireless devices blind detection overhead, the DCI size and bit fields may not change per BWP for a given DCI type.

As the range of the PRB indices may change as the BWP changes, one or more employed bits among the total bit field for RA may be dependent on the employed BWP. A wireless device may use the indicated BWP ID that the resource allocation may be intended to identify the resource allocation bit field.

The DCI size of the BWP may be based on a normal DCI detection without BWP retuning and/or on a DCI detection during the BWP retuning. A DCI format may be independent of the BW of the active DL/UL BWP, which may be called as fallback DCI. At least one of DCI format for DL may be configured to have the same size for a wireless device for one or more configured DL BWPs of a serving cell. At least one of the DCI formats for UL may be configured to have the same size for a wireless device for one or more configured UL BWPs of a serving cell. A BWP-dependent DCI format may be monitored at the same time (e.g. a normal DCI) for both active DL BWP and active UL BWP. A wireless device may monitor both DCI formats at the same time. A base station may assign the fallback DCI format to avoid ambiguity during a transition period in the BWP activation and/or deactivation.

If a wireless device is configured with multiple DL or UL BWPs in a serving cell, an inactive DL and/or UL BWP may be activated by a DCI scheduling a DL assignment or UL grant in the BWP. As the wireless device may be monitoring the PDCCH on the currently active DL BWP, the DCI may comprise an indication to a target BWP that the wireless device may switch to for PDSCH reception or UL transmission. A BWP indication may be inserted in the wireless device-specific DCI format. The bit width of this field may depend on the maximum possible and/or presently configured number of DL and/or UL BWPs. The BWP indication field may be a fixed size based on the maximum number of configured BWPs.

A DCI format size may match the BW of the BWP in which the PDCCH may be received. To avoid an increase in the number of blind decodes, the wireless device may identify the RA field based on the scheduled BWP. For a transition from a small BWP to a larger BWP, the wireless device may identify the RA field as being the LSBs of the required RA field for scheduling the larger BWP.

The same DCI size for scheduling different BWPs may be defined by keeping the same size of resource allocation fields

for one or more configured BWPs. A base station may be aware of a wireless device switching BWPs based on a reception of ACK/NACK from the wireless device. The base station may not be aware of a wireless device switching BWPs, for example, if the base station does not receive at least one response from the wireless device. To avoid such a mismatch between base station and wireless device, a fallback mechanism may be used. The base station may transmit the scheduling DCI for previous BWPs and for newly activated BWP since the wireless device may receive the DCI on either BWP, for example, if there is no response from the wireless device. The base station may confirm the completion of the active BWP switching, for example, after or in response to the base station receiving a response from the wireless device. The base station may not transmit multiple DCIs, for example, if the same DCI size for scheduling different BWPs may be considered and CORESET configuration may be the same for different BWPs. DCI format(s) may be configured user-specifically per cell rather than per BWP. The wireless device may start to monitor pre-configured search-space on the CORESET, for example, if a wireless device synchronizes to a new BWP.

The size of DCI format in different BWPs may vary and may change at least due to different size of RA bitmap on different BWPs. The size of DCI format configured in a cell for a wireless device may be dependent on scheduled BWPs. If the DCI formats may be configured per cell, the corresponding header size in DCI may be relatively small.

The monitored DCI format size on a search-space of a CORESET may be configurable with sufficiently fine granularity and/or the granularity may be predefined. The monitored DCI format size with sufficient granularity may be beneficial, for example, if a base station may freely set the monitoring DCI format size on the search-spaces of a CORESET. The DCI format size may be set such that it may accommodate the largest actual DCI format size variant among one or more BWPs configured in a serving cell.

For a wireless device-specific serving cell, one or more DL BWPs and one or more UL BWPs may be configured by a dedicated RRC for a wireless device. This may be done as part of the RRC connection establishment procedure for a PCell. For an SCell, this may be done via RRC configuration indicating the SCell parameters.

A default DL and/or a default UL BWP may be activated since there may be at least one DL and/or UL BWP that may be monitored by the wireless device depending on the properties of the SCell (DL only, UL only, or both), for example, if a wireless device receives an SCell activation command. The BWP may be activated upon receiving an SCell activation command. The BWP may be informed to the wireless device via the RRC configuration that configured the BWP on this serving cell. For an SCell, RRC signaling for SCell configuration/reconfiguration may be used to indicate which DL BWP and/or UL BWP may be activated if the SCell activation command is received by the wireless device. The indicated BWP may be the initially active DL and/or UL BWP on the SCell. The SCell activation command may activate DL and/or UL BWP.

For an SCell, RRC signaling for the SCell configuration/reconfiguration may be used for indicating a default DL BWP on the SCell. The default DL BWP may be used for fallback purposes. The default DL BWP may be same or different from the initially activated DL and/or UL BWP indicated to wireless device as part of the SCell configuration. A default UL BWP may be configured to a wireless device for transmitting PUCCH for SR, for example, if the PUCCH resources are not configured in every BWP for SR.

An SCell may be for DL only. For a DL only SCell, a wireless device may keep monitoring an initial DL BWP (e.g., initial active or default) until the wireless device receives SCell deactivation command. An SCell may be for UL only. For the UL only SCell, the wireless device may transmit on the indicated UL BWP, for example, if a wireless device receives a grant. The wireless device may not maintain an active UL BWP if wireless device does not receive a grant. A failure to maintain the active UL BWP due to a grant not being received may not deactivate the SCell. An SCell may be for UL and DL. For a UL and DL SCell, a wireless device may keep monitoring an initial DL BWP (e.g., initial active or default) until the wireless device receives an SCell deactivation command. The UL BWP may be used if there may be a relevant grant or an SR transmission.

A BWP deactivation may not result in a SCell deactivation. The active DL and/or UL BWPs may be considered deactivated, for example, if the wireless device receives the SCell deactivation command.

A wireless device may be expected to perform RACH procedure on an SCell during activation. Activation of UL BWP may be needed for the RACH procedure. At an SCell activation, DL only (only active DL BWP) and/or DL/UL (both DL/UL active BWP) may be configured. A wireless device may select default UL BWP based on measurement or the network configures which one in its activation.

One or more BWPs may be semi-statically configured via wireless device-specific RRC signaling. If a wireless device maintains RRC connection with a primary component carrier (CC), the BWP in a secondary CC may be configured via RRC signaling in the primary CC. One or more BWPs may be semi-statically configured to a wireless device via RRC signaling in a PCell. A DCI transmitted in an SCell may indicate a BWP among the one or more configured BWP and grant detailed resource based on the indicated BWP. For cross-CC scheduling, a DCI transmitted in a PCell may indicate a BWP among the one or more configured BWPs, and grants detailed resource based on the indicated BWP.

A DL BWP may be initially activated for configuring CORESET for monitoring the first PDCCH in the SCell, for example, if an SCell may be activated. The DL BWP may serve as a default DL BWP in the SCell. For the wireless device performing initial access via a SS block in PCell, the default DL BWP in an SCell may not be derived from SS block for initial access. The default DL BWP in an SCell may be configured by RRC signaling in the PCell.

An indication indicating which DL BWP and/or which UL BWP are active may be in the RRC signaling for SCell configuration and/or reconfiguration, for example, if an SCell is activated. The RRC signaling for SCell configuration/reconfiguration may be used for indicating which DL BWP and/or which UL BWP are initially activated if the SCell may be activated. An indication indicating which DL BWP and/or which UL BWP are active may be in the SCell activation signaling, for example, if an SCell is activated. SCell activation signaling may be used for indicating which DL BWP and/or which UL BWP are initially activated if the SCell may be activated.

For PCells and SCells, initial default BWPs for DL and UL (e.g., for RMSI reception and PRACH transmission) may be valid until at least one BWP is configured for the DL and UL via RRC wireless device-specific signaling respectively. The initial default DL/UL bandwidth parts may become invalid and new default DL/UL bandwidth parts may take effect. The SCell configuration may comprise default DL/UL bandwidth parts.

An initial BWP on a PCell may be defined by a master information block (MIB). An initial BWP and default BWP may be separately configurable for the SCell. An initial BWP may be the widest configured BWP of the SCell. A wireless device may retune to a default BWP that may be the narrow BWP. The SCell may be active and may be ready to be opened if an additional data burst arrives.

A BWP on SCell may be activated by means of cross-cell scheduling DCI. The cross-cell scheduling may be configured for a wireless device. The base station may activate a BWP on the SCell by indicating CIF and BWPI in the scheduling DCI.

A wireless device and/or base station may perform synchronization tracking within an active DL BWP without a SS block. A tracking reference signal (TRS) and/or the DL BWP configuration may be configured. A DL BWP with a SS block or TRS may be configured as a reference for synchronization tracking.

SS-block based RRM measurements may be decoupled within the BWP framework. Measurement configurations for each RRM and CSI feedback may be independently configured from the BWP configurations. CSI and SRS measurements/transmissions may be performed within the BWP framework.

For a modulation coding scheme (MCS) assignment of the first one or more DL data packets after active DL BWP switching, the network may assign robust MCS to a wireless device for the first one or more DL data packets based on RRM measurement reporting. For a MCS assignment of the first one or more DL data packets after active DL BWP switching, the network may signal to a wireless device by active DL BWP switching DCI to trigger aperiodic CSI measurement/reporting to speed up link adaptation convergence. For a wireless device, periodic CSI measurement outside the active BWP in a serving cell may not be supported. For a wireless device, RRM measurement outside active BWP in a serving cell may be supported. For a wireless device, RRM measurement outside configured BWPs in a serving cell may be supported. RRM measurements may be performed on a SSB and/or CSI-RS. The RRM/RLM measurements may be independent of BWPs.

A wireless device may not be configured with aperiodic CSI reports for non-active DL BWPs. The CSI measurement may be obtained after the BW opening and the wide-band CQI of the previous BWP may be used as starting point for the other BWP on the component carrier.

A wireless device may perform CSI measurements for the BWP before scheduling. Before scheduling on a new BWP, a base station may intend to find the channel quality on the potential new BWPs before scheduling the user on that BWP. The wireless device may switch to a different BWP and measure channel quality for the BWP and then transmit the CSI report. There may be no scheduling needed.

One or more scheduling request (SR) configurations may be configured for a BWP of a cell for a wireless device. A wireless device may use SR resources configured by the SR configurations in a BWP to indicate to the base station the numerology/TTI/service type of a logical channel (LCH) or logical channel group (LCG) that triggered the SR. The maximum number of SR configurations may be the maximum number of logical channels/logical channel groups.

There may be at most one active DL BWP and at most one active UL BWP at a given time for a serving cell. A BWP of a cell may be configured with a specific numerology and/or TTI. For a logical channel and/or logical channel group that triggers a SR transmission while the wireless device oper-

ates in one active BWP, the corresponding SR may remain triggered based on BWP switching.

The logical channel and/or logical channel group to SR configuration mapping may be configured and/or reconfigured based on switching of the active BWP. The RRC dedicated signaling may configure and/or reconfigure the logical channel and/or logical channel group to SR configuration mapping on the new active BWP if the active BWP is switched.

A mapping between a logical channel and/or logical channel group and SR configuration may be configured if a BWP is configured. The RRC may pre-configure mapping between logical channels and/or logical channel groups to SR configurations for the configured BWPs. Based on switching of the active BWP, a wireless device may use the RRC configured mapping relationship for the new BWP. A RRC may configure the mapping between logical channel and SR configurations for the BWP. The sr-ProhibitTimer and SR_COUNTER corresponding to a SR configuration may continue and the value of the sr-ProhibitTimer and the value of the SR_COUNTER may be their values before the BWP switching.

A plurality of logical channel/logical channel group to SR configuration mappings may be configured in a serving cell. A logical channel/logical channel group may be mapped to at most one SR configuration per BWP. A logical channel/logical channel group mapped onto multiple SR configurations in a serving cell may have one SR configuration active at a time, such as that of the active BWP. A plurality of logical channel/logical channel group to SR-configuration mappings may be supported in carrier aggregation (CA). A logical channel/logical channel group may be mapped to one (or more) SR configuration(s) in each of PCell and PUCCH-SCell. A logical channel/logical channel group configured to be mapped to one (or more) SR configuration(s) in each of both PCell and PUCCH-SCell may have two active SR configurations (one on PCell and one on PUCCH-SCell) at a time for CA. The SR resource is received first may be used.

A base station may configure one SR resource per BWP for the same logical channel/logical channel group. If a SR for one logical channel/logical channel group is pending, a wireless device may transmit a SR with the SR configuration in another BWP after BWP switching. The sr-ProhibitTimer and SR_COUNTER for the SR corresponding to the logical channel/logical channel group may continue based on BWP switching. The wireless device may transmit the SR in another SR configuration corresponding to the logical channel/logical channel group in another BWP after BWP switching if a SR for one logical channel/logical channel group may be pending.

If multiple SRs for logical channels/logical channel groups mapped to different SR configurations are triggered, the wireless device may transmit one SR corresponding to the highest priority logical channel/logical channel group. The wireless device may transmit multiple SRs with different SR configurations. SRs triggered at the same time (e.g., in the same NR-UNIT) by different logical channels/logical channel groups mapped to different SR configurations may be merged into a single SR corresponding to the SR triggered by the highest priority logical channel/logical channel group.

If an SR of a first SR configuration is triggered by a first logical channel/logical channel group while an SR procedure triggered by a lower priority logical channel/logical channel group may be on-going on another SR configuration, the later SR may be allowed to trigger another SR procedure on its own SR configuration independently of the

other SR procedure. A wireless device may be allowed to send independently triggered SRs for logical channels/logical channel groups mapped to different SR configurations. A wireless device may be allowed to send triggered SRs for LCHs corresponding to different SR configurations independently.

The dsr-TransMax may be independently configured per SR configuration. The SR_COUNTER may be maintained for each SR configuration independently. A common SR_COUNTER may be maintained for all the SR configurations per BWP.

PUCCH resources may be configured per BWP. The PUCCH resources in the currently active BWP may be used for UCI transmission. PUCCH resources may be configured per BWP. PUCCH resources may be utilized in a BWP not currently active for UCI transmission. PUCCH resources may be configured in a default BWP and BWP switching may be necessary for PUCCH transmission. A wireless device may be allowed to send SR1 in BWP1 even though BWP1 may be no longer active. The network may reconfigure (e.g., pre-configure) the SR resources so that both SR1 and SR2 may be supported in the active BWP. An anchor BWP may be used for SR configuration. In an example, the wireless device may send SR2 as a fallback.

A logical channel/logical channel group mapped to a SR configuration in an active BWP may also be mapped to the SR configuration in another BWP to imply same or different information, such as numerology and/or TTI and priority. A MAC entity can be configured with a plurality of SR configurations within the same BWP. The plurality of the SR configurations may be on the same BWP, on different BWPs, or on different carriers. The numerology of the SR transmission may differ from the numerology that the logical channel/logical channel group that triggered the SR may be mapped to.

The PUCCH resources for transmission of the SR may be on different BWPs or different carriers for a LCH mapped to multiple SR configurations. The selection of which configured SR configuration within the active BWP to transmit one SR may be up to wireless device implementation if multiple SRs are triggered. A single BWP can support multiple SR configurations. Multiple sr-ProhibitTimers (e.g., each for one SR configuration) may be running at the same time. A dsr-TransMax may be independently configured per SR configuration. A SR_COUNTER may be maintained for each SR configuration independently. A single logical channel/logical channel group may be mapped to zero or one SR configurations. A PUCCH resource configuration may be associated with a UL BWP. One or more logical channels may be mapped to none or one SR configuration per BWP in CA.

A BWP may consist of a group of contiguous PRBs in the frequency domain. The parameters for each BWP configuration may include numerology, frequency location, bandwidth size (e.g., in terms of PRBs), CORESET. CORESET may be required for each BWP configuration, such as for a single active DL bandwidth part for a given time instant. One or more BWPs may be configured for each component carrier, for example, if the wireless device is in RRC connected mode.

The configured downlink assignment may be initialized (e.g., if not active) or re-initialized (e.g., if already active) using PDCCH if a new BWP may be activated. For uplink SPS, the wireless device may have to initialize and/or re-initialize the configured uplink grant if switching from one BWP to another BWP. If a new BWP is activated, the

configured uplink grant may be initialized (e.g., if not already active) or re-initialized (e.g., if already active) using PDCCH.

For type 1 uplink data transmission without grant, there may be no L1 signaling to initialize or re-initialize the configured grant. The wireless device may not determine that the type 1 configured uplink grant may be active if the BWP may be switched, for example, even if the wireless device is already active in the previous BWP. The type 1 configured uplink grant may be re-configured using RRC dedicated signaling for switching the BWP. The type 1 configured uplink grant may be re-configured using dedicated RRC signaling if a new BWP is activated.

If SPS is configured on the resources of a BWP and the BWP is subsequently deactivated, the SPS grants or assignments may not continue. All configured downlink assignments and configured uplink grants using resources of this BWP may be cleared, for example, if a BWP is deactivated. The MAC entity may clear the configured downlink assignment or/and uplink grants upon receiving activation and/or deactivation of BWP.

The units of drx-RetransmissionTimer and drx-ULRetransmissionTimer may be OFDM symbol corresponding to the numerology of the active BWP. If a wireless device is monitoring an active DL BWP for a long time without activity, the wireless device may move to a default BWP in order to save power. A BWP inactivity timer may be introduced to switch from an active BWP to the default BWP. Autonomous switching to a DL default BWP may consider both DL BWP inactivity timers and/or DRX timers, such as HARQ RTT and DRX retransmission timers. A DL BWP inactivity timer may be configured per MAC entity. A wireless device may be configured to monitor PDCCH in a default BWP, for example, if a wireless device uses a long DRX cycle.

A power headroom report (PHR) may not be triggered due to the switching of BWP. The support of multiple numerologies/BWPs may not impact PHR triggers. A PHR may be triggered upon BWP activation. A prohibit timer may start upon PHR triggering due to BWP switching. A PHR may not be triggered due to BWP switching while the prohibit timer may be running. A PHR may be reported per activated and/or deactivated BWP.

Packet Data Convergence Protocol (PDCP) duplication may be in an activated state while the wireless device receives the BWP deactivation command. The PDCP duplication may not be deactivated, for example, if the BWP on which the PDCP duplication is operated on is deactivated. The PDCP entity may stop sending the data to the deactivated RLC buffer, for example, even if the PDCP duplication may not be deactivated.

RRC signaling may configure a BWP to be activated, for example, if the SCell is activated. Activation and/or deactivation MAC CE may be used to activate both the SCell and the configured BWP. A HARQ entity can serve different BWP within one carrier.

For a wireless device-specific serving cell, one or more DL BWPs and one or more UL BWPs may be configured by dedicated RRC for a wireless device. A single scheduling DCI may switch the wireless device's active BWP from one to another. An active DL BWP may be deactivated by means of timer for a wireless device to switch its active DL bandwidth part to a default DL bandwidth part. A narrower BWP may be used for DL control monitoring and a wider BWP may be used for scheduled data. Small data may be allowed in the narrower BWP without triggering BWP switching.

A base station may transmit a plurality of beams to a wireless device. A serving beam may be determined, from the plurality of beams, for the wireless communications between the base station and the wireless device. One or more candidate beams may also be determined, from the plurality of beams, for providing the wireless communications if a beam failure event occurs, for example, such that the serving beam becomes unable to provide the desired communications. One or more candidate beams may be determined by a wireless device and/or by a base station. By determining and configuring a candidate beam, the wireless device and base station may continue wireless communications if the serving beam experiences a beam failure event.

Single beam and multi-beam operations may be supported, for example, in a NR (New Radio) system. In a multi-beam example, a base station (e.g., gNB) may perform a downlink beam sweep to provide coverage for downlink Synchronization Signals (SSs) and common control channels. Wireless devices may perform uplink beam sweeps to access a cell. For a single beam, a base station may configure time-repetition transmission within one SS block. This time-repetition may comprise, for example, one or more of a primary synchronization signal (PSS), a secondary synchronization signal (SSS), or a physical broadcast channel (PBCH). These signals may be in a wide beam. In a multi-beam example, a base station may configure one or more of these signals and physical channels, such as in an SS block, in multiple beams. A wireless device may identify, for example, from an SS block, an OFDM symbol index, a slot index in a radio frame, and a radio frame number from an SS block.

In an RRC_INACTIVE state or in an RRC_IDLE state, a wireless device may assume that SS blocks form an SS burst and an SS burst set. An SS burst set may have a given periodicity. SS blocks may be transmitted together in multiple beams (e.g., in multiple beam examples) to form an SS burst. One or more SS blocks may be transmitted via one beam. If multiple SS bursts are transmitted with multiple beams, these SS bursts together may form an SS burst set, such as shown in FIG. 15. A base station 1501 (e.g., a gNB in NR) may transmit SS bursts 1502A to 1502H during time periods 1503. A plurality of these SS bursts may comprise an SS burst set, such as an SS burst set 1504 (e.g., SS bursts 1502A and 1502E). An SS burst set may comprise any number of a plurality of SS bursts 1502A to 1502H. Each SS burst within an SS burst set may be transmitted at a fixed or variable periodicity during time periods 1503.

A wireless device may detect one or more of PSS, SSS, or PBCH signals for cell selection, cell reselection, and/or initial access procedures. The PBCH or a physical downlink shared channel (PDSCH) scheduling system information may be broadcasted by a base station to multiple wireless devices. The PDSCH may be indicated by a physical downlink control channel (PDCCH) in a common search space. The system information may comprise system information block type 2 (SIB2). SIB2 may carry one or more physical random access channel (PRACH) configurations. A base station (e.g., a gNB in NR) may have one or more RACH configurations which may include a PRACH preamble pool, time and/or frequency radio resources, and other power related parameters. A wireless device may select a PRACH preamble from a RACH configuration to initiate a contention-based RACH procedure or a contention-free RACH procedure. A wireless device may perform a 4-step RACH procedure, which may be a contention-based RACH procedure or a contention-free RACH procedure.

A FIG. 16 shows examples of (a) a contention-based four-step RA procedure, (b) a contention free three-step RA procedure, (c) descriptions of a contention-based four-step RA procedure, and (d) a contention free two-step RA procedure. A four-step RA procedure may comprise a RAP transmission in a first step, an RAR transmission in a second step, a scheduled transmission of one or more transport blocks (TBs) in a third step, and contention resolution in a fourth step.

In step 1601, a base station may transmit four-step RA configuration parameters to a wireless device (e.g., a UE). The base station may generate and transmit RA configuration parameters periodically, e.g., based on a timer. The base station may broadcast RA configuration parameters in one or more messages. The wireless device may perform a RAP selection process at step 1602, e.g., after receiving the four-step RA configuration parameters. In a contention-based RA procedure, such as shown in part (a) of FIG. 16, the RA configuration parameters may comprise a root sequence that may be used by the wireless device to generate a RAP. The RAP may be randomly selected by the wireless device, among various RAP candidates generated by the root sequence, during the RAP selection process. The wireless device may perform the RAP selection using one or more RAP selections procedures, such as described herein.

In a first step of the RA procedure, at step 1603, a wireless device may transmit a RAP, e.g., using a configured RA preamble format with a transmission (Tx) beam. A random access channel (RACH) resource may be defined as a time-frequency resource to transmit a RAP. Broadcast system information may indicate whether wireless device should transmit one preamble, or multiple or repeated preambles, within a subset of RACH resources.

A base station may configure an association between a downlink (DL) signal and/or channel, and a subset of RACH resources and/or a subset of RAP indices, for determining the DL transmission in the second step. Based on the DL measurement and the corresponding association, a wireless device may select the subset of RACH resources and/or the subset of RAP indices. Two RAP groups may be informed by broadcast system information and one may be optional. If a base station configures the two groups in the four-step RA procedure, a wireless device may use determine which group from which the wireless device selects an RAP, for example, based on the pathloss and/or a size of the message to be transmitted by the wireless device in the third step. A base station may use a group type to which a RAP belongs as an indication of the message size in the third step and the radio conditions at a wireless device UE. A base station may broadcast the RAP grouping information along with one or more thresholds on system information.

In the second step of the four-step RA procedure, at step 1604, a base station may transmit a random access response (RAR) to the wireless device. The base station may transmit the RAR in response to an RAP that the wireless device may transmit. A wireless device may monitor the PDCCH carrying a DCI, to detect RARs transmitted on a PDSCH in an RA response window. The DCI may be CRC-scrambled by the RA-RNTI (Random Access-Radio Network Temporary Identifier). The RA-RNTI may be used on the PDCCH if Random Access Response messages are transmitted. The RA-RNTI may unambiguously identify which time-frequency resource is used by the MAC entity to transmit the Random Access preamble. The RA response window may start at a subframe that contains the end of an RAP transmission, plus three subframes. The RA response window may have the length indicated by ra-ResponseWindowSize.

A wireless device may determine the RA-RNTI associated with the PRACH in which the wireless device transmits an RAP by the following operation:

$$RA-RNTI = 1 + t_id + 10 * f_id$$

where t_id is the index of the first subframe of the specified PRACH ($0 \leq t_id < 10$), and f_id is the index of the specified PRACH within that subframe, in ascending order of frequency domain ($0 \leq f_id < 6$). Different types of wireless devices, e.g., narrow band-Internet of Things (NB-IoT), bandwidth limited (BL)-UE, and/or UE-Extended Coverage (UE-EC), may use different formulas or operations for determining RA-RNTI. A base station may configure an association between a DL signal or channel, a subset of RACH resources, and/or a subset of RAP indexes. Such an association may be for determining the DL transmission in the second step of the RA procedure, at step 1604 of FIG. 16. Based on the DL measurement and the corresponding association, a wireless device may select the subset of RACH resources and/or the subset of RAP indices. FIG. 18 shows contents of a MAC RAR. For example, FIG. 18A shows the contents of a MAC RAR of a wireless device, FIG. 18B shows the contents of a MAC RAR of a MTC wireless device, and FIG. 18C shows the contents of MAC RAR of a NB-IoT wireless device.

In the third step of the four-step RA procedure (e.g., step 1605 in FIG. 16), a wireless device may adjust an UL time alignment by using the TA value corresponding to the TA command in the received RAR in the second step (e.g., step 1604 in FIG. 16). A wireless device may transmit one or more TBs to a base station using the UL resources assigned in the UL grant in the received RAR. One or more TBs that a wireless device may transmit in the third step (e.g., step 1605 in FIG. 16) may comprise RRC signaling, such as an RRC connection request, an RRC connection Re-establishment request, or an RRC connection resume request. The one or more TBs may also comprise a wireless device identity, e.g., which may be used as part of the contention-resolution mechanism in the fourth step (e.g., step 1606 in FIG. 16).

The fourth step in the four-step RA procedure (e.g., step 1606 in FIG. 16) may comprise a DL message for contention resolution. Based on the second step (e.g., step 1604 in FIG. 16), one or more wireless devices may perform simultaneous RA attempts selecting the same RAP in the first step (e.g., step 1603 in FIG. 16), and/or receive the same RAR with the same TC-RNTI in the second step (e.g., step 1604 in FIG. 16). The contention resolution in the fourth step may be to ensure that a wireless device does not incorrectly use another wireless device identity. The contention resolution mechanism may be based on either a C-RNTI on a PDCCH, or a wireless device Contention Resolution Identity on a DL-SCH, e.g., depending on whether or not a wireless device has a C-RNTI. If a wireless device has a C-RNTI, e.g., if the wireless device detects the C-RNTI on the PDCCH, the wireless device may determine the success of RA procedure. If the wireless device does not have a C-RNTI (e.g., if a C-RNTI is not pre-assigned), the wireless device may monitor a DL-SCH associated with a TC-RNTI, e.g., that a base station may transmit in an RAR of the second step. In the fourth step (e.g., step 1606 in FIG. 16), the wireless device may compare the identity in the data transmitted by the base station on the DL-SCH with the identity that the wireless device transmits in the third step (e.g., step 1605 in FIG. 16). If the wireless determines that two identities are the same or satisfy a threshold similarity, the wireless device may determine that the RA procedure is

successful. If the wireless device determines that the RA is successful, the wireless device may promote the TC-RNTI to the C-RNTI. A TC-RNTI may be an identifier initially assigned to a wireless device when the wireless device first attempts to access a base station. A TC-RNTI may be used for a wireless device in an idle state. After access is allowed by the base station, a C-RNTI may be used for indicating the wireless device. A C-RNTI may be used for a wireless device in an inactive or an active state.

The fourth step in the four-step RA procedure (e.g., step 1606 in FIG. 16) may allow HARQ retransmission. A wireless device may start a mac-ContentionResolution Timer when the wireless device transmits one or more TBs to a base station in the third step (e.g., step 1605 in FIG. 16). The wireless device may restart the mac-ContentionResolution Timer at each HARQ retransmission. When a wireless device receives data on the DL resources identified by C-RNTI or TC-RNTI in the fourth step (e.g., step 1606 in FIG. 16), the wireless device may stop the mac-ContentionResolutionTimer. If the wireless device does not detect the contention resolution identity that matches the identity transmitted by the wireless device in the third step (e.g., step 1605 in FIG. 16), the wireless device may determine that the RA procedure has failed and the wireless device may discard the TC-RNTI. Additionally or alternatively, if the mac-ContentionResolution Timer expires, the wireless device may determine that the RA procedure has failed and the wireless device may discard the TC-RNTI. If the wireless device determines that the contention resolution has failed, the wireless device may flush the HARQ buffer used for transmission of the MAC PDU and the wireless device may restart the four-step RA procedure from the first step (e.g., step 1603 in FIG. 16). The wireless device may delay subsequent RAP transmission, e.g., by a backoff time. The backoff time may be randomly selected, e.g., according to a uniform distribution between 0 and the backoff parameter value corresponding to the BI in the MAC PDU for RAR.

In a four-step RA procedure, the usage of the first two steps may be, for example, to obtain an UL time alignment for a wireless device and/or to obtain an uplink grant. The third and fourth steps may be used to setup RRC connections, and/or resolve contention from different wireless devices.

Part (b) of FIG. 16 shows a three-step contention free RA procedure. A base station may transmit RA configuration parameters to a wireless device (e.g., a UE), in step 1610. In a contention-free RA procedure, such as shown in part (b) of FIG. 16, the configuration parameters may indicate to the wireless device what preamble to send to the base station and when to send the preamble. The base station may also transmit a control command to the wireless device at step 1611. The control command may comprise, e.g., downlink control information. In a first step of the RA procedure, the wireless device may transmit a random access preamble transmission to the base station at step 1612. The RAP transmission may be based on the RA configuration parameters and the control command. In a second step of the RA procedure, the base station may transmit to the wireless device a random access response at step 1613. In a third step of the RA procedure, the wireless device may transmit scheduled transmissions at step 1614. The scheduled transmissions may be based on the RAR. The contention free RA procedure may end with the third step. Thereafter, the base station may transmit a downlink transmission to the wireless device at step 1615. This downlink transmission may comprise, e.g., an acknowledgement (ACK) indication, a non-acknowledgement (NACK) indication, data, or other infor-

mation. Contention-free RA procedures such as described above may have reduced latency compared with contention-based RA procedures. Contention-based RA procedures may involve collisions, such as when more than one wireless device is attempting to communicate with the same base station at the same time.

Part (c) of FIG. 16 shows an example of common language descriptions that may facilitate an understanding of some of the messaging involved in the contention-based four-step RA procedure described above regarding part (a) of FIG. 16. In step 1 of the RA procedure, a wireless device may send a communication to a base station similar to a request such as, "Hello, can I camp on?" (step 1620). If the base station can accommodate the wireless device request, the base station may respond to the wireless device with a message similar to an instruction such as "Send your info & data here" (step 1621). Based on the base station's response, the wireless device may send a message similar to a response such as "Here you are" (step 1622). Based on the information received by the base station, the base station may respond with a message similar to a grant such as "You are now in" (step 1623).

Part (d) of FIG. 16 shows an example of a two-step contention free random access procedure of a wireless device. At step 1630, the wireless device may receive RA configuration parameters from a base station (e.g., from a handover source base station, and/or from a handover target base station via the handover source base station). The RA configuration parameters may comprise one or more parameters indicating a type of a random access process. The type of the random access process may indicate a two-step random access process. At step 1631, the wireless device may transmit an RA preamble and one or more transport blocks as a first step of the procedure, e.g., overlapping in time with each other. In response to the RA preamble and/or the one or more transport blocks, at step 1632, the wireless device may receive an RA response from a base station (e.g., a handover target base station).

PSS, SSS, and/or PBCH may be repeated, for example, for multiple beams for a cell, to support cell selection, reselection, and/or initial access procedures. A RACH process is shown in FIG. 19. For an SS burst, the associated PBCH, or a PDSCH (e.g., indicated by a PPDCCH in common search space), scheduling system information (e.g., a SIB2), may be broadcasted to multiple wireless devices. The system information (e.g., a SIB2) may carry a PRACH configuration for a beam. A base station may have a RACH configuration for a beam, which may include PRACH preamble pool, time and/or frequency radio resources, and/or other power related parameters.

A wireless device may use a PRACH preamble selected from a RACH configuration to initiate a contention-based RACH procedure or a contention-free RACH procedure. The wireless device may perform a 4-step RACH procedure, which may be a contention-based or contention-free RACH procedure. The wireless device may select a beam associated with an SS block that may have the best receiving signal quality. The wireless device may successfully detect a cell identifier that may be associated with the cell and decode system information with a RACH configuration. The wireless device may use one PRACH preamble and select one PRACH resource from RACH resources indicated by the system information associated with the selected beam. A PRACH resource may comprise at least one of: a PRACH index indicating a PRACH preamble, a PRACH format, a PRACH numerology, time and/or frequency radio resource allocation, power setting of a PRACH transmission, and/or

other radio resource parameters. For a contention-free RACH procedure, the PRACH preamble and resource may be indicated in a DCI or other high layer signaling.

FIG. 19 shows an example of a random access procedure (e.g., via a RACH) that may include sending, by a base station, one or more SS blocks. A wireless device 1920 (e.g., a UE) may transmit one or more preambles to a base station 1921 (e.g., a gNB in NR). Each preamble transmission by the wireless device may be associated with a separate random access procedure, such as shown in FIG. 19. The random access procedure may begin at step 1901 with a base station 1921 (e.g., a gNB in NR) sending a first SS block to a wireless device 1920 (e.g., a UE). Any of the SS blocks may comprise one or more of a PSS, SSS, tertiary synchronization signal (TSS), or PBCH signal. The first SS block in step 1901 may be associated with a first PRACH configuration. At step 1902, the base station 1921 may send to the wireless device 1920 a second SS block that may be associated with a second PRACH configuration. At step 1903, the base station 1921 may send to the wireless device 1920 a third SS block that may be associated with a third PRACH configuration. At step 1904, the base station 1921 may send to the wireless device 1920 a fourth SS block that may be associated with a fourth PRACH configuration. Any number of SS blocks may be sent in the same manner in addition to, or replacing, steps 1903 and 1904. An SS burst may comprise any number of SS blocks. For example, SS burst 1910 comprises the three SS blocks sent during steps 1902-1904.

The wireless device 1920 may send to the base station 1921 a preamble, at step 1905, for example, after or in response to receiving one or more SS blocks or SS bursts. The preamble may comprise a PRACH preamble, and may be referred to as RA Msg 1. The PRACH preamble may be transmitted in step 1905 according to or based on a PRACH configuration that may be received in an SS block (e.g., one of the SS blocks from steps 1901-1904) that may be determined to be the best SS block beam. The wireless device 1920 may determine a best SS block beam from among SS blocks it may receive prior to sending the PRACH preamble. The base station 1921 may send a random access response (RAR), which may be referred to as RA Msg2, at step 1906, for example, after or in response to receiving the PRACH preamble. The RAR may be transmitted in step 1906 via a DL beam that corresponds to the SS block beam associated with the PRACH configuration. The base station 1921 may determine the best SS block beam from among SS blocks it previously sent prior to receiving the PRACH preamble. The base station 1921 may receive the PRACH preamble according to or based on the PRACH configuration associated with the best SS block beam.

The wireless device 1920 may send to the base station 1921 an RRCConnectionRequest and/or RRCConnection-ResumeRequest message, which may be referred to as RA Msg3, at step 1907, for example, after or in response to receiving the RAR. The base station 1921 may send to the wireless device 1920 an RRCConnectionSetup and/or RRCConnectionResume message, which may be referred to as RA Msg4, at step 1908, for example, after or in response to receiving the RRCConnectionRequest and/or RRCConnectionResumeRequest message. The wireless device 1920 may send to the base station 1921 an RRCConnectionSetupComplete and/or RRCConnectionResumeComplete message, which may be referred to as RA Msg5, at step 1909, for example, after or in response to receiving the RRCConnectionSetup and/or RRCConnectionResume. An RRC connection may be established between the wireless device 1920 and the base station 1921, and the random access procedure

may end, for example, after or in response to receiving the RRCConnectionSetupComplete and/or RRCConnectionResumeComplete message.

A best beam, including but not limited to a best SS block beam, may be determined based on a channel state information reference signal (CSI-RS). A wireless device may use a CSI-RS in a multi-beam system for estimating the beam quality of the links between the wireless device and a base station. For example, based on a measurement of a CSI-RS, a wireless device may report CSI for downlink channel adaption. A CSI parameter may include a precoding matrix index (PMI), a channel quality index (CQI) value, and/or a rank indicator (RI). A wireless device may report a beam index based on a reference signal received power (RSRP) measurement on a CSI-RS. The wireless device may report the beam index in a CSI resource indication (CRI) for downlink beam selection, and associated with the RSRP value of the beam. A base station may transmit a CSI-RS via a CSI-RS resource, such as via one or more antenna ports, or via one or more time and/or frequency radio resources. A beam may be associated with a CSI-RS. A CSI-RS may comprise an indication of a beam direction. Each of a plurality of beams may be associated with one of a plurality of CSI-RSs. A CSI-RS resource may be configured in a cell-specific way, for example, via common RRC signaling. Additionally or alternatively, a CSI-RS resource may be configured in a wireless device-specific way, for example, via dedicated RRC signaling and/or layer 1 and/or layer 2 (L1/L2) signaling. Multiple wireless devices in or served by a cell may measure a cell-specific CSI-RS resource. A dedicated subset of wireless devices in or served by a cell may measure a wireless device-specific CSI-RS resource. A base station may transmit a CSI-RS resource periodically, using aperiodic transmission, or using a multi-shot or semi-persistent transmission. In a periodic transmission, a base station may transmit the configured CSI-RS resource using a configured periodicity in the time domain. In an aperiodic transmission, a base station may transmit the configured CSI-RS resource in a dedicated time slot. In a multi-shot or semi-persistent (SP) transmission, a base station may transmit the configured CSI-RS resource within a configured period. A base station may transmit one or more SP CSI-RS with a configured periodicity, with a limited or unlimited duration. A base station may configure different CSI-RS resources in different terms for different purposes. Different terms may include, for example, cell-specific, device-specific, periodic, aperiodic, multi-shot, or other terms. Different purposes may include, for example, beam management, CQI reporting, or other purposes.

FIG. 20 shows an example of transmitting CSI-RSs periodically for a beam. A base station 2001 may transmit a beam in a predefined order in the time domain, such as during time periods 2003. Beams used for a CSI-RS transmission, such as for CSI-RS 2004 in transmissions 2002C and/or 2003E, may have a different beam width relative to a beam width for SS-blocks transmission, such as for SS blocks 2002A, 2002B, 2002D, and 2002F-2002H. Additionally or alternatively, a beam width of a beam used for a CSI-RS transmission may have the same value as a beam width for an SS block. Some or all of one or more CSI-RSs may be included in one or more beams. An SS block may occupy a number of OFDM symbols (e.g., 4), and a number of subcarriers (e.g., 240), carrying a synchronization sequence signal. The synchronization sequence signal may identify a cell.

FIG. 21 shows three beams that may be configured for a wireless device, for example, in a wireless device-specific

configuration. Any number of additional beams (e.g., represented by the column of blank squares) or fewer beams may be included. Beam 1 may be allocated with CSI-RS 1 that may be transmitted in some subcarriers in a resource block (RB) of a first symbol. Beam 2 may be allocated with CSI-RS 2 that may be transmitted in some subcarriers in a RB of a second symbol. Beam 3 may be allocated with CSI-RS 3 that may be transmitted in some subcarriers in a RB of a third symbol. All subcarriers in a RB may not necessarily be used for transmitting a particular CSI-RS (e.g., CSI-RS1) on an associated beam (e.g., beam 1) for that CSI-RS. By using frequency division multiplexing (FDM), other subcarriers, not used for beam 1 for the wireless device in the same RB, may be used for other CSI-RS transmissions associated with a different beam for other wireless devices. Additionally or alternatively, by using time domain multiplexing (TDM), beams used for a wireless device may be configured such that different beams (e.g., beam 1, beam 2, and beam 3) for the wireless device may be transmitted using some symbols different from beams of other wireless devices.

A wireless device may perform downlink beam management using a wireless device-specific configured CSI-RS. In a beam management procedure, a wireless device may monitor a channel quality of a beam pair link. The beam pair link may comprise a transmitting beam from a base station (e.g., a gNB in NR) and a receiving beam by the wireless device (e.g., a UE). When multiple CSI-RSs associated with multiple beams are configured, a wireless device may monitor multiple beam pair links between the base station and the wireless device.

A wireless device may transmit one or more beam management reports to a base station. A beam management report may indicate one or more beam pair quality parameters, comprising, for example, one or more beam identifications, RSRP, PMI, CQI, and/or RI, of a subset of configured beams.

A base station and/or a wireless device may perform a downlink L1/L2 beam management procedure. One or more downlink L1/L2 beam management procedures may be performed within one or multiple transmission and receiving points (TRPs).

As shown in FIG. 21, a CSI-RS may be mapped in time and frequency domains. Each square shown in FIG. 21 may represent a resource block within a bandwidth of a cell. Each resource block may comprise a number of subcarriers. A cell may have a bandwidth comprising a number of resource blocks. A base station (e.g., a gNB in NR) may transmit one or more Radio Resource Control (RRC) messages comprising CSI-RS resource configuration parameters for one or more CSI-RS. One or more of the following parameters may be configured by higher layer signaling for each CSI-RS resource configuration: CSI-RS resource configuration identity, number of CSI-RS ports, CSI-RS configuration (e.g., symbol and RE locations in a subframe), CSI-RS subframe configuration (e.g., subframe location, offset, and periodicity in a radio frame), CSI-RS power parameter, CSI-RS sequence parameter, CDM type parameter, frequency density, transmission comb, QCL parameters (e.g., QCL-scrambling identity, crs-portscount, mbsfn-subframeconfiglist, csirs-configZPId, qcl-csi-rs-configNZPId), and/or other radio resource parameters.

CSI-RS may be configured using common parameters, for example, when a plurality of wireless devices receive the same CSI-RS signal. CSI-RS may be configured using wireless device dedicated parameters, for example, when a CSI-RS is configured for a specific wireless device. CSI-RSs

may be included in RRC signaling. A wireless device may be configured, for example, depending on different MIMO beamforming types (e.g., CLASS A or CLASS B), with one or more CSI-RS resource configurations per CSI process. As an example, a wireless device may be configured using at least one of the RRC signaling shown in FIGS. 22A to 22B.

CSI-RS resources may be activated or deactivated (e.g., for some types of MIMO beamforming) by using MAC signaling. A device (e.g., a base station) communicating in a network may activate and/or deactivate configured CSI-RS resources of a serving cell by sending an activation/deactivation CSI-RS resources MAC control element. The configured CSI-RS resources may be initially deactivated, for example, during configuration and/or after a handover.

FIG. 23 part "A" shows an example of an activation/deactivation CSI-RS resources MAC control element. The activation/deactivation CSI-RS resources MAC control element may be identified by a MAC subheader with LCID. The activation/deactivation CSI-RS resources MAC control

the second entry in the list of CSI-RS, and each of A3 through A8 may correspond to the third through eighth entry, respectively, in the list of CSI-RS. The A_i field may be set to "1" to indicate that the i^{th} entry in the list of CSI-RS, which may be specified by `csi-RS-ConfigNZP-ApList`, shall be activated. The A_i field may be set to "0" to indicate that the i^{th} entry in the list shall be deactivated. For each CSI process, the number of A_i fields (e.g., $i=1, 2, \dots, 8$) which are set to 1 may be equal to the value of a higher-layer parameter, such as `activatedResources`.

A wireless device may be triggered with aperiodic CSI reporting, for example, after receiving a RRC for CSI-RS configuration and a MAC layer signaling for CSI-RS activation. The aperiodic CSI reporting may be associated with the CSI-RS resources indicated in a DCI, for example, with DCI format 0C. A CSI request field in DCI format 0C may indicate for which CSI process and/or CSI-RS resource the CSI reporting is configured, such as shown in Table 1 below.

TABLE 1

CSI Request field for PDCCH/EPDCCH with uplink DCI format in wireless device specific search space

Value of CSI request field	Description
000	No aperiodic CSI report is triggered
001	Aperiodic CSI report is triggered for a 1 st set of {CSI process, CSI-RS resource} configured by higher layers for serving cell c
010	Aperiodic CSI report is triggered for a 2 nd set of {CSI process, CSI-RS resource} configured by higher layers for serving cell c
011	Aperiodic CSI report is triggered for a 3 rd set of {CSI process, CSI-RS resource} configured by higher layers for serving cell c
100	Aperiodic CSI report is triggered for a 4 th set of {CSI process, CSI-RS resource} configured by higher layers for serving cell c
101	Aperiodic CSI report is triggered for a 5 th set of {CSI process, CSI-RS resource} configured by higher layers for serving cell c
110	Aperiodic CSI report is triggered for a 6 th set of {CSI process, CSI-RS resource} configured by higher layers for serving cell c
111	Aperiodic CSI report is triggered for a 7 th set of {CSI process, CSI-RS resource} configured by higher layers for serving cell c

element may have a variable size that may be based on the number of CSI processes configured with `csi-RS-NZP-Activation` by RRC (N). The N number of octets, shown in FIG. 23A, each of which may comprise a number of A fields (e.g., A_1 to A_i , described below), may be included in ascending order of a CSI process ID, such as the `CSI-ProcessId`.

FIG. 23 part "B" shows an example of an activation/deactivation CSI-RS command that may activate and/or deactivate CSI-RS resources for a CSI process. For example, for a wireless device that is configured with transmission mode 9, N equals 1. Transmission mode 9 may be a transmission mode in which a base station may transmit data packets with up to 8 layers, for example, if configured with multiple antennas. A wireless device may receive the data packets based on multiple DMRSs (e.g., up to 8 DMRSs (or DMRS ports)). The activation/deactivation CSI-RS resources MAC control element may apply to the serving cell on which the wireless device may receive the activation/deactivation of CSI-RS resources MAC control element.

Activation/deactivation CSI-RS resources MAC control elements may comprise an octet of fields, shown as fields A_1 to A_8 , that may indicate the activation/deactivation status of the CSI-RS resources configured by upper layers for the CSI process. A_1 may correspond to the first entry in a list of CSI-RS, which may be specified by `csi-RS-ConfigNZP-ApList` configured by upper layers. A_2 may correspond to

FIG. 24 shows examples of three beam management procedures, P1, P2, and P3. Procedure P1 may be used to enable a wireless device measurement on different transmit (Tx) beams of a TRP (or multiple TRPs), for example, to support a selection of Tx beams and/or wireless device receive (Rx) beam(s) (shown as shaded ovals in the top row and bottom row, respectively, of P1). Beamforming at a TRP (or multiple TRPs) may include, for example, an intra-TRP and/or inter-TRP Tx beam sweep from a set of different beams (shown, in the top rows of P1 and P2, as unshaded ovals rotated in a counter-clockwise direction indicated by the dashed arrow). Beamforming at a wireless device 2401, may include, for example, a wireless device Rx beam sweep from a set of different beams (shown, in the bottom rows of P1 and P3, as unshaded ovals rotated in a clockwise direction indicated by the dashed arrow). Procedure P2 may be used to enable a wireless device measurement on different Tx beams of a TRP (or multiple TRPs) (shown, in the top row of P2, as unshaded ovals rotated in a counter-clockwise direction indicated by the dashed arrow), for example, which may change inter-TRP and/or intra-TRP Tx beam(s). Procedure P2 may be performed, for example, on a smaller set of beams for beam refinement than in procedure P1. P2 may be a particular example of P1. Procedure P3 may be used to enable a wireless device measurement on the same Tx beam (shown as shaded oval in P3), for example, to change a wireless device Rx beam if the wireless device 2401 uses beamforming.

Based on a wireless device's beam management report, a base station may transmit, to the wireless device, a signal indicating that one or more beam pair links are the one or more serving beams. The base station may transmit PDCCH and/or PDSCH for the wireless device using the one or more serving beams.

A wireless device **2401** (e.g., a UE) and/or a base station **2402** (e.g., a gNB) may trigger a beam failure recovery mechanism. The wireless device **2401** may trigger a beam failure recovery (BFR) request transmission, for example, if a beam failure event occurs. A beam failure event may include, for example, a determination that a quality of beam pair link(s) of an associated control channel is unsatisfactory. A determination of an unsatisfactory quality of beam pair link(s) of an associated channel may be based on the quality falling below a threshold and/or an expiration of a timer.

The wireless device **2401** may measure a quality of beam pair link(s) using one or more reference signals (RS). One or more SS blocks, one or more CSI-RS resources, and/or one or more demodulation reference signals (DM-RSs) of a PBCH may be used as a RS for measuring a quality of a beam pair link. Each of the one or more CSI-RS resources may be associated with a CSI-RS resource index (CRI). A quality of a beam pair link may be based on one or more of an RSRP value, reference signal received quality (RSRQ) value, and/or CSI value measured on RS resources. The base station **2402** may indicate whether an RS resource, for example, that may be used for measuring a beam pair link quality, is quasi-co-located (QCLed) with one or more DM-RSs of a control channel. The RS resource and the DM-RSs of the control channel may be QCLed when the channel characteristics from a transmission via an RS to the wireless device **2401**, and the channel characteristics from a transmission via a control channel to the wireless device, are similar or the same under a configured criterion.

A wireless device may monitor a PDCCH, such as a New Radio PDCCH (NR-PDCCH), on M beam pair links simultaneously, where $M \geq 1$ and the maximum value of M may depend at least on the wireless device capability. Such monitoring may increase robustness against beam pair link blocking. A base station may transmit, and the wireless device may receive, one or more messages configured to cause the wireless device to monitor NR-PDCCH on different beam pair link(s) and/or in different NR-PDCCH OFDM symbol.

A base station may transmit higher layer signaling, and/or a MAC control element (MAC CE), that may comprise parameters related to a wireless device Rx beam setting for monitoring NR-PDCCH on multiple beam pair links. A base station may transmit one or more indications of a spatial QCL assumption between a first DL RS antenna port(s) and a second DL RS antenna port(s). The first DL RS antenna port(s) may be for one or more of a cell-specific CSI-RS, device-specific CSI-RS, SS block, PBCH with DM-RSs of PBCH, and/or PBCH without DM-RSs of PBCH. The second DL RS antenna port(s) may be for demodulation of a DL control channel. Signaling for a beam indication for a NR-PDCCH (e.g., configuration to monitor NR-PDCCH) may be via MAC CE signaling, RRC signaling, DCI signaling, or specification-transparent and/or an implicit method, and any combination thereof.

For reception of unicast DL data channel, a base station may indicate spatial QCL parameters between DL RS antenna port(s) and DM-RS antenna port(s) of DL data channel. A base station may transmit DCI (e.g., downlink grants) comprising information indicating the RS antenna

port(s). The information may indicate the RS antenna port(s) which may be QCLed with DM-RS antenna port(s). A different set of DM-RS antenna port(s) for the DL data channel may be indicated as a QCL with a different set of RS antenna port(s).

If a base station transmits a signal indicating a spatial QCL parameters between CSI-RS and DM-RS for PDCCH, a wireless device may use CSI-RSs QCLed with DM-RS for a PDCCH to monitor beam pair link quality. If a beam failure event occurs, the wireless device may transmit a beam failure recovery request, such as by a determined configuration.

If a wireless device transmits a beam failure recovery request, for example, via an uplink physical channel or signal, a base station may detect that there is a beam failure event, for the wireless device, by monitoring the uplink physical channel or signal. The base station may initiate a beam recovery mechanism to recover the beam pair link for transmitting PDCCH between the base station and the wireless device. The base station may transmit one or more control signals, to the wireless device, for example, after or in response to receiving the beam failure recovery request. A beam recovery mechanism may be, for example, an L1 scheme, or a higher layer scheme.

A base station may transmit one or more messages comprising, for example, configuration parameters for an uplink physical channel and/or a signal for transmitting a beam failure recovery request. The uplink physical channel and/or signal may be based on at least one of the following: a non-contention based PRACH (e.g., a beam failure recovery PRACH or BFR-PRACH), which may use a resource orthogonal to resources of other PRACH transmissions; a PUCCH (e.g., beam failure recovery PUCCH or BFR-PUCCH); and/or a contention-based PRACH resource. Combinations of these candidate signal and/or channels may be configured by a base station.

A base station may send a confirmation message to a wireless device, for example, after or in response to the base station receiving one or multiple BFR requests. The confirmation message may comprise the CRI associated with the candidate beam the wireless may indicate in the one or multiple BFR requests. The confirmation message may comprise an L1 control information.

A primary cell (PCell) may always be activated (e.g., such as in LTE-Advanced carrier aggregation in release 10). A base station may transmit one or more RRC message comprising configuration parameters for one or more secondary cells. Many RRC messages may be used for SCell configuration and/or reconfiguration. A base station may transmit a RRCConnectionReconfiguration message for parameters configuration of one or more secondary cells, such as shown in FIGS. 25A-25F.

A plurality of parameters may be configured to be associated with one or more secondary cell for a wireless device, such as shown in FIGS. 25A-25F. The plurality of parameters may comprise one or more of: a cell ID, an antenna configuration, a CSI-RS configuration, an SRS configuration, a PRACH configuration, etc. The one or more SCells configured by the RRC message may be activated and/or deactivated by at least one MAC control element (MAC CE). The SCell activation and/or deactivation processes may provide battery power savings. If an SCell is deactivated, the wireless device may stop receiving downlink signals and/or the wireless device may stop transmission on the SCell. The default state of an SCell may be deactivated if the SCell has been configured and/or added. Additional activation procedures, for example, using a MAC CE activation command,

may be required to activate the SCell. SCells may be deactivated either by an activation and/or deactivation MAC CE (e.g., activation/deactivation MAC CE) or by an sCellDeactivationTimer. The wireless device and the base station may maintain one sCellDeactivationTimer per SCell with a common value across SCells. The base station may maintain the activation and/or deactivation status of an SCell for a wireless device. The same initial timer value may apply to each instance of the sCellDeactivationTimer. The initial timer may be configured by RRC messaging. sCellDeactivationTimer may be included in a Mac-MainConfig dedicated parameter, for example, in an RRC message such as shown in FIGS. 26A to 26C. The configured SCells may be initially deactivated if an addition occurs and/or after a handover occurs.

The activation/deactivation MAC control element may be used in a variety of ways. The activation/deactivation MAC control element may be identified by a MAC PDU subheader, for example, with a pre-assigned LCID. The activation/deactivation MAC CE may have a fixed size, such as a single octet comprising seven C-fields and one R-field as shown in FIG. 27A and FIG. 27B. The activation/deactivation MAC control element may comprise field indicating by Ci. If there is an SCell configured with SCellIndex i, Ci may indicate the activation/deactivation status of the SCell with SCellIndex i, else the MAC entity may ignore the Ci field. The Ci field may be set to a value of "1" to indicate that the SCell with SCellIndex i may be activated. The Ci field may be set to a value of "0" to indicate that the SCell with SCellIndex i may be deactivated. The field R may correspond to a reserved bit, which may be set to a value of "0". If a wireless device is configured with a larger number of carriers (e.g., more than 5 or 7 carriers), the activation/deactivation MAC CE may comprise more than one byte, which may comprise a longer bitmap such as shown in FIG. 27B.

Deactivation timer management processes may be performed. For example, if a PDCCH on the activated SCell indicates an uplink grant or a downlink assignment; or if a PDCCH on a serving cell scheduling the activated SCell indicates an uplink grant or a downlink assignment for the activated SCell: the wireless device may restart an sCellDeactivationTimer associated with the SCell. A MAC entity may (e.g., for each TTI and for each configured SCell) perform certain functions related to activation and/or deactivation of one or more SCells. If the MAC entity receives an activation/deactivation MAC CE activating the SCell in a TTI, the MAC entity may: activate the SCell; start or restart the sCellDeactivationTimer associated with the SCell; and/or trigger PHR. If the MAC entity receives an activation/deactivation MAC CE deactivating the SCell in a TTI, or if the sCellDeactivationTimer associated with the activated SCell expires in the TTI, the MAC entity may: deactivate the SCell; stop the sCellDeactivationTimer associated with the SCell; and/or flush all HARQ buffers associated with the SCell.

If a wireless device activates the SCell, the wireless device may perform SCell operations including: SRS transmissions on the SCell; CQI, PMI, rank indicator (RI), and/or precoding type indicator (PTI) reporting for the SCell; PDCCH monitoring on the SCell; and/or PDCCH monitoring for the SCell. If the SCell is deactivated, a wireless device may perform the following actions: not transmit SRSs on the SCell; not report CQI, PMI, RI, and/or PTI for the SCell; not transmit on an UL-SCH on the SCell; not transmit on a RACH on the SCell; not monitor the PDCCH on the SCell; not monitor the PDCCH for the SCell. For an

SCell that is self-scheduled (e.g., by a PDCCH transmitted on the SCell), the wireless device may not monitor the PDCCH on the SCell if the SCell is deactivated. For an SCell that is cross-carrier scheduled by a serving cell (e.g., a cell other than the SCell), the wireless device may not monitor the PDCCH for the SCell if the SCell is deactivated. If an SCell is deactivated, the ongoing random access procedure on the SCell, if any, may be aborted.

If a wireless device receives a MAC activation command for a secondary cell in subframe n, the corresponding actions in the MAC layer may be applied no later than a minimum time period (e.g., such as indicated in 3GPP TS 36.133) and no earlier than a maximum time period (e.g., subframe n+8), except for the following: the actions related to CSI reporting and the actions related to the sCellDeactivationTimer associated with the secondary cell, which may be applied in the maximum time period (e.g., subframe n+8). If a wireless device receives a MAC deactivation command for a secondary cell or the sCellDeactivationTimer associated with the secondary cell expires in subframe n, the corresponding actions in the MAC layer may apply no later than the minimum time period (e.g., such as indicated in 3GPP TS 36.133), except for the actions related to CSI reporting which may be applied in the maximum time period (e.g., subframe n+8).

If a wireless device receives a MAC activation command for a secondary cell in subframe n, the actions related to CSI reporting and the actions related to the sCellDeactivationTimer associated with the secondary cell, may be applied in subframe n+8. If a wireless device receives a MAC deactivation command for a secondary cell or other deactivation conditions are met (e.g., the sCellDeactivationTimer associated with the secondary cell expires) in subframe n, the actions related to CSI reporting may be applied in subframe n+8.

FIG. 28 shows an example timeline for a wireless device receiving a MAC activation command. The wireless device may start or restart the sCellDeactivationTimer in the (n+8)th subframe, if the wireless device receives a MAC activation command in the nth subframe, such as shown in parts "(a)" and "(b)" of FIG. 28. The wireless device may start reporting invalid (e.g., as shown in part "(a)") or valid (e.g., as shown in part "(b)") CSI for the SCell at the (n+8)th subframe, if the wireless device receives a MAC activation command in the nth subframe. A wireless device (e.g., having slow activation) may report an invalid CSI (e.g., out-of-range CSI) at the (n+8)th subframe, such as shown in part "(a)" of FIG. 28. The wireless device may start reporting a valid CSI for the SCell as a later subframe, such as subframe n+8+k, as shown in part "(a)". A wireless device (e.g., having a quick activation) may report a valid CSI at the (n+8)th subframe, such as shown in part "(b)" of FIG. 28.

If a wireless device receives a MAC activation command for an SCell in subframe n, the wireless device may start reporting CQI, PMI, RI, and/or PTI for the SCell at subframe n+8, and/or the wireless device may start or restart the sCellDeactivationTimer associated with the SCell at subframe n+8. The sCellDeactivationTimer may be maintained in both the base station and the wireless device, wherein both wireless device and base station may stop, start, and/or restart this timer in the same TTI. Without such maintaining of the timer, the sCellDeactivationTimer in the wireless device may not be in-sync with the corresponding sCellDeactivationTimer in the base station. The base station may start monitoring and/or receiving CSI (e.g., CQI, PMI, RI, and/or PTI) according to a predefined timing in the same TTI and/or after wireless device starts transmitting the CSI. If the

CSI timings in wireless device and base station are not coordinated, for example, based on a common standard or air interface signaling, the network operation may result in inefficient operations and/or errors.

A base station may transmit, via a PDCCH, a DCI for scheduling decision and power-control commands. The DCI may comprise one or more of: downlink scheduling assignments, uplink scheduling grants, or power-control commands. The downlink scheduling assignments may comprise

less device-specific search spaces. A PDCCH wireless device-specific search space at CCE aggregation level L E {1, 2, 4, 8} may be defined by a set of PDCCH candidates for CCE aggregation level L. For a DCI format, a wireless device may be configured per serving cell by one or more higher layer parameters a number of PDCCH candidates per CCE aggregation level L.

TABLE 2

example DCI format and usage			
	DCI format	Example size (Bits)	Usage
Uplink	0	45	Uplink scheduling grant
	4	53	Uplink scheduling grant with spatial multiplexing
	6-0A, 6-0B	46, 36	Uplink scheduling grant for eMTC devices
Downlink	1C	31	Special purpose compact assignment
	1A	45	Contiguous allocation only
	1B	46	Codebook-based beamforming using CRS
	1D	46	MU-MIMO using CRS
	1	55	Flexible allocations
	2A	64	Open-loop spatial multiplexing using CRS
	2B	64	Dual-layer transmission using DM-RS (TM8)
	2C	66	Multi-layer transmission using DM-RS (TM9)
	2D	68	Multi-layer transmission using DM-RS (TM9)
	2	67	Closed-loop spatial multiplexing using CRS
	6-1A, 6-1B	46, 36	Downlink scheduling grants for eMTC devices
	3, 3A	45	Power control commands
Special	5		Sidelink operation
	6-2		Paging/direct indication for eMTC devices

one or more of: PDSCH resource indication, transport format, HARQ information, control information related to multiple antenna schemes, or a command for power control of the PUCCH used for transmission of ACK/NACK based on or in response to downlink scheduling assignments. The uplink scheduling grants may comprise one or more of: PUSCH resource indication, transport format, HARQ related information, or a power control command of the PUSCH.

Different types of control information may correspond to different DCI message sizes. Supporting spatial multiplexing with noncontiguous allocation of RBs in the frequency domain may require a larger scheduling message in comparison with an uplink grant that may allow only frequency-contiguous allocation. The DCI may be categorized into different DCI formats (e.g., such as in an LTE system), where a format may correspond to a certain message size and/or usage. Table 2 below provides a summary of example DCI formats, including the size for an example of a 20 MHz FDD operation with two Tx antennas at the base station 2602 and no carrier aggregation.

A wireless device may monitor one or more PDCCH to detect one or more DCI with one or more DCI formats. The one or more PDCCH may be transmitted in common search space or wireless device-specific search space. The wireless device may monitor PDCCH with only a limited set of DCI format, for example, to save power consumption. The wireless device may not be required to detect a DCI with DCI format 6, which may be used for an eMTC wireless device. The wireless device may consume more power based on the number of DCI formats to be detected. The more DCI formats to be detected, the more power may be consumed by the wireless device.

The one or more PDCCH candidates that a wireless device monitors may be defined in terms of PDCCH wire-

Information in the DCI formats that may be used for downlink scheduling may be organized into different groups. One or more fields of the DCI formats may comprise one or more of: resource information, such as a carrier indicator (e.g., 0 or 3 bits) and/or a RB allocation; a HARQ process number; an MCS, new data indicator (NDI), and/or RV (e.g., for the first TB and/or for the second TB); MIMO related information such as PMI, precoding information, a transport block swap flag, a power offset between PDSCH and a reference signal, a reference-signal scrambling sequence, a number of layers, and/or a number of antenna ports for a transmission; PDSCH resource-element mapping and/or QCI; downlink assignment index (DAI); a transmit power control (TPC) for PUCCH; a SRS request (e.g., 1 bit), that may comprise an indication of or trigger for a one-shot SRS transmission; an ACK and/or NACK offset; a DCI format indication, for example, which may be used to differentiate between DCI format 1A and DCI format 0 or other formats that may have the same message size; and/or padding (e.g., if necessary).

Information in the DCI formats that may be used for uplink scheduling may be organized into different groups. One or more fields of the DCI formats may comprise one or more of: resource information, such as a carrier indicator, resource allocation type, and/or a RB allocation; an MCS and/or NDI (e.g., for the first TB and/or for the second TB); a phase rotation of the uplink demodulation reference signal (DMRS); precoding information; a CSI request, a request for an aperiodic CSI report; a SRS request (e.g., 2 bits), that may comprise an indication of or a trigger for an aperiodic SRS transmission that may use one of up to three preconfigured settings; an uplink index/DAI; a TPC for PUSCH; a DCI format indication, for example, which may be used to differentiate between DCI format 1A and DCI format 0; and/or padding (e.g., if necessary).

61

A base station may perform CRC scrambling on a DCI, for example, before transmitting the DCI via a PDCCH. The base station may perform CRC scrambling, for example, by bit-wise addition (or, e.g., modulo-2 addition or exclusive OR (XOR) operation) of multiple bits of at least one wireless device identifier (e.g., C-RNTI, TC-RNTI, SI-RNTI, RA-RNTI, and the like) with the CRC bits of the DCI. The wireless device may check the CRC bits of the DCI, if detecting the DCI. The wireless device may receive the DCI if the CRC is scrambled by a sequence of bits that is the same as (or indicates a match with) the at least one wireless device identifier.

A base station may transmit one or more PDCCH in different control resource sets, for example, which may support wide bandwidth operation. The base station may transmit one or more RRC message comprising configuration parameters for one or more control resource sets. At least one of the one or more control resource sets may comprise one or more of: a first OFDM symbol (e.g., CORESET_StartSymbol); a number of consecutive OFDM symbols (e.g., CORESET_NumSymbol); a set of resource blocks (e.g., CORESET_RBSet); a CCE-to-REG mapping (e.g., CORESET_mapping); and/or a REG bundle size, such as for interleaved CCE-to-REG mapping (e.g., CORESET_REG_bundle). A wireless device may monitor PDCCH to detect a DCI on a subset of control resource sets (e.g., if control resource sets are configured). Such monitoring may reduce power consumption by the wireless.

A base station may transmit one or more messages comprising configuration parameters for one or more active bandwidth parts (BWPs). The one or more active BWPs may have different numerologies. The base station may transmit, to a wireless device, control information for cross-BWP scheduling.

FIG. 29 shows an example of multiple BWP configurations. One or more BWPs may overlap with one or more other BWPs in a frequency domain. For example, BWP 1 may overlap BWP 3, both of which may overlap BWP 4 and BWP 5; BWP 2 may overlap BWP 4; and/or BWP 4 may overlap BWP 5. One or more BWPs may have a same central frequency with one or more other BWPs. For example, BWP 1 may have a same central frequency as BWP 3.

A base station may transmit one or more messages comprising configuration parameters for one or more DL BWPs and/or one or more UL BWPs for a cell. The one or more BWPs may comprise at least one BWP as the active DL BWP or the active UL BWP, and/or zero or one BWP as the default DL BWP or the default UL BWP. For a PCell, the active DL BWP may be the DL BWP on which the wireless device may monitor one or more PDCCH and/or receive PDSCH. The active UL BWP may be the UL BWP on which the wireless device may transmit an uplink signal. For an SCell, the active DL BWP may be the DL BWP on which the wireless device may monitor one or more PDCCH and receive PDSCH when the SCell is activated, for example, by receiving an activation/deactivation MAC CE. The active UL BWP may be the UL BWP on which the wireless device may transmit PRACH and/or PUCCH (e.g., if configured) and/or PUSCH if the SCell is activated, for example, by receiving an activation/deactivation MAC CE.

Configuration of multiple BWPs may be used to reduce a wireless device power consumption. A wireless device configured to use an active BWP and a default BWP may switch to the default BWP, for example, if there is no activity on the active BWP. A default BWP may be configured to use a narrow bandwidth, and/or an active BWP may be configured

62

to use a wide bandwidth. If there is no signal transmitting on or receiving from an active BWP, the wireless device may switch the BWP to the default BWP, which may reduce power consumption.

Switching a BWP may be triggered by a DCI and/or a timer. If a wireless device receives a DCI indicating DL BWP switching from an active BWP to a new BWP, the wireless device may monitor PDCCH and/or receive PDSCH on the new BWP, for example, after or in response to receiving the DCI. If the wireless device receives a DCI indicating UL BWP switching from an active BWP to a new BWP, the wireless device may transmit PUCCH (e.g., if configured) and/or PUSCH on the new BWP, for example, after or in response to receiving the DCI.

A base station may transmit, to a wireless device, one or more RRC messages comprising a BWP inactive timer. The wireless device may start the timer, for example, if it switches its active DL BWP to a DL BWP other than the default DL BWP. The wireless device may restart the timer to the initial value, for example, if it successfully decodes a DCI to schedule PDSCH(s) in its active DL BWP. The wireless device may switch its active DL BWP to the default DL BWP, for example, if the BWP timer expires.

FIG. 30 shows an example of BWP switching associated with a BWP inactive timer. A wireless device may receive one or more RRC messages comprising parameters for an SCell and one or more BWP configuration associated with the SCell. Among the one or more BWPs, at least one BWP may be configured as the first active BWP (e.g., BWP 1), and/or one BWP may be configured as the default BWP (e.g., BWP 0). The wireless device may receive a MAC CE to activate the SCell at the n^{th} subframe. The wireless device may start or restart the sCellDeactivationTimer and/or start action related to CSI reporting for the SCell, or for the initial active BWP of the SCell, at the $(n+8)^{th}$ subframe. The wireless device may start the BWP inactive timer and/or restart the sCellDeactivationTimer (e.g., if the wireless device receives a DCI indicating switching a BWP from BWP 1 to BWP 2), at the $(n+8+k)^{th}$ subframe. If the wireless device receives a PDCCH indicating a DL scheduling on BWP 2, for example, at the $(n+8+k+m)^{th}$ subframe, the wireless device may restart the BWP inactive timer and/or the wireless device may restart the sCellDeactivationTimer. The wireless device may switch back to the default BWP (e.g., BWP 0) if the BWP inactive timer expires, for example, at the $(n+8+k+m)^{th}$ subframe. The wireless device may deactivate the SCell if the sCellDeactivationTimer expires, for example, at the $(n+8+k+m+l+o)^{th}$ subframe.

The BWP inactive timer may be applied in a PCell. A base station may transmit one or more RRC messages comprising a BWP inactive timer to a wireless device. The wireless device may start the timer if the wireless device switches its active DL BWP to a DL BWP other than the default DL BWP. The wireless device may restart the timer to the initial value if it successfully decodes a DCI to schedule PDSCH(s) in its active DL BWP. The wireless device may switch its active DL BWP to the default DL BWP if the BWP timer expires.

The BWP inactive timer may be used to reduce wireless device power consumption, for example, if the wireless device is configured with multiple cells and at least one cell has a wide bandwidth. For example, the wireless device may transmit on or receive from a narrow-bandwidth BWP on the PCell or SCell if there is no activity on an active BWP. The

wireless device may deactivate the SCell, which may be triggered by sCellDeactivationTimer expiring, if there is no activity on the SCell.

CSI reporting and/or semi-persistent (SP) CSI reporting may be activated and/or deactivated by a MAC CE. The MAC CE may comprise a BWP identifier which may reduce activation time delay that may otherwise occur for BWP switching. The MAC CE comprising a BWP identifier may enable a base station flexibility in transmitting the MAC. The MAC CE comprising a BWP identifier may reduce downlink data scheduling delay. The MAC CE comprising a BWP identifier may increase spectrum efficiency of downlink data transmission. If activation of CSI reporting is transmitted after activation of a cell, the activation may be slow. For example, a MAC CE may not be transmitted in a time urgent manner, which may result in a delayed CSI report activation. Activation of BWPs, however, may be required to be relatively quick. For example, BWP activation and/or BWP switching may be based on physical layer switching. BWPs may be used for time-sensitive communications. A base station may determine a CSI reporting delay above a threshold may be unacceptable for scheduling on a BWP. By indicating a BWP in a MAC CE prior to BWP activation and/or BWP switching, CSI reporting activation delay may be reduced. A base station may transmit, to a wireless device, a MAC CE on a first BWP to activate one or more CSI resources on a second BWP indicated by the BWP identifier.

A base station may transmit, to a wireless device, one or more RRC messages comprising configuration parameters for a cell. The cell may comprise a PCell, an SCell (e.g., an SCell of a plurality of SCells). The configuration parameters may comprise one or more BWPs comprising at least a first BWP, and/or one or more CSI report configurations comprising at least a first CSI report configuration. The one or more CSI report configurations may be associated with a semi-persistent CSI reporting on a physical uplink control channel (PUCCH). The at least a first BWP may be associated with one or more of: a first parameter for a frequency location, a second parameter for a bandwidth, a third parameter for a subcarrier spacing, and/or a fourth parameter for a cyclic prefix. A value associated with the second parameter for a bandwidth may be less than a value associated with a bandwidth of the cell.

The base station may transmit, to the wireless device, a first MAC CE comprising: a BWP identifier field indicating the first BWP; and a channel state information (CSI) report configuration activation/deactivation field indicating activation of the first CSI report configuration. The activation/deactivation field may comprise an activation command and/or a deactivation command. Additionally or alternatively, the MAC CE may comprise a BWP identifier field indicating the first BWP; a semi-persistent (SP) CSI reference signal (CSI-RS) resource set; and/or an indicator indicating activation of the SP CSI-RS resource set. The BWP identifier may comprise any number of bits, such as, for example, 1, 2, 3, or 4 bits. The MAC CE may comprise a fixed and/or predetermined length. The at least a first CSI report configuration may be associated with one or more of: a report configuration type indicator (e.g., indicating a periodic, semi-persistent, or aperiodic report configuration); reference signal resource configuration parameters; report quantity parameters; frequency domain configuration parameters; and/or time domain configuration parameters. The one or more CSI reports may be based on: one or more reference signal resources indicated by the one or more reference signal resource parameters; and/or one or more

frequency configuration parameters indicated by the one or more report frequency domain configuration parameters. The one or more CSI reports may comprise at least one of the one or more report quantities indicated by the one or more report quantity parameters.

The wireless device may receive the one or more RRC messages. The wireless device may receive the first MAC CE. The wireless device may activate the first CSI report configuration for the first BWP, for example, after or in response to receiving the first MAC CE. The wireless device may activate the first CSI report configuration via a BWP, for example, including via a BWP for the CSI reporting or via any other BWP. Additionally or alternatively, the wireless device may activate the SP CSI-RS resource set for the first BWP, for example, after or in response to receiving the first MAC CE. The wireless device may activate the SP CSI-RS resource set via a BWP, for example, including via a BWP for the SP CSI-RS resource set or via any other BWP. The wireless device may activate SP CSI reporting via a physical uplink control channel (PUCCH). The wireless device may transmit, to the base station, one or more CSI reports based on the first CSI report configuration. Additionally or alternatively, the wireless device may transmit, to the base station, one or more CSI reports based on the SP CSI-RS resource set. The wireless device may transmit the one or more CSI reports via an uplink control channel (e.g., a physical uplink control channel) and/or via a physical uplink shared channel. The uplink control channel and/or the physical uplink shared channel may be associated with the first CSI report configuration. The wireless device may transmit the one or more CSI reports with periodic, semi-persistent, or aperiodic transmission indicated by the report configuration type indicator. The wireless device may transmit, via an uplink control channel, the one or more CSI reports with semi-persistent transmission based on or in response to the report configuration type indicator indicating semi-persistent transmission. The wireless device may transmit, via a physical uplink shared channel, the one or more CSI reports with aperiodic transmission based on or in response to the report configuration type indicator indicating aperiodic transmission. The one or more CSI reports may comprise one or more of: a first parameter associated with a channel quality indicator; a second parameter associated with a precoding matrix index; a third parameter associated with a rank indicator; and/or a fourth parameter associated with a layer 1 reference signal received power.

The base station may transmit, to the wireless device, a second MAC CE comprising a second BWP identifier field indicating the first BWP; and a CSI report configuration activation/deactivation field indicating a deactivation of the first CSI report configuration. Additionally or alternatively, the base station may transmit, to the wireless device, a second MAC CE comprising a second BWP identifier field indicating the first BWP; an SP CSI-RS resource set; and an indicator indicating a deactivation of the SP CSI-RS resource set. The wireless device may receive the second MAC CE. The wireless device may deactivate the first CSI report configuration for the first BWP, for example, after or in response to receiving the second MAC CE. The wireless device may deactivate the first CSI report configuration via a BWP, for example, including via a BWP for the deactivation of the first CSI report configuration or via any other BWP. Additionally or alternatively, the wireless device may deactivate the SP CSI-RS resource set for the first BWP, for example, after or in response to receiving the second MAC CE. The wireless device may deactivate the SP CSI-RS resource set via a BWP, for example, including via a BWP

for the deactivation of the SP CSI-RS resource set or via any other BWP. The wireless device may stop the transmission of the one or more CSI reports, for example, after or in response to the deactivating.

A base station may transmit one or more RRC message comprising one or more CSI configuration parameters. The one or more CSI parameters may comprise one or more: CSI-RS resource setting; CSI reporting settings and/or CSI measurement setting. A CSI-RS resource setting may comprise one or more CSI-RS resource sets. A CSI-RS resource set may be provided for a periodic CSI-RS, (P CSI-RS) and/or a semi-persistent (SP) CSI-RS. A base station may transmit one or more P CSI-RS and/or SP CSI-RS with a configured periodicity in a time domain. The base station may transmit the one or more SP CSI-RS with a limited transmission duration that may be configured by the base station. The base station may transmit the one or more SP CSI-RS for the wireless device, for example, prior to or until the base station deactivates the one or more SP CSI-RS. The base station may deactivate and/or stop transmission of the one or more SP CSI-RS, for example, by transmitting a SP CSI-RS deactivation MAC CE and/or DCI.

A CSI-RS resource set may comprise one or more of: CSI-RS type (e.g., periodic, aperiodic, semi-persistent); CSI-RS resources (e.g., comprising a CSI-RS resource configuration identity and/or a number of CSI-RS ports); CSI RS configuration (e.g., a symbol and/or RE locations in a subframe); CSI RS subframe configuration (e.g., subframe location, offset, and/or periodicity in a radio frame); CSI-RS power parameter; CSI-RS sequence parameter; CDM type parameter; frequency density; transmission comb; and/or QCL parameters.

One or more CSI-RS resources may be transmitted periodically, using aperiodic transmission, using a multi-shot transmission, or using a semi-persistent transmission. In a periodic transmission, the configured CSI-RS resource may be transmitted using a configured periodicity in a time domain. In an aperiodic transmission, the configured CSI-RS resource may be transmitted, for example, in a dedicated time slot or subframe. In a multi-shot transmission or semi-persistent transmission, the configured CSI-RS resource may be transmitted within a configured period.

One or more CSI reporting settings may comprise one or more of: a report configuration identifier; a report type; reported CSI parameter(s); CSI type (e.g., a type I or a type II); codebook configuration; time-domain behavior; frequency granularity for CQI and/or PMI; and/or measurement restriction configurations. The report type may indicate a time domain behavior of the report (e.g., aperiodic, semi-persistent, or periodic). The one or more CSI reporting settings may comprise one or more of: a periodicity parameter; a duration parameter; and/or an offset (e.g., in unit of slots and/or subframes), for example, if the report type in a CSI reporting setting is a periodic report or a semi-persistent report. The periodicity parameter may indicate the periodicity of a CSI report. The duration parameter may indicate a duration of CSI report transmission. The offset parameter may indicate a value of a timing offset of a CSI report.

An SP CSI report may comprise multiple CSI reporting settings. An SP CSI report may comprise one CSI resource set for an SP CSI-RS. A CSI measurement setting may comprise one or more links comprising one or more link parameters. A link parameter may comprise one or more of: a CSI reporting setting indication, a CSI-RS resource setting indication, and/or measurement parameters. A base station

may trigger a CSI reporting, for example, by transmitting an RRC message, a MAC CE, and/or a DCI, such as shown in Table 3 below.

TABLE 3

Triggering of CSI Reporting for the CSI-RS Configurations			
CSI-RS Configuration	P-CSI Reporting	SP-CSI Reporting	A-CSI Reporting
P CSI-RS	Higher layers	MAC CE and/or DCI	DCI
SP CSI-RS	Not Supported	MAC CE and/or DCI	DCI
Ap. CSI-RS	Not Supported	Not Supported	DCI

A wireless device may transmit one or more SP-CSI reporting. The one or more SP-CSI reporting may be transmitted with a transmission periodicity. The one or more SP-CSI reporting may be triggered by the wireless device receiving a MAC CE, and/or DCI. The MAC CE or the DCI may indicate an SP-CSI reporting on one or more periodic (P) CSI-RS resources. The MAC CE or the DCI may indicate an SP-CSI reporting on one or more SP CSI-RS resources.

FIG. 31 shows an example of SP-CSI reporting. A base station may transmit, to a wireless device, one or RRC messages comprising configuration parameters. The configuration parameters may comprise, for example, one or more SP-CSI RS configurations. The base station may transmit, to the wireless device (e.g., at subframe n) a MAC CE and/or a DCI. The MAC CE and/or the DCI may comprise an indication of an SP CSI reporting activation. The wireless device may perform CSI measurement, for example, at subframe $n+k$. The base station may start transmitting (e.g., at the start of a CSI-RS transmission window) one or more SP CSI-RS at subframe $n+k$, for example, if the base station transmits at subframe n a MAC CE and/or DCI to trigger an SP CSI reporting. The value k may be zero, or an integer greater than zero. The value k may be configured in an RRC message and/or the value k may be predefined as a fixed value. The wireless device may transmit (e.g., during an SP-CSI RS transmission period) SP CSI reporting at subframe $n+k+m$, $n+k+m+1$, and/or $n+k+m+2 \cdot l$, $n+k+m+3 \cdot l$, etc., for example, with a periodicity of l subframes. The wireless device may stop transmitting SP CSI reporting, for example, after or in response to receiving a MAC CE and/or DCI for deactivating SP CSI reporting (e.g., which may end a CSI-RS transmission window). The value “ m ” may be configured with a RRC and/or may be predefined as a fixed value.

FIG. 32 shows an example of SP CSI reporting. A wireless device may transmit one or more SP CSI reports, for example, with a transmission periodicity and a transmission duration (e.g., SP CSI reporting duration). The transmission periodicity and/or the transmission duration may be configured by RRC messaging. The wireless device may transmit the one or more SP-CSI reports at subframe $n+k+m$, $n+k+m+1$, $n+k+m+2 \cdot l$, and/or $n+k+m+3 \cdot l$, etc., for example, with a periodicity of l subframes, within the transmission duration (e.g., SP-CSI RS transmission). The wireless device may stop transmitting the one or more SP-CSI reports, for example, if the transmission duration expires. Stopping SP CSI reporting that is triggered by a transmission duration expiring, may reduce signaling overhead relative to stopping SP CSI reporting that is triggered by explicitly sending a CSI deactivation MAC CE and/or DCI. A base station may transmit, to the wireless device, one or more MAC CE comprising one or more parameters indicating activation

67

and/or deactivation of one or more CSI resource set, for example, for aperiodic CSI reporting.

FIG. 33 shows an example of a SP CSI activation MAC CE transmission after a DCI for BWP switching. For example, the base station may transmit a DL BWP-switching DCI (e.g., at subframe n , indicating switching from BWP 0 to BWP 2), before transmitting an SP CSI activation MAC CE and/or an aperiodic CSI activation MAC CE (e.g., at subframe $n+k$, indicating CSI reporting on a new BWP such as BWP 2). A MAC CE transmitted after a BWP switching DCI may comprise a smaller size than a MAC CE transmitted before BWP switching, for example, because the MAC CE transmitted before BWP may additionally comprise a BWP identifier and/or one or more BWP parameters for the BWP switching. Multiple BWPs in a cell may be active, and the MAC CE (e.g., transmitted after a BWP switching DCI) may indicate which BWP is active. The configuration described regarding FIG. 33 may be advantageous for BWP activation and/or switching of a narrowband BWP which may benefit from a smaller size MAC CE. A wireless device configured with multiple BWPs may switch an active DL BWP to a new DL BWP, for example, if the wireless device receives a DCI indicating a DL BWP switching, and/or the wireless device may switch to a default DL BWP, for example, if a BWP inactive timer expires. The base station may transmit one or more RRC messages comprising configuration parameters. The configuration parameters may comprise CSI for BWPs. The wireless device may receive a PDCCH signal and/or a PDSCH signal on an active DL BWP (e.g., BWP 0).

In subframe n , the base station may transmit a DCI indicating active BWP switching (e.g., BWP 0 $\rightarrow$ 2, or switching from BWP 0 to BWP 2). In order to perform dynamic DL scheduling for DL BWP 2, the base station may transmit a MAC CE to activate an SP or aperiodic CSI reporting (e.g., at subframe $n+k$) on the new DL BWP (e.g., BWP 2). A HARQ mechanism may be used for the transmission of the MAC CE, for example, which may increase the likelihood that a wireless device correctly receives the MAC CE. The wireless device may transmit the first SP CSI report at subframe $n+k+m$ for DL BWP 2. The value “ m ” (e.g., a time offset between transmitting or receiving the MAC CE for CSI activation and a first CSI reporting) may be configured based on a capability of the wireless device and/or the network, or the value “ m ” be fixed (e.g., such as a predefined value). The value “ m ” may be determined based on the time used for a MAC CE transmission and/or reception with possible retransmissions using a HARQ mechanism. The value of “ m ” may be determined based on the time used for RF chain retuning at the wireless device’s receiver and/or the time used for measuring the CSI RS. BWP switching indicated by a DCI may require a quick CSI report for DL data scheduling. For example, it may not be efficient to transmit the MAC CE for CSI activation after transmitting a DCI for BWP switching.

FIG. 34 shows an example of SP CSI activation MAC CE transmission before a DCI for BWP switching. A MAC CE transmitted before a BWP switching DCI may comprise a larger size than a MAC CE transmitted after BWP switching, for example, because the MAC CE transmitted before BWP may additionally comprise a BWP identifier and/or one or more BWP parameters for the BWP switching. However, a MAC CE transmitted before a DCI for BWP switching may provide faster BWP switching, for example, by reducing CSI reporting activation delay that may otherwise occur after a BWP switching DCI and before a MAC CE. A base station may transmit a MAC CE for CSI activation before trans-

68

mitting the DCI for BWP switching. The base station may transmit the MAC CE for SP or aperiodic CSI activation for a new DL BWP (e.g., BWP 2) at subframe n . The base station may transmit a DCI for DL BWP switching (BWP 1 $\rightarrow$ 2, or switching from BWP 1 to BWP 2) at subframe $n+k$. The wireless device may transmit a first SP or aperiodic CSI reporting at subframe $n+k+o$. The value “ o ” (e.g., a time offset between receiving the DCI for BWP switching and a first CSI reporting) may be configured based on a capability of the wireless device and/or a network, or the value may be fixed (e.g., such as a predefined value). The value “ o ” may be determined based on the time used for detecting DCI, the time used for RF chain retuning, and/or the time used for measuring the one or more CSI RS. A wireless device may not use a HARQ mechanism for receiving the DCI such that the CSI reporting may be very quick, for example, relative to SP CSI activation MAC CE transmission after a DCI for BWP switching such as described above regarding FIG. 33. SP CSI activation MAC CE transmission may be transmitted after a DCI for BWP switching.

A MAC CE for activation/deactivation of a CSI report may not comprise a field indicating for which cell is the CSI report (e.g., such as in some 3GPP MAC standard specifications). The MAC CE activating a CSI report may be transmitted on an activated SCell (or PCell). A wireless device may transmit the CSI report for the activated SCell (or PCell) after or in response to receiving the MAC CE on the activated SCell (or PCell). The MAC CE may be transmitted, for example, only after the SCell is activated. CSI reports triggered by MAC CE activation may require a different implementation than CSI reports triggered by DCI. For example, a DCI triggering CSI may include uplink grants or downlink assignment, however, a MAC CE triggering CSI may not include uplink grants or downlink assignment. MAC CE CSI (e.g. SP CSI, CSI, A-CSI, and/or the like) activation/deactivation may cause transmission delay of CSI report on a BWP if multiple bandwidth parts are configured. A MAC CE based activation/deactivation mechanism for a CSI report of a BWP may provide enhanced activation/deactivation for CSI reporting. Enhancements may be applied to MAC CE based activation/deactivation mechanism for CSI reporting on a BWP. A MAC CE based activation/deactivation mechanism for CSI reporting may reduce BWP switching delay. A MAC CE based activation/deactivation mechanism for CSI reporting may enable a base station to flexibly transmit the MAC CE. A MAC CE based activation/deactivation mechanism for CSI reporting may reduce downlink data scheduling delay.

The MAC CE transmission and BWP-switching DCI described above regarding FIG. 33 and FIG. 34 may be performed on any PCell and/or on any SCell. The MAC CE transmission and BWP-switching DCI described above regarding FIG. 33 may be performed on a PCell (e.g., with a smaller MAC CE and/or for a narrowband BWP) and the MAC CE transmission and BWP-switching DCI described above regarding FIG. 34 may be performed on an SCell (e.g., with a larger MAC CE and/or for a BWP that may not be a narrowband BWP). Additionally or alternatively, the MAC CE transmission and BWP-switching DCI described above regarding FIG. 33 may be performed on a PCell (e.g., with a smaller MAC CE and/or for a narrowband BWP) and the MAC CE transmission and BWP-switching DCI described above regarding FIG. 34 may be performed on a PCell (e.g., with a larger MAC CE and/or for a BWP that may not be a narrowband BWP).

A base station may transmit a MAC CE for CSI (e.g., SP or aperiodic) activation with a BWP index, for example, to

indicate on which BWP (e.g. a DL BWP, an UL BWP) the CSI report may be configured. For example, as shown in FIG. 34, the base station may transmit a CSI activation/deactivation MAC CE for CSI activation with a BWP index (e.g., BWP 2). The base station may transmit the CSI activation/deactivation MAC CE on the active BWP (e.g., BWP 1), which may be different from the new BWP (e.g., BWP 2). Unlike a DCI comprising a BWP index and/or including uplink grant and downlink assignment, the MAC CE for CSI activation may not cause BWP switching delay for data transmission via the BWP.

Transmitting CSI activation MAC CE with a BWP index may allow a base station to flexibly schedule the MAC CE transmission. Otherwise, if the CSI activation MAC CE does not comprise a BWP index, it may be necessary to transmit the MAC CE on the new BWP, which is the target BWP to which a wireless device switches. Transmitting the MAC CE on the new BWP (e.g., without the above process described regarding FIG. 34) may cause a CSI reporting delay. Enhancements may be provided for a MAC CE based CSI report for a BWP. The enhanced MAC CE may comprise a BWP identifier indicating a BWP on which the CSI report is triggered, which may reduce BWP switching delay, enable a base station flexibility in transmitting the MAC CE, and/or reduce downlink data scheduling delay.

A wireless device may receive the MAC CE for CSI activation. The wireless device may receive a DCI for BWP switching. For example, after or in response to receiving the MAC CE comprising a field indicating BWP 2, the wireless device may measure the CSI RS on BWP 2, according to the BWP index indicated in the MAC CE. The wireless device may transmit one or more CSI report for the BWP 2. In such a manner, a wireless device may be able to transmit CSI of a specific BWP with reduced processing and delay requirements.

FIG. 35 shows an example of separate SP/aperiodic CSI activation/deactivation MAC CEs with a BWP index. A base station may transmit a CSI (e.g., SP) activation/deactivation MAC CE comprising one or more of: a BWP index or a BWP indicator; a CSI activation/deactivation indication (e.g., Act/Deact in part "A" of FIG. 35); a CSI report configuration indication (e.g., SP CSI report configuration ID); a cell index, and/or reserve bits. The BWP index may indicate on which BWP the CSI reporting is triggered. Including BWP index in the MAC CE provides more flexibility in scheduling MAC CE and reduces CSI transmission delay. The CSI activation/deactivation indication may indicate whether the CSI reporting is activated or deactivated. One bit may be used for an indication of CSI activation/deactivation (e.g., "1" may indicate CSI activation and/or CSI reporting being triggered and "0" may indicate CSI deactivation and/or CSI reporting being stopped). The CSI report configuration indication may indicate a CSI reporting setting comprising one or more CSI report parameters. The CSI activation/deactivation MAC CE may comprise one or more parameters indicating one or more PUCCH and/or PUSCH resources for the one or more CSI reports. The MAC CE may comprise multiple activation/deactivation bits, for example, if multiple SP CSI-RS resource sets are configured for a SP CSI. An activation/deactivation bit may indicate whether the resource set is activated or deactivated. The MAC CE may comprise multiple CSI report configuration indications, for example, if multiple CSI report configurations are supported for SP CSI.

A base station may transmit a CSI (e.g., aperiodic CSI) activation/deactivation MAC CE comprising at least one of: a BWP index or a BWP indicator; and/or multiple aperiodic

CSI resource activation/deactivation commands, such as shown in part "B" of FIG. 35. The BWP index may indicate the BWP on which the CSI RS resources are configured. The one or more bits for aperiodic CSI resource activation/deactivation commands in a first byte (e.g., Oct 1) may indicate the resource set(s) activated if the bit is "1", or deactivated if "0", in the first CSI resource setting. The first CSI resource setting may correspond to the first entry of CSI resource settings list comprised in a RRC message. The first bit in the CSI resource activation/deactivation commands may correspond to a first subset of the one or more CSI RS resource settings. The second bit in the CSI resource activation/deactivation commands may correspond to a second subset of the one or more CSI RS resource settings, and so on. The third byte (e.g., Oct 2) may correspond to a second CSI resource setting. Oct N byte may correspond to the Nth CSI resource setting. The number of CSI activation/deactivation command bytes in the MAC CE may depend on the number of CSI resource settings for aperiodic CSI. For example, FIG. 35 part "B" shows N bytes may be used for CSI commands, in addition to a byte for BWP index, if N CSI resource settings are configured (e.g., in an RRC message). The example herein may enhance MAC CE based CSI reporting for a BWP. The enhanced MAC CE may comprise a BWP identifier indicating a BWP on which the CSI report is triggered. Having a separate MAC CE for SP CSI activation/deactivation and aperiodic CSI activation/deactivation may allow a base station flexibility to activate/deactivate a SP CSI report and/or a aperiodic CSI report.

FIG. 36 shows an example of joint SP/aperiodic CSI activation/deactivation MAC CE with a BWP index. A base station may transmit a CSI activation/deactivation MAC CE to activate/deactivate SP CSI and a CSI. The MAC CE may comprise one or more of: a BWP index or a BWP indicator; a CSI activation/deactivation indication (e.g., for SP CSI); a SP CSI report configuration index; and/or one or more bytes for aperiodic CSI activation/deactivation. Having a joint MAC CE for SP CSI activation/deactivation and aperiodic CSI activation/deactivation may reduce signaling overhead (e.g., MAC CE transmission and HARQ feedback) for activating/deactivating a SP CSI report and/or a aperiodic CSI report.

FIG. 37 shows an example of aperiodic CSI activation MAC CE with multiple BWPs. A base station may transmit a CSI activation/deactivation MAC CE to activate/deactivate multiple CSI resource sets for multiple BWPs. The MAC CE may comprise one or more parameters. The one or more parameters may comprise one or more of: multiple BWP indexes (e.g., BWP index 1 and BWP index 2); and/or multiple CSI activation/deactivation commands (e.g., Oct 1 and Oct 2). An activation/deactivation command in Oct 1 may indicate whether the CSI resource set is activated (e.g., if the bit is "1") or deactivated (e.g., if the bit is "0") for the BWP associated with BWP index 1. The first bit in Oct 1 may correspond to the first resource set indicated in the RRC message, and the second bit may correspond to the second resource set, etc. A MAC CE indicating activation/deactivation of CSI report on multiple BWPs may enable a base station activate/deactivate CSI report on the multiple BWPs by transmitting one MAC CE, for example, instead of multiple MAC CEs. In this manner, a MAC CE indicating activation/deactivation of CSI report on multiple BWPs may reduce signal overhead for activation/deactivation of CSI report on multiple BWPs.

FIG. 38 shows an example of base station and wireless device transmissions for activating and/or deactivating CSI reporting for a BWP. A base station 3802 may transmit, to a

71

wireless device **3801**, one or more RRC messages **3803**. The one or more RRC messages **3803** may comprise RRC configuration parameters. The RRC configuration parameters may comprise parameters for one or more BWPs and/or one or more CSI report configurations. The base station **3802** may transmit, to the wireless device **3801**, a first MAC CE **3804**. The first MAC CE may **3804** comprise an indication of activation of a CSI report configuration for a BWP. The MAC CE **3804** may comprise a CSI activation/deactivation MAC CE on an active BWP, indicating activation of CSI (e.g., SP and/or aperiodic CSI) on a new BWP indicated by a BWP index comprised in the MAC CE. The wireless device **3801** may activate a CSI report configuration for the BWP indicated by the first MAC CE **3804**. The wireless device **3801** may measure reference signals according to the activated CSI report configuration. The wireless device **3801** may activate the CSI report configuration, and/or measure the reference signals, after or in response to the first MAC CE **3804**. The wireless device **3801** may switch to a new BWP indicated in the first MAC CE **3804**, and/or measure the CSI RS on the new BWP, for example, if the wireless device **3801** receives a DCI for BWP switching to the new BWP. The wireless device **3801** may transmit, to the base station **3802**, one or more CSI reports for the new BWP **3806**. The base station **3802** may transmit, to the wireless device **3801**, a second MAC CE **3807**. The second MAC CE **3807** may comprise an indication of a deactivation of the CSI report configuration for the BWP. After transmitting a DCI for BWP switching (e.g., from a first DL BWP to a second DL BWP), the base station **3802** may transmit, to the wireless device **3801**, a CSI activation/deactivation MAC CE via the second BWP (or via another BWP), indicating deactivation of CSI (e.g., SP and/or aperiodic CSI) for the first BWP associated with a BWP index indicated in the MAC CE. After transmitting a DCI for BWP switching (e.g., from a first DL BWP to a second DL BWP), the base station **3802** may not transmit a CSI activation/deactivation MAC CE via the second BWP (or via another BWP), to deactivate CSI (e.g., SP and/or aperiodic CSI) for the first BWP. The base station **3802** may deactivate the CSI for the first BWP, for example, after or in response to transmitting the DCI for BWP switching (e.g., from the first BWP to the second BWP). The wireless device **3801** may deactivate the CSI report configuration for the BWP indicated in the first MAC CE **3804**, for example, after or in response to receiving the second MAC CE **3807**. The wireless device **3801** may stop transmitting the CSI reports, for example, after or in response to receiving the second MAC CE **3807**.

A wireless device may deactivate the CSI (e.g., SP and/or aperiodic CSI) on a first BWP, for example, after or in response to receiving a DCI for BWP switching (e.g., from the first BWP to a second BWP). The base station may transmit a CSI activation/deactivation MAC CE, for example, after transmitting a DCI for BWP switching. The base station may transmit the MAC CE without a BWP index. A wireless device may activate/deactivate one or more CSI reporting according to the MAC CE, on a BWP on which the wireless device receives the MAC CE.

FIG. 39 shows an example of wireless device processes for activating CSI reporting for a BWP. At step **3901**, a wireless device may receive one or more RRC messages comprising one or more configuration parameters. The one or more configuration parameters may comprise one or more of: CSI-RS resource settings; CSI reporting settings, and/or a CSI measurement setting. The one or more RRC messages may comprise one or more BWP configuration parameters.

72

At step **3902**, the wireless device may receive a MAC CE comprising one or more of: a BWP identifier, index, and/or indicator; a CSI activation/deactivation indication; and/or a CSI report configuration indication. The MAC CE may comprise an indication to activate a CSI report (e.g., a SP CSI report) configuration for a BWP that is indicated by the MAC CE. At step **3903**, the wireless device may activate the CSI report configuration for the BWP indicated by the MAC CE. The wireless device may receive one or more DCI comprising one or more parameters indicating DL BWP switching from a first DL BWP to a second DL BWP. The wireless device may determine one or more CSI resources and/or reporting configurations on the second DL BWP, for example, if the BWP index in the MAC CE corresponds to the second DL BWP. The wireless device may measure the one or more CSI resources for the second BWP. At step **3904**, the wireless device may transmit one or more CSI reports based on the CSI report configuration indicated by the MAC CE. The wireless device may transmit one or more CSI measurements using the at least one CSI report configuration. The wireless device may transmit the one or more CSI measurement in one or more CSI reports (e.g., SP CSI reports) prior to or until the wireless device is notified to stop the transmission. After step **3904**, the process may end.

FIG. 40 shows an example of wireless device processes for deactivating CSI reporting for a BWP. The processes of FIG. 40 may be performed by a wireless device after one or more of the steps described above regarding FIG. 39. At step **4001**, the wireless device may receive a second MAC CE for deactivating CSI reporting for a BWP. The MAC CE may comprise a BWP identifier and/or an indication for deactivating a CSI report configuration for the BWP indicated by the BWP identifier. At step **4002**, the wireless device may deactivate the CSI report configuration for the BWP, for example, after or in response to receiving the second MAC CE. At step **4003**, the wireless device may stop transmitting the CSI reports (e.g., SP CSI reports) associated with the CSI report configuration for the BWP, for example, after or in response to deactivating the CSI report configuration for the BWP. After step **4003**, the process may end.

FIG. 41 shows an example of base station processes for activating CSI (e.g., SP CSI) reporting for a BWP. At step **4101**, a base station may transmit, to a wireless device, configuration parameters. The configuration parameters may comprise parameters for one or more BWPs and/or one or more CSI report configurations. At step **4102**, the base station may transmit, to the wireless device, a MAC CE. The MAC CE may comprise a CSI activation/deactivation MAC CE comprising one or more of: a CSI activation/deactivation indication; a CSI report configuration indication; and/or multiple aperiodic CSI resource activation/deactivation commands. The CSI activation/deactivation MAC CE may comprise one or more parameters indicating one or more PUCCH and/or PUSCH resources for the one or more CSI reports (e.g., SP CSI). The MAC CE may comprise a BWP identifier. The MAC CE may comprise an indication of a CSI report configuration for the BWP indicated by the BWP identifier. At step **4103**, the base station may switch the CSI report configuration into active state for the wireless device. At step **4104**, the base station may receive, from the wireless device, CSI reports that may be provided by the wireless device based on the activated CSI report configuration. After step **4104**, the process may end.

FIG. 42 shows an example of base station processes for deactivating CSI (e.g., SP CSI) reporting for a BWP. The processes of FIG. 42 may be performed by a base station after one or more of the steps described above regarding FIG. 41.

At step 4201, a base station may transmit, to a wireless device, a second MAC CE. The second MAC CE may comprise a BWP identifier. The second MAC CE may comprise an indication for deactivating, for the wireless device, a CSI report configuration for the BWP indicated by the BWP identifier. At step 4202, the base station may switch the CSI report configuration into a deactivated or inactive state for the wireless device. At step 4203, the base station may stop receiving CSI reports from the wireless device, for example, based on the wireless device receiving the indication for deactivating. After step 4203, the process may end.

A wireless device may perform a contention-based (CB) random access (RA) (e.g., RACH) procedure. A wireless device may perform a contention-free (CF) RA procedure. A contention-free random access procedure may be used, for example, to reestablish uplink synchronization upon downlink data arrival, handover, or positioning, or upon uplink synchronization of a secondary carrier. A wireless device may transmit a preamble via a random access channel resource on a cell indicated by a DCI, for example, if performing a CF RA procedure. If configured with multiple BWPs in the cell, a wireless device may be limited to perform RA on an active BWP on which the wireless device is currently operating (e.g., transmitting control, data, and/or a reference signal). If a base station indicates a RA on a new BWP other than the active BWP, the base station implementing may transmit a first DCI indicating switching from the active BWP to the new BWP to the wireless device and then transmit a second DCI indicating a RA on the new BWP. However, transmitting two DCIs may cause RA delay, increase battery power of the wireless device, increase signal overhead of a base station, and/or reduce spectrum efficiency of transmission. DCI initiating a random access process may use a DCI format different from a DCI used for BWP switching or a DCI for downlink assignment or uplink grant. Transmission of a MAC CE for initiating a preamble transmission in a wireless device may not be efficient, because MAC CE may introduce delay and overhead and may not be suitable for initiating a RA procedure in the wireless device. Enhancements may be provided for CF RA procedure and a DCI format if multiple BWPs are configured for use in a cell.

A contention-free RA procedure may be performed on a BWP other than an active BWP, for example, for purposes such as offloading the RA procedure on the active BWP and/or preparing BWP switching. A RA procedure may be delayed when performing BWP switching. A new PDCCH order may be used in place of a combination of: a DCI for BWP switching and a PDCCH order transmitted on a new BWP. The new PDCCH order may reduce delay for performing BWP switching. The new PDCCH order may comprise an indication of both BWP switching and a PRACH preamble. By triggering a BWP switching (e.g., via the indication of BWP switching such as a BWP identifier) and a random access procedure (e.g., via the PRACH) using a single command (e.g., the new PDCCH order), signaling overhead may be reduced and/or transmission delay may be reduced. The PDCCH order may comprise a BWP identifier and a RACH resource, wherein the BWP identifier may indicate a BWP on which a wireless device may transmit a preamble for a RA procedure. Delay of RA may be reduced for switching a BWP, power consumption of the wireless device may be reduced for performing a RA on a BWP, and spectrum efficiency of a system may be increased.

A base station may transmit, to a wireless device, one or more messages comprising configuration parameters for a first UL BWP of a cell. The cell may comprise, for example, a primary cell or a secondary cell of a plurality of cells. The

cell may comprise a plurality of UL BWPs. The plurality of UL BWPs may comprise the first UL BWP. The first UL BWP may be indicated by a first UL BWP index. The plurality of UL BWPs may comprise a second UL BWP. The second UL BWP may be indicated by a second UL BWP index. The first UL BWP index may be different from the second UL BWP index. A first bandwidth value associated with the first UL BWP may be less than a second bandwidth value associated with the cell. The configuration parameters may comprise one or more of: an UL BWP index indicating the first UL BWP; preamble indices one or more preambles; and/or RACH parameters indicating one or more RA resources (e.g., RACH resources). The configuration parameters for the first UL BWP may comprise one or more of: frequency resource parameters, an indication of subcarrier spacing, and/or a cyclic prefix. The RACH parameters may comprise one or more of: a preamble format, a preamble transmission numerology, a time and/or frequency radio resource parameter, and/or a power setting parameter.

The base station may transmit, to the wireless device, a first DCI, for example, to initiate a random access process (e.g., contention-free or contention based). The first DCI may comprise one or more of: a first field indicating the UL BWP index (e.g., 1, 2, 3, or 4 bits); a second field indicating a preamble of the one or more preambles; and/or a third field indicating a RA resource (e.g., RACH resource) of the one or more RA resources. The DCI may comprise a fourth field indicating a cell identifier of the cell. The base station may transmit, to the wireless device, a DCI comprising first fields indicating a plurality of UL BWP indices, a preamble identifier indicating a preamble, and/or a RACH resource indicator indicating a RACH resource. Providing an UL BWP index in a DCI indicating a RA may reduce a number of downlink DCIs and downlink overhead (e.g., one DCI for BWP switching to a new BWP, and another DCI for RA on the new BWP). Providing the UL BWP index in the DCI may avoid a wireless device blind decoding two DCIs (e.g., one for BWP switching and another for CFRA). Delay of CFRA may be reduced if switching a BWP. Power consumption of the wireless device may be reduced if performing a CFRA on a BWP. Spectrum efficiency of a system may be increased. A DCI instead of a MAC CE may be used for initiating a random access process in the wireless device to reduce the delays and overhead associated with the MAC CE.

The wireless device may receive a MAC CE indicating activation of a cell. The wireless device may receive the one or more messages comprising the configuration parameters. The wireless device may receive the first DCI. The wireless device may receive a DCI comprising an indication of an UL BWP index, a preamble identifier indicating a preamble, and/or an indication of a RACH resource. The wireless device may activate the cell, for example, after or in response to receiving a MAC CE indicating activation of the cell. The wireless device may activate the cell, for example, after or in response to receiving a second DCI indicating activation of the cell. The wireless device may receive a second DCI indicating activation of the cell. The wireless device may activate the second UL BWP as an active UL BWP. The wireless device may activate the second UL BWP after or in response to receiving a command indicating activation of the second UL BWP. The command may comprise one or more of: an RRC message, a MAC CE, and/or a DCI.

The wireless device may switch from a second UL BWP to a first UL BWP as an active UL BWP after or in response to receiving the first DCI. The wireless device may transmit

the preamble via the RA resource for the first UL BWP indicated by the UL BWP index. The wireless device may transmit the preamble via an RA resource (e.g., a RACH resource) for the first UL BWP of a cell identified by a cell identifier. The wireless device may transmit the preamble via the RACH resource for a first UL BWP indicated by the UL BWP index. This enhanced random access process and DCI format may reduce random access delay.

The base station may transmit, to the wireless device, a first DCI to initiate a random access process (e.g., contention-free or contention based). The first DCI may comprise first fields indicating a plurality of UL BWP indices, a preamble identifier indicating a preamble, and/or a RACH resource indicator indicating a RACH resource. The wireless device may select (e.g., after receiving or in response to the first DCI) an UL BWP from a plurality of UL BWPs indicated by a plurality of UL BWP indices. The wireless device may transmit, to the base station, the preamble (e.g., the preamble indicated in the first DCI). The wireless device may transmit the preamble via the RACH resource for the selected UL BWP.

The wireless device may monitor a downlink control channel for a second DCI after or in response to transmitting the preamble. The wireless device may receive the second DCI that may indicate a downlink resource of an RA response (RAR). The wireless device may receive the RAR via the downlink resource, for example, after or in response to the second DCI.

A base station may transmit, for example, RRC signaling or a PDCCH order, to trigger a wireless device to initiate a contention-free RA procedure. A PDCCH order may be a specific message transmitted on the PDCCH, which may comprise information about when to initiate a RA procedure, and for a contention-free RA procedure, the preamble to use. PDCCH orders may be used for various purposes. A DCI (e.g., corresponding to a PDCCH order) may be carried by a DCI format 1A in an LTE_A system or by any other format in any other system. The DCI may comprise one or more of: a carrier indicator (e.g., 0 to 3 bits), and/or a flag. One or more flags may comprise a flag for format0/format1A differentiation and/or a flag for format0A/format1A differentiation. A flag may comprise 1-bit, for example, where a value of "0" may indicate format 0 or format 0A, and a value "1" may indicate a format 1A or a format 1A is used. A flag may be used to indicate an RA procedure initiated by a PDCCH order, for example, only if format 1A CRC is scrambled with a C-RNTI and all the remaining fields are set as follows: localized and/or distributed VRB assignment flag may comprise 1 bit set to a value of "0"; a resource block assignment may comprise all bits set to a value of "1"; preamble index may comprise, for example, 6 bits; a PRACH mask index may comprise, for example, 4 bits; and/or all remaining bits in format 1A for compact scheduling assignment of one PDSCH codeword may be set to zero.

A preamble index and resource index (e.g., indicated by the PRACH mask index) may be indicated in a PDCCH order. A wireless device may transmit a preamble associated with a preamble index on a time-frequency resource associated with a PRACH mask index, after or in response to receiving a PDCCH order. The association between preamble index and preamble, and/or the association between time-frequency resource and PRACH mask index, may be indicated in one or more RRC messages. The base station may transmit a PDCCH order with a carrier indicator field indicating a secondary cell on which a wireless device may transmit the preamble, for example, if one or more second-

ary cells is activated but not uplink synchronized. The wireless device may decode a PDCCH with DCI format 1A scrambled by a C-RNTI on a common and/or a wireless device specific search space, for example, if the wireless device is configured by higher layers to decode PDCCHs with the CRC scrambled by the C-RNTI and/or if the wireless device is configured to receive random access procedures initiated by PDCCH orders.

Multiple UL BWPs, in a cell that is configured with multiple UL BWPs, may have different bandwidths and/or different numerologies. Different wireless devices may have different capabilities, for example, in terms of bandwidth, numerology, etc., and it may be beneficial to allow wireless devices to initiate an (e.g., CF based) RA process on different UL BWPs. For example, different UL BWPs may have different interference and/or load conditions. Wireless devices may be limited, for example, to always initiate the RA process on an initial UL BWP configured in an RRC message, or on an active UL BWP on which the wireless devices are operating. The initial and/or active UL BWP may be heavy-loaded and/or high-interfered. Some RA processes may increase delay, for example, by transmitting a first DCI for BWP switching, and a second DCI indicating the RA process. Some RA processes may increase power consumption of a wireless device. An RA process may be enhanced, for example, if multiple BWPs are configured. A base station may be enabled to dynamically indicate a BWP on which a wireless device may initiate an RA process if the BWP is light-loaded and/or low-interfered. A base station may dynamically change an UL BWP on which a wireless device may initiate a contention-free RA process. Delay of CFRA may be reduced when switching a BWP. Power consumption of the wireless device may be reduced if performing a CFRA on a BWP. Spectrum efficiency of a system may be increased.

A base station may transmit one or more RRC messages comprising configuration parameters for a plurality of cells. The configuration parameters for at least one of the plurality of cells may comprise one or more UL BWPs. The one or more UL BWPs may be associated with one or more BWP parameters. The one or more BWP parameters may comprise one or more of: an UL BWP identifier; a radio resource configuration; and/or a PRACH configuration. The one or more PRACH configurations may comprise one or more of: PRACH preambles each associated with a preamble index; a PRACH format; a PRACH numerology; time and/or frequency radio resources each associated with a resource index; power setting of PRACH transmission; and/or other radio resource parameters.

A base station may transmit one or more DCI comprising one or more of: a preamble index, a preamble resource indication, and/or an UL BWP indication to a wireless device for initiating a RA process (e.g., contention-free or contention based). The preamble index may be selected from the one or more PRACH preambles indicated in the one or more RRC messages. The preamble resource indication may be selected from the one or more time and/or frequency radio resources. The UL BWP indication may indicate an UL BWP, from the one or more UL BWPs, on which a wireless device may transmit the preamble. A wireless device may switch to the UL BWP as an active BWP, for example, after or in response to receiving the one or more DCI. A wireless device may transmit the preamble on a time and/or frequency radio resource indicated by the preamble resource indication on a UL BWP indicated by the UL BWP indication, for example, after or in response to receiving the one or more DCI. Providing an UL BWP indication in a DCI

indicating a RA may reduce a number of DCIs, one for BWP switching to a new BWP, and another for RA on the new BWP. Providing the UL BWP indication in the DCI may reduce blind decoding of a wireless device. Delay of RA may be reduced if switching a BWP. Power consumption of the wireless device may be reduced if performing a RA on a BWP. Spectrum efficiency of a system may be increased.

FIG. 43 shows an example embodiment of random access procedure (e.g., contention-free or contention based) for a BWP. A base station 4302 may transmit, to a wireless device 4301, one or more RRC messages 4303. The one or more RRC messages 4303 may comprise an RRC message configuring 4 BWPs (e.g., BWP 0, BWP 1, BWP 2, and BWP 3), among which at least one (e.g., two) of the BWPs may be configured as active BWPs (e.g., BWP 0 and BWP 2). The base station 4302 may configure one, two, three, or four of the four BWPs as active BWPs. The base station 4302 may transmit, to the wireless device 4301, a DCI (or a PDCCH order) 4304 indicating a random access procedure on an active UL BWP (e.g., BWP 0).

The base station 4302 may transmit the DCI 4304 indicating an RA procedure on an active BWP (e.g., BWP 0), with a preamble index and a preamble resource indication. The wireless device 4301 may transmit, to the base station 4302, a response message 4305 (e.g., preamble 4305) on the active BWP (e.g., BWP 0) such as a PRACH transmission on the active UL BWP indicated in the PDCCH order. The wireless device 4301 may transmit a preamble via the time and/or frequency resource for the active UL BWP (e.g., UL BWP 0) indicated in the DCI 4304. The wireless device 4301 may monitor a PDCCH to detect the RAR corresponding to the preamble the wireless device 4301 transmits. The base station 4302 may transmit, to the wireless device 4301, an RAR transmission 4306, after or in response to receiving the preamble 4305. Providing an UL BWP indicator in the PDCCH order may help a wireless device determine on which UL BWP the wireless device may transmit a preamble for the RA procedure. Collision of RA procedure on a BWP of a cell may be reduced, power consumption of the wireless device may be reduced if performing a RA on a BWP, and/or spectrum efficiency of a system may be increased.

FIG. 44 shows an example of RA (e.g., contention-free or contention based) on a BWP of an SCell for a wireless device. A base station 4402 may transmit, to a wireless device 4401, a DCI indicating an UL BWP on a secondary cell on which a wireless device 4401 may initiate a contention-free RA procedure. The base station 4402 may transmit one, to the wireless device 4401, or more RRC messages 4403 comprising configuration parameters for at least one secondary cell. The configuration parameters may comprise one or more of: UL BWPs (e.g., BWP 0, BWP 1, BWP 2, and BWP 3), among which at least one BWP (e.g., BWP 0 and BWP 2) may be configured as active BWPs. The base station 4402 may configure one, two, three, or four of the four BWPs as active BWPs.

The base station 4402 may transmit, to the wireless device 4401, one or more messages 4404 for an SCell activation. For example, the base station 4402 may transmit a MAC CE and/or DCI to activate an SCell. The base station 4402 may transmit, to the wireless device 4401, a DCI (or a PDCCH order) 4404 indicating a random access procedure on an active UL BWP (e.g., BWP 0).

The base station 4402 may transmit a DCI 4404 indicating an RA procedure on an active BWP (e.g., BWP 0). The DCI may comprise one or more of: a carrier index indicating an SCell, an UL BWP index, a preamble index, and/or a preamble resource indication. The wireless device 4401 may

transmit, to the base station 4402, a preamble 4406 on the active BWP (e.g., BWP 0) such as a PRACH transmission on the active UL BWP indicated in the PDCCH order. The wireless device 4401 may transmit a preamble via the time and/or frequency resource for the active UL BWP (e.g., UL BWP 0) of the SCell. The wireless device 4401 may monitor a PDCCH to detect an RAR corresponding to the preamble the wireless device 4401 transmits. The base station 4402 may transmit, to the wireless device 4401, an RAR transmission 4407, after or in response to receiving the preamble 4406. Providing an UL BWP indicator in the PDCCH order may help a wireless device determine on which UL BWP of the SCell the wireless device may transmit a preamble for the RA procedure. Collision of RA procedure on a BWP of a SCell may be reduced, power consumption of the wireless device may be reduced if performing a RA on a BWP, and/or the spectrum efficiency of a system may be increased.

A wireless device may receive one or more RRC message comprising: configuration parameters for a plurality of cells. The plurality of cells may comprise a primary cell and at least one secondary cell. The configuration parameters for at least one of the plurality of cells may comprise one or more of: UL BWPs associated with one or more BWP parameters. The BWP parameters may comprise: an UL BWP identifier; one or more PRACH parameters; and/or one or more radio resource configuration (e.g., frequency location, bandwidth, subcarrier spacing, and/or cyclic prefix). The one or more PRACH parameters may comprise one or more of: a PRACH index indicating a PRACH preamble, a PRACH format, a PRACH numerology, time and/or frequency radio resource allocation, power setting of PRACH transmission, and/or other radio resource parameters.

The wireless device may receive one or more MAC CE or DCI comprising: one or more parameters indicating activation and/or deactivation of one or more secondary cells. The wireless device may receive one or more DCI indicating one or more of: a parameter indicating one or more preamble indices; a parameter indicating one or more RACH mask identifiers; and/or an UL BWP identifier indicating an active UL BWP. For example, if four UL BWPs (e.g., BWP 0, BWP 1, BWP 2, and BWP 3) are configured in the RRC message, two bits may be used to indicate which BWP may be used for preamble transmission (e.g., a value of "00" may correspond to BWP 0, a value of "01" may correspond to BWP 1, a value "10" may correspond to BWP 2, and a value "11" may correspond to BWP 3). If the base station indicates a wireless device to transmit a preamble on BWP 0, the indication in the DCI may be a bit string, such as "00". The wireless device may determine one or more radio resource for transmitting the one or more preambles, for example, based on the one or more RACH mask identifiers and/or the UL BWP identifier, for a secondary cell indicated by the one or more MAC CE. The wireless device may transmit one or more preambles via the one or more radio resource on an UL BWP indicated by the UL BWP identifier. The one or more DCI may comprise a SCell index.

The wireless device may receive one or more DCI indicating one or more of: a parameter indicating one or more preamble indices; a parameter indicating one or more RACH mask identifiers; and an UL BWP identifier indicating one or more active UL BWPs. The wireless device may select one UL BWP from the multiple UL BWPs to initiate a contention-free RA procedure, for example, if multiple UL BWPs are indicated in a DCI. The multiple UL BWPs indicated in the DCI may a subset of the one or more UL BWPs configured in the RRC message. Providing multiple UL BWPs in the PDCCH order may enable a wireless device

selecting from the multiple UL BWPs an UL BWP on which the wireless device may transmit a preamble for the RA procedure. Collision of RA procedure on a BWP may be reduced and/or spectrum efficiency of a system may be increased.

An indication of multiple UL BWPs may comprise a bit-map or similar method. For example, if four UL BWPs (e.g., BWP 0, BWP 1, BWP 2, and/or BWP 3) are configured in an RRC message, and if the base station allows a wireless device to transmit a preamble on BWP 0 and BWP 2, the indication in the DCI may be a bit string "1010", where the value of "1" in the far left may indicate the first BWP (e.g., BWP 0) on which the wireless device is allowed to transmit a preamble, the value of "0" in the second bit may indicate that the second BWP (e.g., BWP 1) is not allowed for transmission of a preamble, the value of "1" in the third bit may indicate that the third BWP (e.g., BWP 2) is allowed for transmission of a preamble, and the "0" in the fourth bit may indicate that the fourth BWP (e.g., BWP 3) is not allowed for transmission of a preamble. If multiple UL BWPs are indicated in the DCI, a preamble index and/or a preamble resource indication may be associated with one of the multiple UL BWPs. For example, the DCI may comprise one or more bits associated with Preamble_index 1, Preamble_index 2, Preamble_resource_indicator 1, and/or Preamble_resource_indicator 2, where, the Preamble_index 1 may correspond to the first BWP (e.g., BWP 0), the Preamble_index 2 may correspond to the third BWP (e.g., BWP 2). The Preamble_resource_indicator 1 may correspond to the first BWP (e.g., BWP 0) and the Preamble_resource_indicator 2 may correspond to the third BWP (e.g., BWP 2).

FIG. 45 shows an example embodiment of RA procedure (e.g., contention-free or contention based) on multiple BWPs for multiple wireless devices. A wireless device 4501 may correspond to the wireless device 4301; a base station 4502 may correspond to the base station 4302; and messages 4504, 4507, 4509, and 4511 may correspond to messages 4303, 4304, 4305, and 4306; as described above regarding FIG. 43. In addition, the base station 4502 may communicate with a plurality of wireless devices comprising at least the wireless device 4501 and a wireless device 4503. The base station 4502 may configure both the wireless device 4501 (e.g., as described with respect to the wireless device 4301 in FIG. 43) and the wireless device 4503.

The base station 4502 may transmit, to the wireless device 4503, one or more RRC messages 4506. The one or more RRC messages 4506 may comprise an RRC message configuring 4 BWPs (e.g., BWP 0, BWP 1, BWP 2, and BWP 3), among which at least one (e.g., two) of the BWPs may be configured as active BWPs (e.g., BWP 0 and BWP 2). The base station 4502 may configure one, two, three, or four of the four BWPs as active BWPs. The base station 4502 may transmit the one or more RRC messages 4506 to the wireless device 4503 after, before, or simultaneous with transmitting the one or more RRC messages 4504 (e.g., corresponding to message 4303 described regarding FIG. 43) to the wireless device 4501, including, for example, before the base station 4502 receives a response 4509 (e.g., corresponding to message 4305 described regarding FIG. 43) from the wireless device 4501. The base station 4502 may transmit, to the wireless device 4503, a DCI (such as a PDCCH order) 4508 indicating a random access procedure on an active UL BWP (e.g., BWP 2). The active UL BWP (e.g., BWP 2) indicated in the DCI 4508 to the wireless device 4503 may comprise a different active UL BWP than the active UL BWP (e.g., BWP 0) indicated in the DCI (or PDCCH order) 4507 to the wireless device 4501. The base

station 4502 may transmit, to the wireless device 4501, a DCI (or PDCCH order) 4507 indicating an RA procedure on an active BWP (e.g., BWP 0), with a preamble index and a preamble resource indication. The base station 4502 may transmit the DCI 4508 to the wireless device 4503 after, before, or simultaneous with transmitting the DCI 4507 (e.g., corresponding to DCI 4304 described regarding FIG. 43) to the wireless device 4501, including, for example, before the base station 4502 receives a preamble 4509 (e.g., corresponding to preamble 4305 described regarding FIG. 43) from the wireless device 4501.

The wireless device 4503 may transmit, to the base station 4502, a preamble 4510 on the active BWP (e.g., BWP 2) such as a PRACH transmission on the active UL BWP indicated in the PDCCH order. The wireless device 4503 may transmit a preamble via the time and/or frequency resource for the active UL BWP (e.g., UL BWP 2) indicated in the DCI 4508. The wireless device 4503 may transmit the preamble 4510 to the base station 4502 after, before, or simultaneous with transmitting the preamble 4509 (e.g., corresponding to preamble 4305 described regarding FIG. 43) from the wireless device 4501, including, for example, before the base station 4502 receives the preamble 4509 from the wireless device 4501. The wireless device 4503 may monitor a PDCCH to detect the RAR corresponding to the preamble the wireless device 4503 transmits. The base station 4502 may transmit, to the wireless device 4503, an RAR transmission 4512, after or in response to receiving the response message 4510.

The configuring an initial active UL BWP for a wireless device described above regarding FIG. 43 and/or FIG. 45 may be performed on any PCell and/or on any SCell. The configuring an initial active UL BWP for a wireless device described above regarding FIG. 43 and/or FIG. 45 may be performed on a PCell (e.g., with a smaller MAC CE and/or for a narrowband BWP) and the MAC CE transmission and BWP-switching DCI described above regarding FIG. 33 and/or FIG. 34 may be performed on an SCell (e.g., with a larger MAC CE and/or for a BWP that may not be a narrowband BWP). Additionally or alternatively, the configuring an initial active UL BWP for a wireless device described above regarding FIG. 43 and/or FIG. 45 may be performed on an SCell (e.g., with a smaller MAC CE and/or for a narrowband BWP) and the MAC CE transmission and BWP-switching DCI described above regarding FIG. 33 and/or FIG. 34 may be performed on a PCell (e.g., with a larger MAC CE and/or for a BWP that may not be a narrowband BWP). One or both of the configuring an initial active UL BWP for a wireless device described above regarding FIG. 43 and/or FIG. 45 and the MAC CE transmission and BWP-switching DCI described above regarding FIG. 33 and/or FIG. 34 may be performed on any cell and/or by any wireless device, base station, or combination thereof.

FIG. 46 shows an example of wireless device processes for random access procedure (e.g., contention-free or contention based) on an active UL BWP. At step 4601, a wireless device may receive one or more RRC messages. The one or more RRC messages may comprise configuration parameters for a plurality of cells. The configuration parameters for at least one of the plurality of cells may comprise one or more of: an UL BWP associated with one or more BWP parameters. The one or more BWP parameters may comprise an UL BWP identifier; PRACH parameters; a radio resource configuration (e.g., frequency location, bandwidth, subcarrier spacing, and/or cyclic prefix). The one or more PRACH parameters may comprise one or more of: a PRACH index indicating a PRACH preamble, a PRACH

81

format, a PRACH numerology, time and/or frequency radio resource allocation, power setting of PRACH transmission, and/or other radio resource parameters. At step 4602, the wireless device may receive one or more DCI. The one or more DCI may indicate one or more of: a preamble (e.g., a parameter indicating one or more preamble indices); a RACH resource (e.g., a parameter indicating one or more RACH mask identifiers); and/or an UL BWP identifier indicating an active BWP. The wireless device may determine one or more radio resource for transmitting the one or more preambles, for example, based on the one or more RACH mask identifiers and/or the UL BWP identifier. At step 4603, the wireless device may transmit one or more preambles via the one or more radio resource on an UL BWP indicated by the UL BWP identifier. After step 4603, the process may end.

FIG. 47 shows an example of base station processes for random access procedure (e.g., contention-free or contention based) on an active UL BWP. At step 4701, a base station may transmit, to a wireless device, one or more configuration parameters. The configuration parameters may comprise one or more indications of an UL BWP (e.g., an UL BWP identifier) and/or one or more RACH resources. At step 4702, the base station may transmit, to the wireless device, one or more DCI. The one or more DCI may comprise one or more of: an indication of an UL BWP (e.g., an UL BWP identifier such as included at step 4701), a preamble (e.g., a parameter indicating one or more preamble indices), and/or a RACH resource of the one or more RACH resources. At step 4703, the base station may receive, from the wireless device, a preamble via the RACH resource (e.g., the RACH resource included at step 4702) on the UL BWP (e.g., the UL BWP indicated at step 4702). After step 4703, the process may end.

FIG. 48 shows general hardware elements that may be used to implement any of the various computing devices discussed herein, including, e.g., the base station 401, the wireless device 406, or any other base station, wireless device, or computing device described herein. The computing device 4800 may include one or more processors 4801, which may execute instructions stored in the random access memory (RAM) 4803, the removable media 4804 (such as a Universal Serial Bus (USB) drive, compact disk (CD) or digital versatile disk (DVD), or floppy disk drive), or any other desired storage medium. Instructions may also be stored in an attached (or internal) hard drive 4805. The computing device 4800 may also include a security processor (not shown), which may execute instructions of one or more computer programs to monitor the processes executing on the processor 4801 and any process that requests access to any hardware and/or software components of the computing device 4800 (e.g., ROM 4802, RAM 4803, the removable media 4804, the hard drive 4805, the device controller 4807, a network interface 4809, a GPS 4811, a Bluetooth interface 4812, a WiFi interface 4813, etc.). The computing device 4800 may include one or more output devices, such as the display 4806 (e.g., a screen, a display device, a monitor, a television, etc.), and may include one or more output device controllers 4807, such as a video processor. There may also be one or more user input devices 4808, such as a remote control, keyboard, mouse, touch screen, microphone, etc. The computing device 4800 may also include one or more network interfaces, such as a network interface 4809, which may be a wired interface, a wireless interface, or a combination of the two. The network interface 4809 may provide an interface for the computing device 4800 to communicate with a network 4810 (e.g., a

82

RAN, or any other network). The network interface 4809 may include a modem (e.g., a cable modem), and the external network 4810 may include communication links, an external network, an in-home network, a provider's wireless, coaxial, fiber, or hybrid fiber/coaxial distribution system (e.g., a DOCSIS network), or any other desired network. Additionally, the computing device 4800 may include a location-detecting device, such as a global positioning system (GPS) microprocessor 4811, which may be configured to receive and process global positioning signals and determine, with possible assistance from an external server and antenna, a geographic position of the computing device 4800.

The example in FIG. 48 may be a hardware configuration, although the components shown may be implemented as software as well. Modifications may be made to add, remove, combine, divide, etc. components of the computing device 4800 as desired. Additionally, the components may be implemented using basic computing devices and components, and the same components (e.g., processor 4801, ROM storage 4802, display 4806, etc.) may be used to implement any of the other computing devices and components described herein. For example, the various components described herein may be implemented using computing devices having components such as a processor executing computer-executable instructions stored on a computer-readable medium, as shown in FIG. 48. Some or all of the entities described herein may be software based, and may co-exist in a common physical platform (e.g., a requesting entity may be a separate software process and program from a dependent entity, both of which may be executed as software on a common computing device).

One or more features of the disclosure may be implemented in a computer-usable data and/or computer-executable instructions, such as in one or more program modules, executed by one or more computers or other devices. Generally, program modules include routines, programs, objects, components, data structures, etc. that perform particular tasks or implement particular abstract data types when executed by a processor in a computer or other data processing device. The computer executable instructions may be stored on one or more computer readable media such as a hard disk, optical disk, removable storage media, solid state memory, RAM, etc. The functionality of the program modules may be combined or distributed as desired. The functionality may be implemented in whole or in part in firmware or hardware equivalents such as integrated circuits, field programmable gate arrays (FPGA), and the like. Particular data structures may be used to more effectively implement one or more features of the disclosure, and such data structures are contemplated within the scope of computer executable instructions and computer-usable data described herein.

Many of the elements in examples may be implemented as modules. A module may be an isolatable element that performs a defined function and has a defined interface to other elements. The modules may be implemented in hardware, software in combination with hardware, firmware, wetware (i.e., hardware with a biological element) or a combination thereof, all of which may be behaviorally equivalent. For example, modules may be implemented as a software routine written in a computer language configured to be executed by a hardware machine (such as C, C++, Fortran, Java, Basic, Matlab or the like) or a modeling/simulation program such as Simulink, Stateflow, GNU Octave, or LabVIEWMathScript. Additionally or alternatively, it may be possible to implement modules using

physical hardware that incorporates discrete or programmable analog, digital and/or quantum hardware. Examples of programmable hardware may comprise: computers, microcontrollers, microprocessors, application-specific integrated circuits (ASICs); field programmable gate arrays (FPGAs); and complex programmable logic devices (CPLDs). Computers, microcontrollers, and microprocessors may be programmed using languages such as assembly, C, C++ or the like. FPGAs, ASICs, and CPLDs may be programmed using hardware description languages (HDL), such as VHSIC hardware description language (VHDL) or Verilog, which may configure connections between internal hardware modules with lesser functionality on a programmable device. The above-mentioned technologies may be used in combination to achieve the result of a functional module.

A non-transitory tangible computer readable media may comprise instructions executable by one or more processors configured to cause operations of multi-carrier communications described herein. An article of manufacture may comprise a non-transitory tangible computer readable machine-accessible medium having instructions encoded thereon for enabling programmable hardware to cause a device (e.g., a wireless device, wireless communicator, a wireless device, a base station, and the like) to allow operation of multi-carrier communications described herein. The device, or one or more devices such as in a system, may include one or more processors, memory, interfaces, and/or the like. Other examples may comprise communication networks comprising devices such as base stations, wireless devices or user equipment (wireless device), servers, switches, antennas, and/or the like. A network may comprise any wireless technology, including but not limited to, cellular, wireless, WiFi, 4G, 5G, any generation of 3GPP or other cellular standard or recommendation, wireless local area networks, wireless personal area networks, wireless ad hoc networks, wireless metropolitan area networks, wireless wide area networks, global area networks, space networks, and any other network using wireless communications. Any device (e.g., a wireless device, a base station, or any other device) or combination of devices may be used to perform any combination of one or more of steps described herein, including, e.g., any complementary step or steps of one or more of the above steps.

Although examples are described above, features and/or steps of those examples may be combined, divided, omitted, rearranged, revised, and/or augmented in any desired manner. Various alterations, modifications, and improvements will readily occur to those skilled in the art. Such alterations, modifications, and improvements are intended to be part of this description, though not expressly stated herein, and are intended to be within the spirit and scope of the disclosure. Accordingly, the foregoing description is by way of example only, and is not limiting.

What is claimed is:

1. A wireless device comprising:
 - one or more processors; and
 - memory storing instructions that, when executed by the one or more processors, cause the wireless device to:
 - receive one or more radio resource configuration (RRC) messages comprising:
 - one or more channel state information (CSI) reporting settings, wherein a first CSI reporting setting comprises:
 - a report configuration identifier;
 - a report type indicating a semi-persistent (SP) report type; and

- a periodicity parameter indicating a reporting periodicity; and
 - configuration parameters of a plurality of bandwidth parts (BWPs) of a cell;
- receive an SP CSI activation/deactivation medium access control (MAC) control element (CE) comprising:
 - a BWP indicator indicating a first BWP, of the plurality of BWPs of the cell, for which CSI reporting is triggered; and
 - an SP CSI activation/deactivation bit indicating activation of the CSI reporting based on the first CSI reporting setting;
 - receive a first downlink control information (DCI) indicating switching to the first BWP as an active BWP;
 - activate the first BWP of the plurality of BWPs; and
 - based on the SP CSI activation/deactivation MAC CE indicating the activation of the CSI reporting and based on the first BWP being the active BWP, transmit one or more SP CSI reports based on the first CSI reporting setting.
2. The wireless device of claim 1, wherein the instructions further cause the wireless device to not transmit any SP CSI reports in response to the SP CSI activation/deactivation MAC CE before receiving the first DCI.
 3. The wireless device of claim 2, wherein the instructions further cause the wireless device to:
 - receive a second DCI indicating switching from the first BWP to a second BWP as the active BWP; and
 - in response to switching from the first BWP to the second BWP as the active BWP:
 - deactivate the first BWP; and
 - stop transmission of further SP CSI reports based on the first CSI reporting setting.
 4. The wireless device of claim 2, wherein the instructions further cause the wireless device to:
 - receive a second SP CSI activation/deactivation MAC CE indicating deactivation of the CSI reporting based on the first CSI reporting setting; and
 - stop transmission of further SP CSI reports based on the first CSI reporting setting.
 5. The wireless device of claim 4, wherein the first DCI switches both a downlink BWP and an uplink BWP based on an association between the downlink BWP and the uplink BWP that allows one activation/deactivation command to switch both the DL BWP and the UL BWP.
 6. The wireless device of claim 5, wherein the first BWP is the downlink BWP.
 7. The wireless device of claim 6, wherein the first CSI reporting setting comprises an offset parameter indicating a reporting timing offset.
 8. The wireless device of claim 4, wherein the SP CSI activation/deactivation MAC CE comprises a plurality of SP CSI activation/deactivation bits, wherein for each of the plurality of SP CSI activation/deactivation bits:
 - a value of 1 indicates activation of CSI reporting based on a CSI reporting setting; and
 - a value of 0 indicates deactivation of CSI reporting based on the CSI reporting setting.
 9. The wireless device of claim 8, wherein the SP CSI activation/deactivation MAC CE indicates the report configuration identifier of the first CSI reporting setting.
 10. The wireless device of claim 1, wherein the instructions further cause the wireless device to:
 - receive a second DCI indicating switching from the first BWP to a second BWP as the active BWP; and

85

in response to switching from the first BWP to the second BWP as the active BWP;
deactivate the first BWP; and
stop transmission of further SP CSI reports based on the first CSI reporting setting.

11. The wireless device of claim 10, wherein the plurality of BWPs comprises a plurality of downlink BWPs and a plurality of uplink BWPs.

12. The wireless device of claim 11, wherein the first BWP is an uplink BWP.

13. The wireless device of claim 10, wherein the SP CSI activation/deactivation MAC CE comprises a plurality of SP CSI activation/deactivation bits, wherein for each of the plurality of SP CSI activation/deactivation bits:

a value of 1 indicates activation of CSI reporting based on a CSI reporting setting; and

a value of 0 indicates deactivation of CSI reporting based on the CSI reporting setting.

14. The wireless device of claim 13, wherein the BWP indicator indicates the first BWP for which SP CSI reporting is triggered by the SP CSI activation/deactivation MAC CE.

15. The wireless device of claim 10, wherein the first DCI switches both a downlink BWP and an uplink BWP based on an association between the downlink BWP and the uplink BWP that allows one activation/deactivation command to switch both the DL BWP and the uplink BWP.

16. The wireless device of claim 15, wherein the first BWP is the downlink BWP.

17. The wireless device of claim 16, wherein the one or more RRC messages further comprise:

one or more CSI-reference signal (CSI-RS) resource settings; and

a CSI measurement setting.

18. The wireless device of claim 17, wherein the first CSI reporting setting comprises an offset parameter indicating a reporting timing offset.

19. The wireless device of claim 17, wherein the SP CSI activation/deactivation MAC CE is for multiple CSI reporting settings when multiple CSI reporting settings are supported for SP CSI.

20. The wireless device of claim 1, wherein the SP CSI activation/deactivation MAC CE indicates one or more physical uplink control channel (PUCCH) resources for the one or more SP CSI reports.

21. The wireless device of claim 20, wherein the instructions further cause the wireless device to not transmit any SP CSI reports in response to the SP CSI activation/deactivation MAC CE before receiving the first DCI.

22. The wireless device of claim 20, wherein the instructions further cause the wireless device to:

86

receive a second DCI indicating switching from the first BWP to a second BWP as the active BWP; and
in response to switching from the first BWP to the second BWP as the active BWP:

deactivate the first BWP; and

stop transmission of further SP CSI reports based on the first CSI reporting setting.

23. The wireless device of claim 22, wherein the instructions further cause the wireless device to:

receive a second SP CSI activation/deactivation MAC CE indicating deactivation of the CSI reporting based on the first CSI reporting setting; and

stop transmission of further SP CSI reports based on the first CSI reporting setting.

24. The wireless device of claim 23, wherein the one or more RRC messages further comprise:

one or more CSI-reference signal (CSI-RS) resource settings; and

a CSI measurement setting.

25. The wireless device of claim 24, wherein the BWP indicator indicates the first BWP for which SP CSI reporting is triggered by the SP CSI activation/deactivation MAC CE.

26. The wireless device of claim 24, wherein:

the first DCI switches both a downlink BWP and an uplink BWP based on an association between the downlink BWP and the uplink BWP that allows one activation/deactivation command to switch both the DL BWP and the UL BWP; and

the first BWP is the uplink BWP.

27. The wireless device of claim 26, wherein the instructions further cause the wireless device to measure one or more CSI-RSs based on one or more CSI-RS resource settings.

28. The wireless device of claim 27, wherein the SP CSI activation/deactivation MAC CE is for multiple CSI reporting settings when multiple CSI reporting settings are supported for SP CSI.

29. The wireless device of claim 26, wherein the SP CSI activation/deactivation MAC CE indicates the report configuration identifier of the first CSI reporting setting.

30. The wireless device of claim 20, wherein the SP CSI activation/deactivation MAC CE comprises a plurality of SP CSI activation/deactivation bits, wherein for each of the plurality of SP CSI activation/deactivation bits:

a value of 1 indicates activation of CSI reporting based on a CSI reporting setting; and

a value of 0 indicates deactivation of CSI reporting based on the CSI reporting setting.

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